



PVT Report
for
Conoco Phillips Australia
Pharos-1 MDT and SCAR
AFL 2014028

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Pharos-1 MDT

AFL 2014028

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Section A - Summary of Analysis Methods

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Section B - Summary of Samples Received and Validation Data

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Reported Well and Sampling Information - MDT
Reservoir and Well Information

Field.....	
Well.....	Pharos-1
Reservoir Fluid.....	Gas
Formation.....	-
Reservoir Pressure	
Reservoir Temperature.....	
Installation.....	-
Test.....	MDT
Perforations.....	-

Sampling Information

Date sampled.....	13-July-14
Time sampled	Various
Type of samples.....	MDT
Sampling company.....	Schlumberger
Sampling point.....	Bottom-hole
Sampling Depth.....	5000.2 & 5021.0 mMDRT
Choke.....	-
Status of well.....	-
Bottomhole pressure.....	-
Bottomhole temperature.....	-
Wellhead flowing pressure.....	-
Wellhead flowing temperature.....	-
Separator pressure	-
Separator temperature	-
Pressure base.....	14.696 psia
Temperature base	15.6°C
Water flowrate.....	-
Gas gravity (Air = 1).....	-
H2S.....	-
CO2.....	-
BS&W.....	-
Oil gravity at 60°F	-

Note: Well and Sampling conditions for MDT samples 1.01 through 1.09

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Summary of Downhole Gas Samples Received and Validation Data

MDT - Gas										
Sample no.	MPSR no.	Sample Depth m MDRT	Sampling Pressure (psia)	Temp. (°C)	Laboratory opening :- Pressure Temp. (psig) (°C)		Free Water Recov'd (cm ³)	Sample Volume (cm ³)*	SBM Contam. Flashed Cond. (wt%)**	SBM Contam. Whole Fluid (wt%)**
1.01	2668	5000.2	7273	158.6	2874	19.1	15	260	9	0.2
1.02	3351	5000.2	7273	158.6	2926	18.0	15	255	8	0.2
1.03	2642	5000.2	7273	159.2	2656	18.7	15	245	14	0.4
1.04	5055	5000.2	7273	159.2	2597	18.4	10	240	13	0.3
1.05	5049	5021.0	7280	159.3	2931	18.8	15	260	27	0.9
1.06	2489	5021.0	7280	159.3	4291	18.8	15	330	24	0.8
1.08	3374	5021.0	7280	159.3	2764	19.0	12	245	24	0.8
1.09	5048	5021.0	7280	159.3	2674	18.8	12	240	25	0.9

Notes: Evaluation Only. Created with Aspose.Cells for Python via .NET. Copyright 2003 - 2025 Aspose

* Sample volumes at 10000 psig and 93°C.

** Estimated SBM contamination. See sections E and F for more information.

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Summary of Downhole Gas Samples Received and Validation Data
SCAR and Slickline - Gas

Sample no.	Oilphase Cylinder no.	Sample Depth m MDRT	Sampling Pressure (psia)	Temp. (°C)	Laboratory opening :- Pressure (psig)	Temp. (°C)	Free Water Recov'd (cm ³)	Sample Volume (cm ³)*	SBM Contam. Flashed Cond. (wt%)**	SBM Contam. Whole Fluid (wt%)**
SCAR										
1.65	14952-IB	4837.4			8234	22.0	0	185	2	<0.1
1.66	28985-IB	4837.4			8571	18.9	0	185	2	<0.1
1.67	5059-EA	4837.4			8281	18.4	2	185	2	<0.1
1.68	5142-EA	4837.4			8504	18.6	2	185	3	<0.1
1.69	14242-IB	4837.4			7578	19.7	2	185	2	<0.1
1.70	4990-EA	4837.4			7867	19.9	2	185	3	<0.1
1.71	1473-EA	4829.5			9207	19.8	0	185	2	<0.1
1.72	9083-MA	4829.5			8402	20.0	0	185	2	<0.1
1.73	14809-QA	4829.5			7921	20.0	0	185	2	<0.1
1.74	18974-QA	4829.5			8774	19.4	0	185	3	<0.1
Slickline										
1.77	18969-QA	4986.0			8882	19.9	2	365	2	<0.1
1.78	28960-IB	4986.0			8628	19.8	2	325	2	<0.1

Notes:

* Sample volumes at 10000 psig and 120°C.

** Estimated SBM contamination. See sections E and H for more information.

*** the flashed condensate samples from 1.69 and 1.73 both contain additional contamination in the C20 to C30 range. This represents approximately 5 to 10% by weight of the flashed fluid and could not be backed out of the uncontaminated compositions in section H.

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Section C - Compositional Analysis Data - MDT Gas

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**Compositional Analysis of Gas Sample No. 1.01, MPSR 2668 to C36+
 Depth: 5000.2m MDRT**

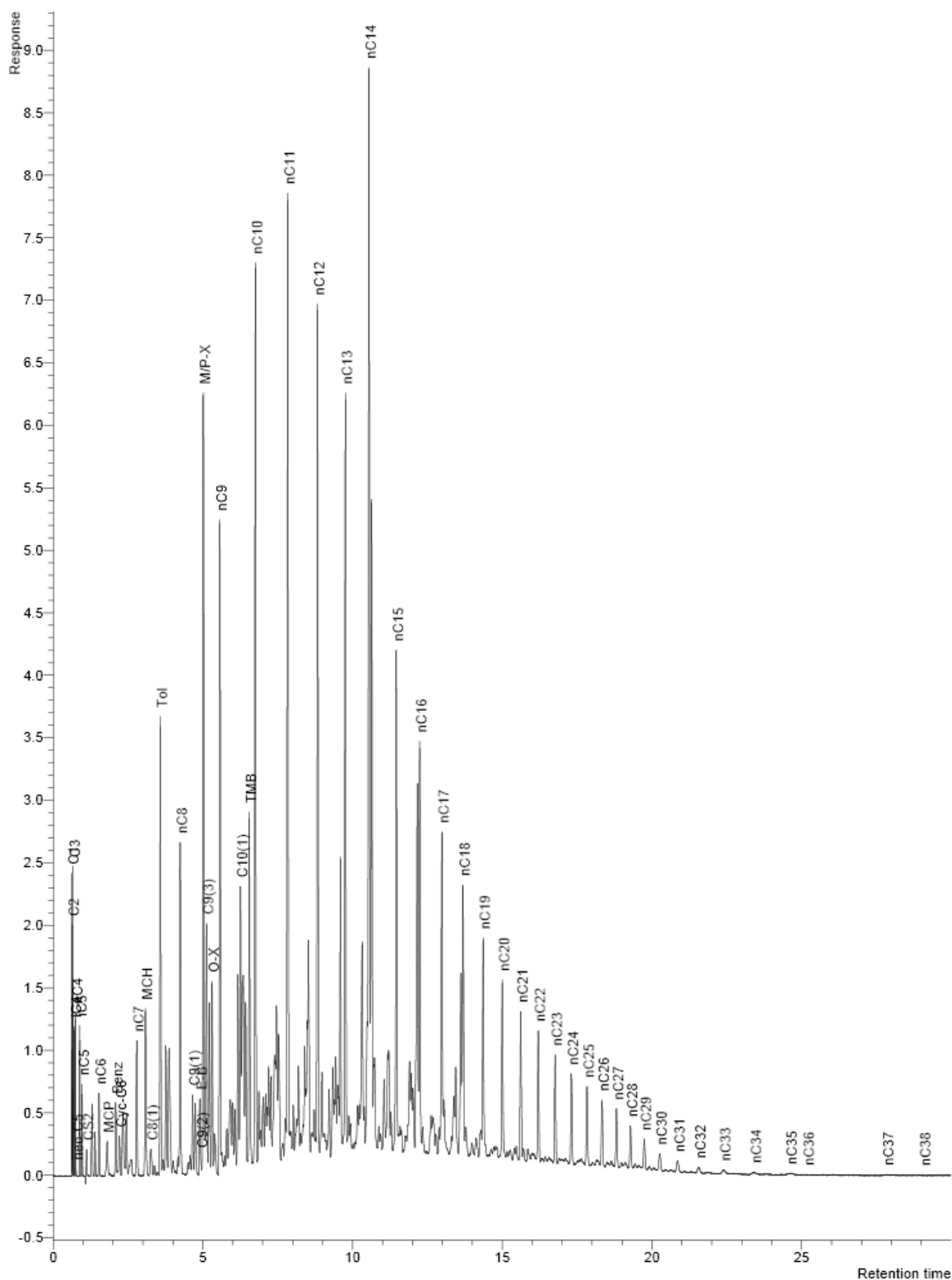
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.848	13.798	26.793
N ₂	Nitrogen	0.000	0.000	0.322	0.321	0.396
C ₁	Methane	1.273	0.123	77.806	77.528	54.878
C ₂	Ethane	0.618	0.112	4.864	4.849	6.433
C ₃	Propane	0.636	0.169	1.347	1.345	2.616
iC ₄	i-Butane	0.331	0.116	0.354	0.354	0.908
nC ₄	n-Butane	0.394	0.138	0.323	0.323	0.829
C ₅	neo-Pentane	0.014	0.006	0.003	0.003	0.011
iC ₅	i-Pentane	0.467	0.203	0.208	0.209	0.664
nC ₅	n-Pentane	0.318	0.138	0.118	0.119	0.377
C ₆	Hexanes	1.075	0.558	0.206	0.209	0.795
	Me-Cyclo-pentane	0.335	0.170	0.033	0.034	0.127
	Benzene	0.512	0.241	0.053	0.055	0.188
	Cyclo-hexane	0.308	0.156	0.059	0.060	0.221
C ₇	Heptanes	1.975	1.192	0.103	0.110	0.484
	Me-Cyclo-hexane	1.146	0.678	0.052	0.056	0.244
	Toluene	2.575	1.429	0.100	0.109	0.444
C ₈	Octanes	4.268	2.937	0.065	0.080	0.404
	Ethyl-benzene	0.399	0.255	0.005	0.007	0.031
	Meta/Para-xylene	4.383	2.803	0.038	0.054	0.254
	Ortho-xylene	1.002	0.641	0.006	0.009	0.044
C ₉	Nonanes	6.402	4.946	0.045	0.068	0.383
	Tri-Me-benzene	1.699	1.230	0.000	0.006	0.033
C ₁₀	Decanes	10.498	8.998	0.024	0.062	0.392
C ₁₁	Undecanes	10.162	8.999	0.009	0.046	0.299
C ₁₂	Dodecanes	8.496	8.240	0.004	0.035	0.250
C ₁₃	Tridecanes	7.498	7.904	0.003	0.030	0.230
C ₁₄	Tetradecanes	9.799	11.216	0.001	0.037	0.307
C ₁₅	Pentadecanes	4.796	5.951	0.000	0.017	0.158
C ₁₆	Hexadecanes	4.849	6.484	0.000	0.018	0.172
C ₁₇	Heptadecanes	2.520	3.598	0.000	0.009	0.096
C ₁₈	Octadecanes	2.908	4.397	0.000	0.011	0.117
C ₁₉	Nonadecanes	1.677	2.657	0.000	0.006	0.071
C ₂₀	Eicosanes	1.254	2.078	0.000	0.005	0.055
C ₂₁	Heneicosanes	1.003	1.759	0.000	0.004	0.047
C ₂₂	Docosanes	0.845	1.553	0.000	0.003	0.041
C ₂₃	Tricosanes	0.658	1.261	0.000	0.002	0.033
C ₂₄	Tetracosanes	0.568	1.133	0.000	0.002	0.030
C ₂₅	Pentacosanes	0.459	0.954	0.000	0.002	0.025
C ₂₆	Hexacosanes	0.388	0.839	0.000	0.001	0.022
C ₂₇	Heptacosanes	0.323	0.728	0.000	0.001	0.019
C ₂₈	Octacosanes	0.258	0.602	0.000	0.001	0.016
C ₂₉	Nonacosanes	0.207	0.502	0.000	0.001	0.013
C ₃₀	Triacontanes	0.168	0.420	0.000	0.001	0.011
C ₃₁	Hentriacontanes	0.135	0.349	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.096	0.257	0.000	0.000	0.007
C ₃₃	Trtriacontanes	0.084	0.231	0.000	0.000	0.006
C ₃₄	Tetratriacontanes	0.066	0.189	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.056	0.164	0.000	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.098	0.296	0.000	0.000	0.008
Totals :		100.000	100.000	100.000	100.000	100.000

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Compositional Analysis of Gas Sample No. 1.01, MPSR 2668 to C36+
Depth: 5000.2m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	94.874	0.601	0.942
Molecular Weight (g mol-1)	172	101.3	128
Density at 15.6°C (g cm-3)	0.8104	0.7760	0.7887
C₁₁₊			
Mole%	59.373	0.017	0.232
Molecular Weight (g mol-1)	203	158	200
Density at 15.6°C (g cm-3)	0.8307	0.7979	0.8287
C₂₀₊			
Mole%	6.667		0.023
Molecular Weight (g mol-1)	332		332
Density at 15.6°C (g cm-3)	0.8815		0.8815
C₃₆₊			
Mole %	0.098		0.000
Molecular Weight (g mol-1)	514		514
Density at 15.6°C (g cm-3)	0.9216		0.9216
Whole Sample Properties			
Average mole weight (g mol-1)	166 (Calculated)	22.2	22.7
Density (g cm-3 at 15.6°C)	0.8041 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7669	
GOR Ratio (scf/bbl) from flash at 19.5°C			176483.1
CGR Ratio (bbl/MMscf) from flash at 19.5°C			5.7
Gas Mole Fraction			0.9964
Liquid Mole Fraction			0.0036

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Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.01, MPSR 2668**
Depth: 5000.2m MDRTPharos-1 S/F (1,1)
Acquired Friday, 19 September 2014 8:00:01 PM

wb14092014,as_2.10061,1,1,1



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**Compositional Analysis of Gas Sample No. 1.02, MPSR 3351 to C36+
 Depth: 5000.2m MDRT**

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.886	13.834	26.876
N ₂	Nitrogen	0.000	0.000	0.335	0.334	0.413
C ₁	Methane	1.431	0.140	77.808	77.527	54.899
C ₂	Ethane	0.682	0.125	4.864	4.848	6.435
C ₃	Propane	0.699	0.188	1.328	1.325	2.580
iC ₄	i-Butane	0.361	0.128	0.350	0.350	0.899
nC ₄	n-Butane	0.443	0.157	0.314	0.315	0.808
C ₅	neo-Pentane	0.011	0.005	0.004	0.004	0.014
iC ₅	i-Pentane	0.498	0.219	0.204	0.205	0.654
nC ₅	n-Pentane	0.336	0.148	0.113	0.114	0.363
C ₆	Hexanes	1.128	0.593	0.205	0.208	0.792
	Me-Cyclo-pentane	0.351	0.180	0.033	0.034	0.126
	Benzene	0.495	0.236	0.052	0.053	0.184
	Cyclo-hexane	0.329	0.169	0.058	0.059	0.220
C ₇	Heptanes	1.993	1.218	0.100	0.107	0.475
	Me-Cyclo-hexane	1.149	0.688	0.051	0.055	0.239
	Toluene	2.540	1.427	0.100	0.109	0.443
C ₈	Octanes	4.279	2.981	0.062	0.078	0.391
	Ethyl-benzene	0.406	0.263	0.005	0.007	0.031
	Meta/Para-xylene	4.352	2.818	0.036	0.052	0.246
	Ortho-xylene	1.005	0.651	0.006	0.010	0.045
C ₉	Nonanes	6.469	5.060	0.043	0.066	0.376
	Tri-Me-benzene	1.696	1.243	0.000	0.006	0.033
C ₁₀	Decanes	10.629	9.222	0.025	0.064	0.405
C ₁₁	Undecanes	10.331	9.261	0.011	0.049	0.318
C ₁₂	Dodecanes	8.602	8.446	0.005	0.037	0.260
C ₁₃	Tridecanes	7.589	8.099	0.002	0.030	0.231
C ₁₄	Tetradecanes	9.492	10.998	0.000	0.035	0.295
C ₁₅	Pentadecanes	4.808	6.040	0.000	0.018	0.162
C ₁₆	Hexadecanes	4.740	6.417	0.000	0.018	0.172
C ₁₇	Heptadecanes	2.514	3.633	0.000	0.009	0.097
C ₁₈	Octadecanes	2.841	4.348	0.000	0.011	0.117
C ₁₉	Nonadecanes	1.652	2.649	0.000	0.006	0.071
C ₂₀	Eicosanes	1.249	2.095	0.000	0.005	0.056
C ₂₁	Heneicosanes	0.970	1.722	0.000	0.004	0.046
C ₂₂	Docosanes	0.782	1.455	0.000	0.003	0.039
C ₂₃	Tricosanes	0.630	1.222	0.000	0.002	0.033
C ₂₄	Tetracosanes	0.538	1.086	0.000	0.002	0.029
C ₂₅	Pentacosanes	0.424	0.892	0.000	0.002	0.024
C ₂₆	Hexacosanes	0.353	0.773	0.000	0.001	0.021
C ₂₇	Heptacosanes	0.285	0.649	0.000	0.001	0.017
C ₂₈	Octacosanes	0.224	0.531	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.175	0.429	0.000	0.001	0.012
C ₃₀	Triacontanes	0.144	0.365	0.000	0.001	0.010
C ₃₁	Hentriacontanes	0.112	0.293	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.081	0.218	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.065	0.182	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.047	0.134	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.034	0.100	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.034	0.104	0.000	0.000	0.003
Totals :		100.000	100.000	100.000	100.000	100.000

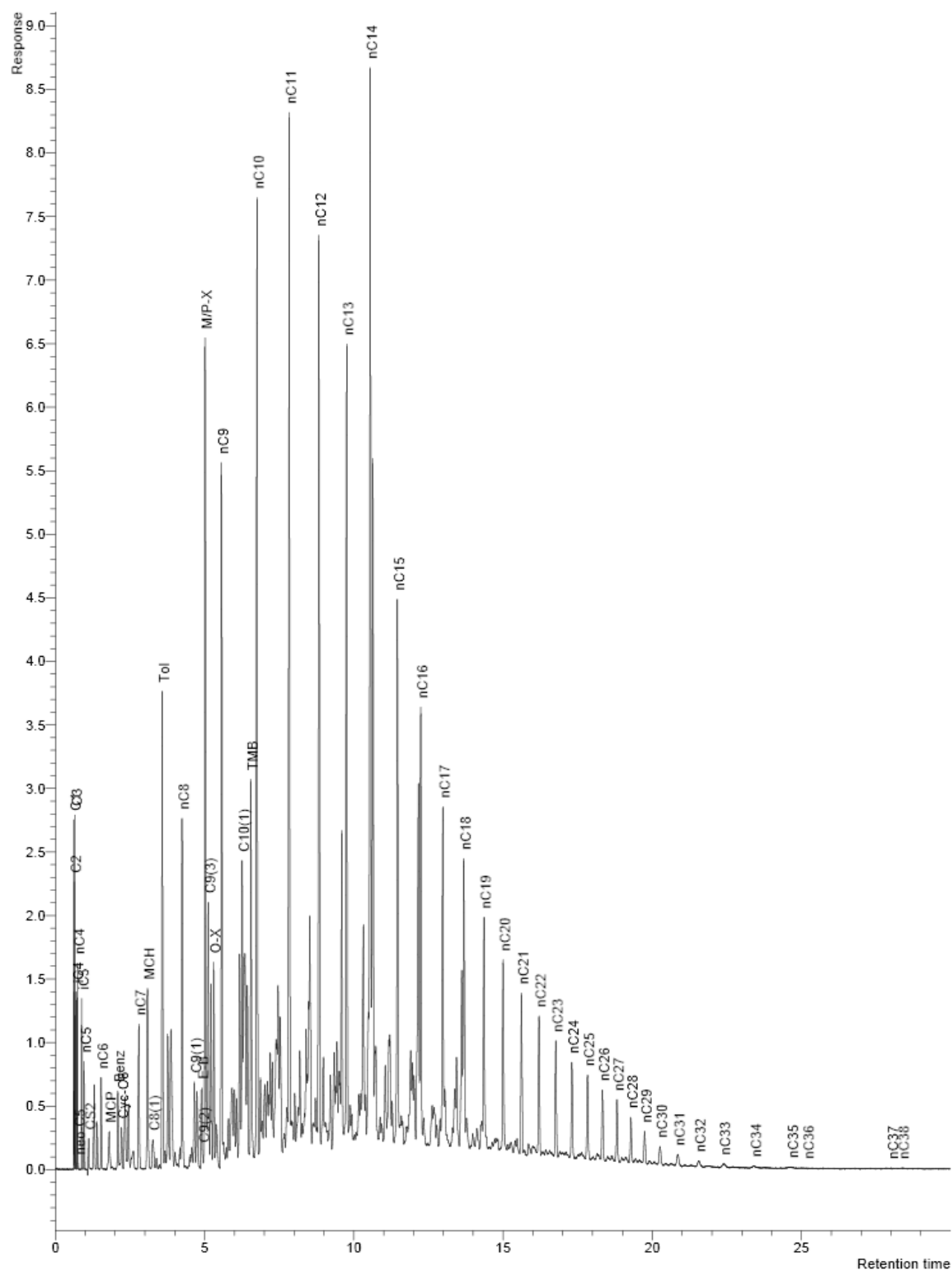
Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.02, MPSR 3351 to C36+
Depth: 5000.2m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	94.410	0.589	0.936
Molecular Weight (g mol-1)	171	101.1	127
Density at 15.6°C (g cm-3)	0.8090	0.7762	0.7885
C₁₁₊			
Mole%	58.715	0.017	0.236
Molecular Weight (g mol-1)	202	154	198
Density at 15.6°C (g cm-3)	0.8295	0.7946	0.8274
C₂₀₊			
Mole%	6.147		0.023
Molecular Weight (g mol-1)	327		327
Density at 15.6°C (g cm-3)	0.8799		0.8799
C₃₆₊			
Mole %	0.034		0.000
Molecular Weight (g mol-1)	509		509
Density at 15.6°C (g cm-3)	0.9207		0.9207
Whole Sample Properties			
Average mole weight (g mol-1)	164 (Calculated)	22.1	22.7
Density (g cm-3 at 15.6°C)	0.8021 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7664	
GOR Ratio (scf/bbl) from flash at 18.8°C			174429.9
CGR Ratio (bbl/MMscf) from flash at 18.8°C			5.7
Gas Mole Fraction			0.9963
Liquid Mole Fraction			0.0037

Fingerprint Profile of Flashed Condensate, Sample No .1.02, MPSR 3351
Depth: 5000.2m MDRT

Pharos-1 S/F (1,1)
Acquired Thursday, 18 September 2014 8:04:30 PM

wb14092014,as_2.10040,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Gas Sample No. 1.03, MPSR 2642 to C36+
 Depth: 5000.2m MDRT**

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.642	13.590	26.533
N ₂	Nitrogen	0.000	0.000	0.368	0.366	0.455
C ₁	Methane	1.712	0.165	78.153	77.869	55.416
C ₂	Ethane	0.592	0.107	4.809	4.793	6.394
C ₃	Propane	0.589	0.156	1.318	1.315	2.573
iC ₄	i-Butane	0.312	0.109	0.345	0.345	0.890
nC ₄	n-Butane	0.381	0.133	0.312	0.312	0.804
C ₅	neo-Pentane	0.014	0.006	0.005	0.005	0.015
iC ₅	i-Pentane	0.445	0.193	0.195	0.196	0.629
nC ₅	n-Pentane	0.305	0.132	0.109	0.109	0.350
C ₆	Hexanes	1.043	0.540	0.193	0.196	0.751
	Me-Cyclo-pentane	0.330	0.167	0.031	0.032	0.119
	Benzene	0.462	0.217	0.049	0.050	0.174
	Cyclo-hexane	0.307	0.155	0.055	0.056	0.208
C ₇	Heptanes	1.867	1.124	0.095	0.102	0.451
	Me-Cyclo-hexane	1.073	0.633	0.049	0.052	0.229
	Toluene	2.370	1.312	0.094	0.103	0.420
C ₈	Octanes	4.019	2.758	0.058	0.073	0.371
	Ethyl-benzene	0.384	0.245	0.005	0.006	0.029
	Meta/Para-xylene	4.094	2.611	0.036	0.051	0.242
	Ortho-xylene	0.952	0.607	0.006	0.009	0.043
C ₉	Nonanes	6.074	4.680	0.041	0.064	0.362
	Tri-Me-benzene	1.670	1.206	0.000	0.006	0.033
C ₁₀	Decanes	10.038	8.579	0.024	0.062	0.388
C ₁₁	Undecanes	9.864	8.710	0.009	0.046	0.297
C ₁₂	Dodecanes	8.316	8.042	0.001	0.033	0.233
C ₁₃	Tridecanes	7.275	7.647	0.000	0.027	0.212
C ₁₄	Tetradecanes	11.833	13.505	0.000	0.044	0.374
C ₁₅	Pentadecanes	4.618	5.714	0.000	0.017	0.158
C ₁₆	Hexadecanes	5.571	7.429	0.000	0.021	0.206
C ₁₇	Heptadecanes	2.449	3.487	0.000	0.009	0.097
C ₁₈	Octadecanes	3.323	5.010	0.000	0.012	0.139
C ₁₉	Nonadecanes	1.620	2.560	0.000	0.006	0.071
C ₂₀	Eicosanes	1.211	2.001	0.000	0.005	0.055
C ₂₁	Heneicosanes	0.936	1.636	0.000	0.004	0.045
C ₂₂	Docosanes	0.783	1.435	0.000	0.003	0.040
C ₂₃	Tricosanes	0.601	1.148	0.000	0.002	0.032
C ₂₄	Tetracosanes	0.515	1.024	0.000	0.002	0.028
C ₂₅	Pentacosanes	0.413	0.855	0.000	0.002	0.024
C ₂₆	Hexacosanes	0.338	0.729	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.285	0.640	0.000	0.001	0.018
C ₂₈	Octacosanes	0.224	0.523	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.182	0.440	0.000	0.001	0.012
C ₃₀	Triacontanes	0.150	0.374	0.000	0.001	0.010
C ₃₁	Hentriacontanes	0.122	0.314	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.088	0.236	0.000	0.000	0.007
C ₃₃	Trtriacontanes	0.077	0.211	0.000	0.000	0.006
C ₃₄	Tetratriacontanes	0.061	0.174	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.051	0.148	0.000	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.058	0.173	0.000	0.000	0.005
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.03, MPSR 2642 to C36+
Depth: 5000.2m MDRT

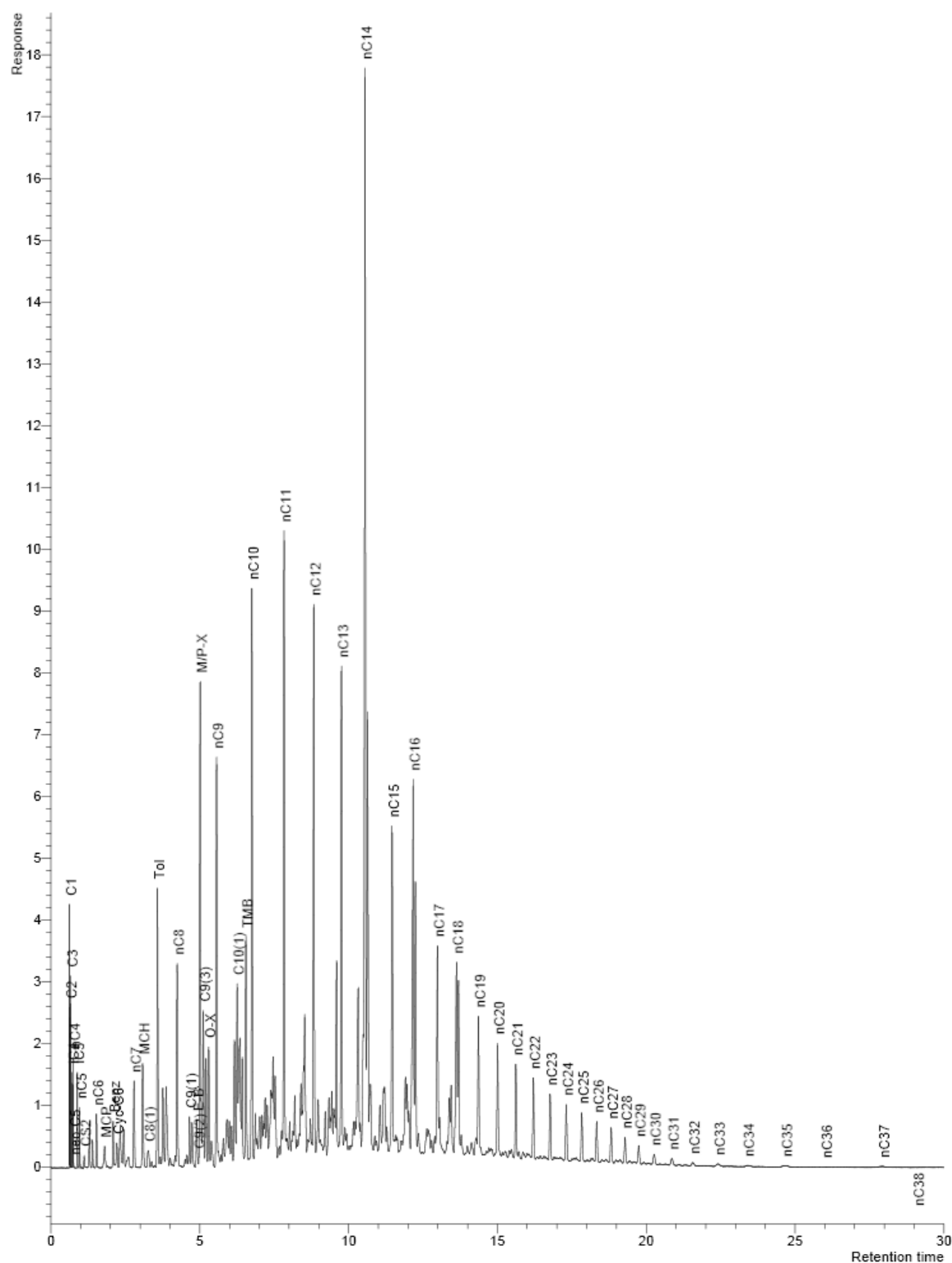
Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	94.606	0.552	0.904
Molecular Weight (g mol-1)	173	100.5	129
Density at 15.6°C (g cm-3)	0.8110	0.7760	0.7902
C₁₁₊			
Mole%	60.963	0.010	0.238
Molecular Weight (g mol-1)	203	149	200
Density at 15.6°C (g cm-3)	0.8300	0.7907	0.8288
C₂₀₊			
Mole%	6.095		0.023
Molecular Weight (g mol-1)	330		329
Density at 15.6°C (g cm-3)	0.8808		0.8808
C₃₆₊			
Mole %	0.058		0.000
Molecular Weight (g mol-1)	515		507
Density at 15.6°C (g cm-3)	0.9221		0.9206
Whole Sample Properties			
Average mole weight (g mol-1)	167 (Calculated)	22.0	22.5
Density (g cm-3 at 15.6°C)	0.8043 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7619	
GOR Ratio (scf/bbl) from flash at 20.4°C			170121.3
CGR Ratio (bbl/MMscf) from flash at 20.4°C			5.9
Gas Mole Fraction			0.9962
Liquid Mole Fraction			0.0038

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.03, MPSR 2642**
Depth: 5000.2m MDRT

Pharos-1 S/F (1,1)

Acquired Monday, 15 September 2014 11:19:29 PM

wb14092014,as_2.10010,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Gas Sample No. 1.04, MPSR 5055 to C36+
 Depth: 5000.2m MDRT**

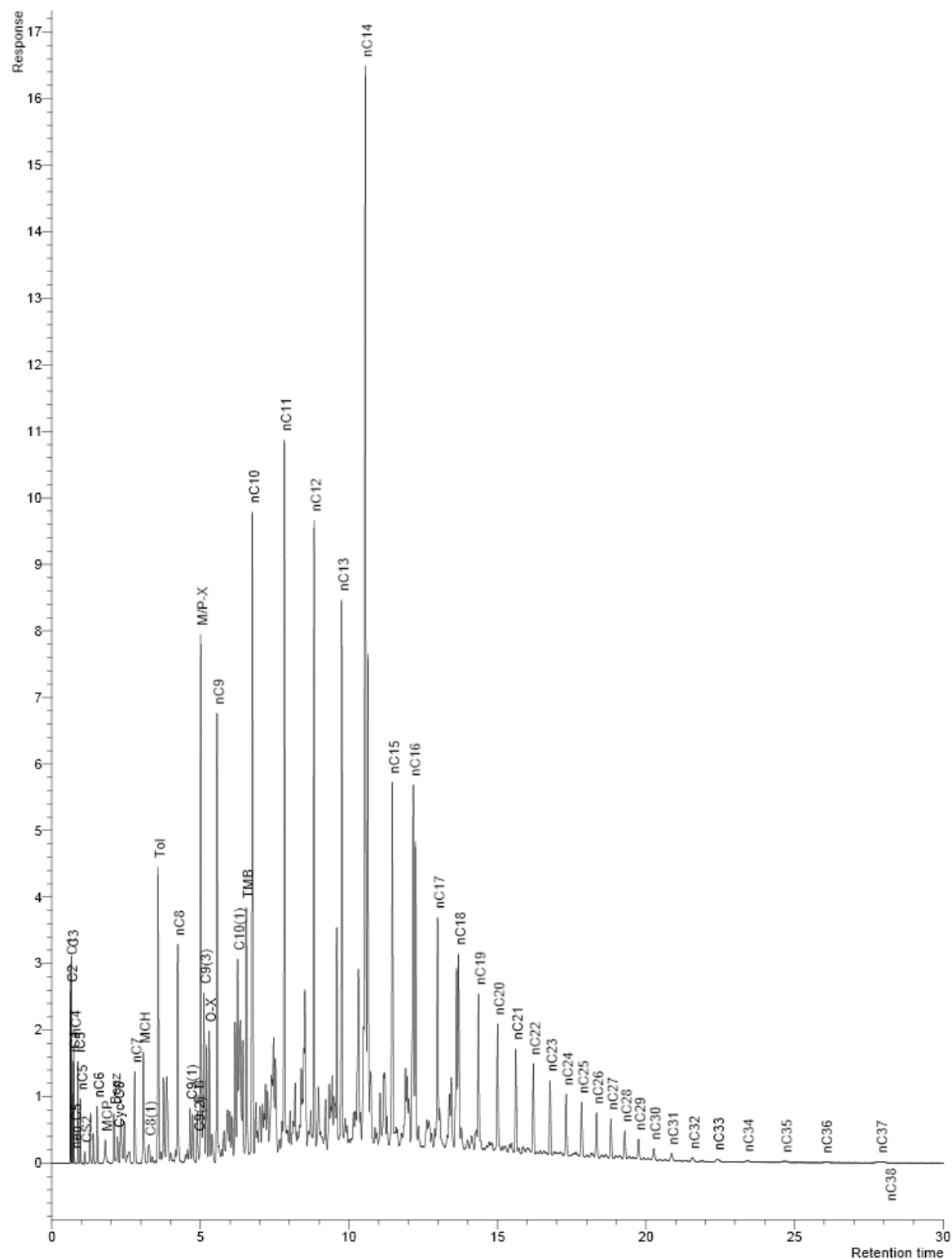
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.968	13.920	27.052
N ₂	Nitrogen	0.000	0.000	0.296	0.295	0.364
C ₁	Methane	1.153	0.111	77.786	77.523	54.917
C ₂	Ethane	0.604	0.109	4.836	4.822	6.403
C ₃	Propane	0.612	0.162	1.328	1.325	2.581
iC ₄	i-Butane	0.318	0.111	0.351	0.350	0.899
nC ₄	n-Butane	0.390	0.136	0.317	0.317	0.814
C ₅	neo-Pentane	0.016	0.007	0.004	0.004	0.014
iC ₅	i-Pentane	0.453	0.196	0.208	0.209	0.667
nC ₅	n-Pentane	0.310	0.134	0.114	0.115	0.367
C ₆	Hexanes	1.027	0.531	0.204	0.207	0.788
	Me-Cyclo-pentane	0.315	0.159	0.033	0.034	0.126
	Benzene	0.452	0.212	0.051	0.053	0.182
	Cyclo-hexane	0.301	0.152	0.058	0.059	0.220
C ₇	Heptanes	1.825	1.097	0.100	0.106	0.470
	Me-Cyclo-hexane	1.059	0.624	0.051	0.055	0.237
	Toluene	2.330	1.288	0.094	0.101	0.413
C ₈	Octanes	3.957	2.712	0.071	0.084	0.424
	Ethyl-benzene	0.334	0.213	0.005	0.006	0.029
	Meta/Para-xylene	4.079	2.598	0.037	0.050	0.237
	Ortho-xylene	0.947	0.603	0.006	0.009	0.043
C ₉	Nonanes	6.101	4.695	0.041	0.062	0.352
	Tri-Me-benzene	1.718	1.239	0.000	0.006	0.032
C ₁₀	Decanes	10.240	8.741	0.023	0.059	0.369
C ₁₁	Undecanes	10.285	9.071	0.009	0.045	0.290
C ₁₂	Dodecanes	8.696	8.400	0.003	0.033	0.237
C ₁₃	Tridecanes	7.541	7.917	0.002	0.028	0.214
C ₁₄	Tetradecanes	11.472	13.077	0.002	0.042	0.349
C ₁₅	Pentadecanes	4.798	5.930	0.000	0.017	0.151
C ₁₆	Hexadecanes	5.436	7.240	0.000	0.019	0.184
C ₁₇	Heptadecanes	2.511	3.570	0.000	0.009	0.091
C ₁₈	Octadecanes	3.167	4.769	0.000	0.011	0.121
C ₁₉	Nonadecanes	1.609	2.539	0.000	0.006	0.065
C ₂₀	Eicosanes	1.236	2.039	0.000	0.004	0.052
C ₂₁	Heneicosanes	0.926	1.616	0.000	0.003	0.041
C ₂₂	Docosanes	0.771	1.411	0.000	0.003	0.036
C ₂₃	Tricosanes	0.592	1.129	0.000	0.002	0.029
C ₂₄	Tetracosanes	0.503	0.998	0.000	0.002	0.025
C ₂₅	Pentacosanes	0.415	0.858	0.000	0.001	0.022
C ₂₆	Hexacosanes	0.331	0.712	0.000	0.001	0.018
C ₂₇	Heptacosanes	0.274	0.615	0.000	0.001	0.016
C ₂₈	Octacosanes	0.207	0.483	0.000	0.001	0.012
C ₂₉	Nonacosanes	0.172	0.416	0.000	0.001	0.011
C ₃₀	triacontanes	0.137	0.342	0.000	0.000	0.009
C ₃₁	Hentriacontanes	0.104	0.269	0.000	0.000	0.007
C ₃₂	Dotriacontanes	0.075	0.201	0.000	0.000	0.005
C ₃₃	Trtriacontanes	0.066	0.181	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.050	0.142	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.041	0.120	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.042	0.125	0.000	0.000	0.003
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.04, MPSR 5055 to C36+
Depth: 5000.2m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	95.116	0.586	0.913
Molecular Weight (g mol-1)	173	101.2	127
Density at 15.6°C (g cm-3)	0.8103	0.7744	0.7877
C₁₁₊			
Mole%	61.457	0.016	0.229
Molecular Weight (g mol-1)	201	158	198
Density at 15.6°C (g cm-3)	0.8292	0.7982	0.8274
C₂₀₊			
Mole%	5.942		0.019
Molecular Weight (g mol-1)	327		327
Density at 15.6°C (g cm-3)	0.8800		0.88
C₃₆₊			
Mole %	0.042		0.000
Molecular Weight (g mol-1)	507		507
Density at 15.6°C (g cm-3)	0.9205		0.9205
Whole Sample Properties			
Average mole weight (g mol-1)	167 (Calculated)	22.2	22.5
Density (g cm-3 at 15.6°C)	0.8043 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7670	
GOR Ratio (scf/bbl) from flash at 19.7°C			184332.3
CGR Ratio (bbl/MMscf) from flash at 19.7°C			5.4
Gas Mole Fraction			0.9965
Liquid Mole Fraction			0.0035

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.04, MPSR 5055**
Depth: 5000.2m MDRTPharos-1 S/F (1,1)
Acquired Friday, 12 September 2014 9:40:07 PM

wb07092014,as_2.10002,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Gas Sample No. 1.05, MPSR 5049 to C36+
 Depth: 5021.0m MDRT**

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.903	13.840	26.721
N ₂	Nitrogen	0.000	0.000	0.372	0.370	0.455
C ₁	Methane	1.296	0.122	77.777	77.434	54.496
C ₂	Ethane	0.533	0.094	4.859	4.840	6.384
C ₃	Propane	0.530	0.137	1.320	1.316	2.546
iC ₄	i-Butane	0.279	0.095	0.349	0.348	0.888
nC ₄	n-Butane	0.349	0.119	0.316	0.316	0.806
C ₅	neo-Pentane	0.009	0.004	0.003	0.003	0.009
iC ₅	i-Pentane	0.411	0.174	0.205	0.206	0.651
nC ₅	n-Pentane	0.286	0.121	0.113	0.114	0.361
C ₆	Hexanes	0.997	0.504	0.204	0.207	0.783
	Me-Cyclo-pentane	0.318	0.157	0.033	0.034	0.125
	Benzene	0.456	0.209	0.052	0.053	0.183
	Cyclo-hexane	0.298	0.147	0.058	0.059	0.218
C ₇	Heptanes	1.803	1.060	0.099	0.107	0.469
	Me-Cyclo-hexane	1.036	0.597	0.051	0.056	0.240
	Toluene	2.316	1.252	0.088	0.098	0.397
C ₈	Octanes	3.755	2.516	0.074	0.090	0.453
	Ethyl-benzene	0.340	0.212	0.005	0.006	0.029
	Meta/Para-xylene	3.818	2.378	0.036	0.053	0.245
	Ortho-xylene	0.875	0.545	0.005	0.009	0.042
C ₉	Nonanes	5.457	4.106	0.042	0.066	0.373
	Tri-Me-benzene	1.476	1.041	0.000	0.007	0.035
C ₁₀	Decanes	8.891	7.421	0.023	0.063	0.392
C ₁₁	Undecanes	8.423	7.263	0.010	0.048	0.311
C ₁₂	Dodecanes	7.132	6.736	0.004	0.036	0.255
C ₁₃	Tridecanes	6.180	6.344	0.002	0.029	0.226
C ₁₄	Tetradecanes	17.809	19.850	0.000	0.080	0.670
C ₁₅	Pentadecanes	4.118	4.976	0.000	0.019	0.167
C ₁₆	Hexadecanes	7.444	9.694	0.000	0.034	0.326
C ₁₇	Heptadecanes	2.174	3.023	0.000	0.010	0.102
C ₁₈	Octadecanes	4.274	6.293	0.000	0.019	0.212
C ₁₉	Nonadecanes	1.431	2.207	0.000	0.006	0.074
C ₂₀	Eicosanes	1.100	1.774	0.000	0.005	0.060
C ₂₁	Heneicosanes	0.841	1.435	0.000	0.004	0.048
C ₂₂	Docosanes	0.784	1.402	0.000	0.004	0.047
C ₂₃	Tricosanes	0.524	0.978	0.000	0.002	0.033
C ₂₄	Tetracosanes	0.443	0.860	0.000	0.002	0.029
C ₂₅	Pentacosanes	0.352	0.712	0.000	0.002	0.024
C ₂₆	Hexacosanes	0.289	0.609	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.260	0.571	0.000	0.001	0.019
C ₂₈	Octacosanes	0.181	0.413	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.149	0.351	0.000	0.001	0.012
C ₃₀	triacontanes	0.121	0.295	0.000	0.001	0.010
C ₃₁	Hentriacontanes	0.096	0.242	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.070	0.183	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.067	0.179	0.000	0.000	0.006
C ₃₄	Tetratriacontanes	0.051	0.142	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.056	0.159	0.000	0.000	0.005
C ₃₆₊	Hexatriacontanes plus	0.102	0.298	0.000	0.000	0.010
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.05, MPSR 5049 to C36+
Depth: 5021.0m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	95.310	0.581	1.006
Molecular Weight (g mol-1)	176	101.3	134
Density at 15.6°C (g cm-3)	0.8135	0.7729	0.7919
C₁₁₊			
Mole%	64.469	0.016	0.305
Molecular Weight (g mol-1)	204	154	201
Density at 15.6°C (g cm-3)	0.8305	0.7948	0.8290
C₂₀₊			
Mole%	5.485		0.024
Molecular Weight (g mol-1)	330		330
Density at 15.6°C (g cm-3)	0.8809		0.8809
C₃₆₊			
Mole %	0.102		0.000
Molecular Weight (g mol-1)	514		514
Density at 15.6°C (g cm-3)	0.9217		0.9217
Whole Sample Properties			
Average mole weight (g mol-1)	171 (Calculated)	22.1	22.8
Density (g cm-3 at 15.6°C)	0.8078 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7664	
GOR Ratio (scf/bbl) from flash at 21.1°C			138958.5
CGR Ratio (bbl/MMscf) from flash at 21.1°C			7.2
Gas Mole Fraction			0.9955
Liquid Mole Fraction			0.0045

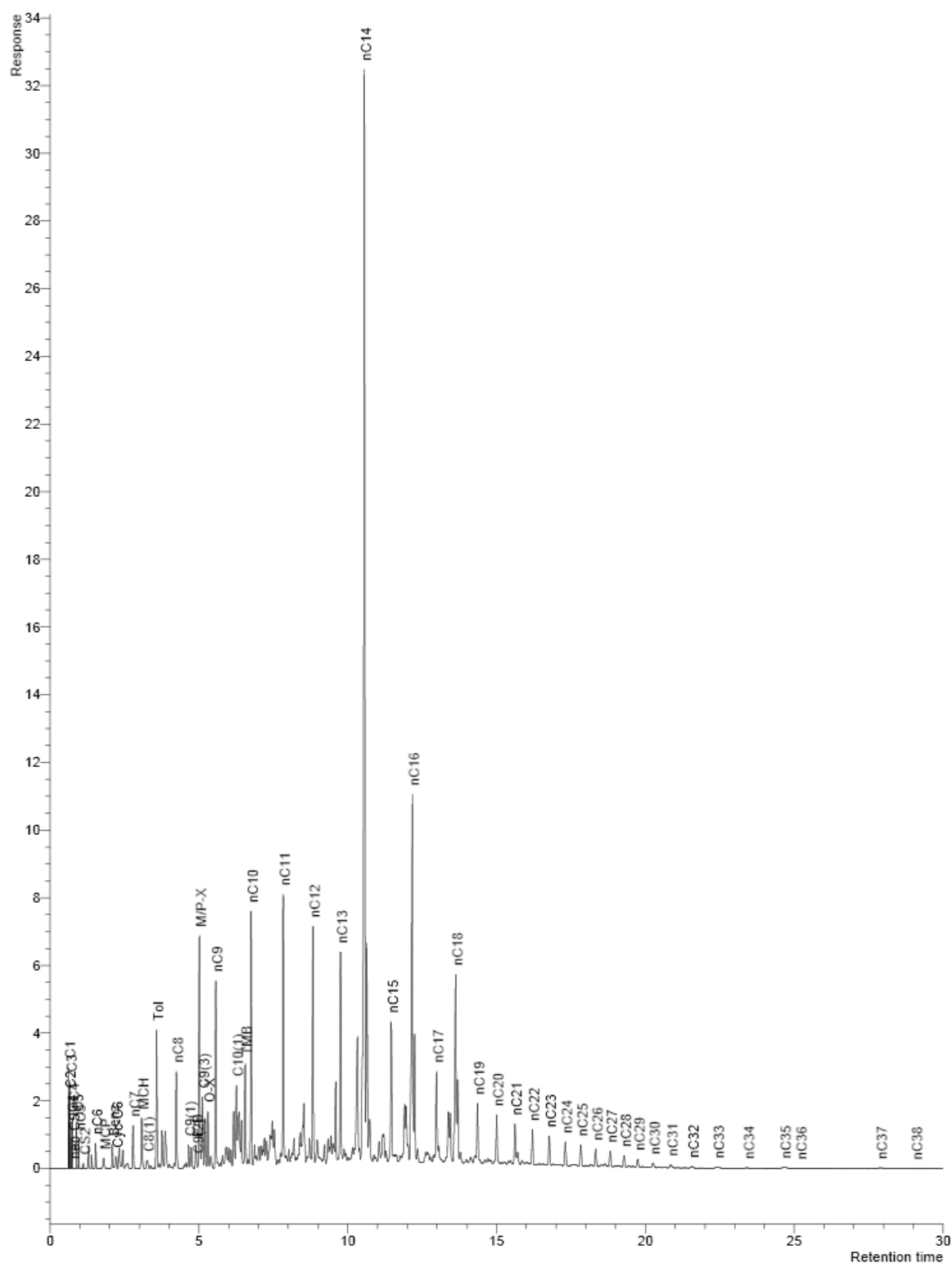
Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Fingerprint Profile of Flashed Condensate, Sample No .1.05, MPSR 5049
Depth: 5021.0m MDRT

Pharos-1 S/F (1,1)
 Acquired Friday, 19 September 2014 6:16:54 PM

wb14092014,as_2_10054,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Gas Sample No. 1.06 , MPSR 2489 to C36+
 Depth: 5021.0m MDRT**

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.877	13.813	26.642
N ₂	Nitrogen	0.000	0.000	0.322	0.320	0.393
C ₁	Methane	1.200	0.113	77.806	77.455	54.456
C ₂	Ethane	0.476	0.084	4.877	4.856	6.400
C ₃	Propane	0.487	0.126	1.332	1.328	2.566
iC ₄	i-Butane	0.261	0.089	0.352	0.351	0.895
nC ₄	n-Butane	0.323	0.110	0.318	0.318	0.810
C ₅	neo-Pentane	0.014	0.006	0.004	0.004	0.014
iC ₅	i-Pentane	0.390	0.165	0.204	0.205	0.649
nC ₅	n-Pentane	0.272	0.115	0.113	0.114	0.359
C ₆	Hexanes	0.995	0.503	0.204	0.208	0.784
	Me-Cyclo-pentane	0.312	0.154	0.033	0.034	0.125
	Benzene	0.430	0.197	0.052	0.054	0.183
	Cyclo-hexane	0.290	0.143	0.058	0.059	0.217
C ₇	Heptanes	1.721	1.012	0.100	0.108	0.472
	Me-Cyclo-hexane	1.012	0.583	0.050	0.055	0.236
	Toluene	2.255	1.219	0.099	0.109	0.442
C ₈	Octanes	3.704	2.483	0.061	0.078	0.390
	Ethyl-benzene	0.348	0.217	0.005	0.007	0.031
	Meta/Para-xylene	3.825	2.383	0.036	0.054	0.250
	Ortho-xylene	0.881	0.549	0.005	0.009	0.044
C ₉	Nonanes	5.549	4.176	0.043	0.069	0.386
	Tri-Me-benzene	1.496	1.055	0.000	0.007	0.036
C ₁₀	Decanes	9.129	7.622	0.026	0.068	0.426
C ₁₁	Undecanes	8.652	7.463	0.012	0.051	0.331
C ₁₂	Dodecanes	7.362	6.955	0.004	0.037	0.264
C ₁₃	Tridecanes	6.366	6.537	0.004	0.033	0.256
C ₁₄	Tetradecanes	17.295	19.282	0.002	0.082	0.679
C ₁₅	Pentadecanes	4.266	5.157	0.000	0.020	0.178
C ₁₆	Hexadecanes	7.336	9.557	0.000	0.034	0.329
C ₁₇	Heptadecanes	2.249	3.128	0.000	0.010	0.108
C ₁₈	Octadecanes	4.219	6.214	0.000	0.019	0.214
C ₁₉	Nonadecanes	1.481	2.286	0.000	0.007	0.079
C ₂₀	Eicosanes	1.116	1.801	0.000	0.005	0.062
C ₂₁	Heneicosanes	0.854	1.458	0.000	0.004	0.050
C ₂₂	Docosanes	0.781	1.398	0.000	0.004	0.048
C ₂₃	Tricosanes	0.532	0.992	0.000	0.002	0.034
C ₂₄	Tetracosanes	0.458	0.889	0.000	0.002	0.031
C ₂₅	Pentacosanes	0.356	0.720	0.000	0.002	0.025
C ₂₆	Hexacosanes	0.294	0.619	0.000	0.001	0.021
C ₂₇	Heptacosanes	0.259	0.568	0.000	0.001	0.020
C ₂₈	Octacosanes	0.182	0.415	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.150	0.353	0.000	0.001	0.012
C ₃₀	triacontanes	0.116	0.283	0.000	0.001	0.010
C ₃₁	Hentriacontanes	0.094	0.236	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.065	0.170	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.055	0.149	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.037	0.102	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.036	0.102	0.000	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.021	0.062	0.000	0.000	0.002
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.06 , MPSR 2489 to C36+
Depth: 5021.0m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	95.583	0.591	1.028
Molecular Weight (g mol-1)	176	101.8	134
Density at 15.6°C (g cm-3)	0.8130	0.7767	0.7933
C₁₁₊			
Mole%	64.631	0.021	0.317
Molecular Weight (g mol-1)	203	159	200
Density at 15.6°C (g cm-3)	0.8300	0.7989	0.8283
C₂₀₊			
Mole%	5.405		0.024
Molecular Weight (g mol-1)	325		325
Density at 15.6°C (g cm-3)	0.8795		0.8795
C₃₆₊			
Mole %	0.021		0.000
Molecular Weight (g mol-1)	512		512
Density at 15.6°C (g cm-3)	0.9214		0.9214
Whole Sample Properties			
Average mole weight (g mol-1)	170 (Calculated)	22.1	22.8
Density (g cm-3 at 15.6°C)	0.8077 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7667	
GOR Ratio (scf/bbl) from flash at 20.0°C			135745.1
CGR Ratio (bbl/MMscf) from flash at 20.0°C			7.4
Gas Mole Fraction			0.9954
Liquid Mole Fraction			0.0046

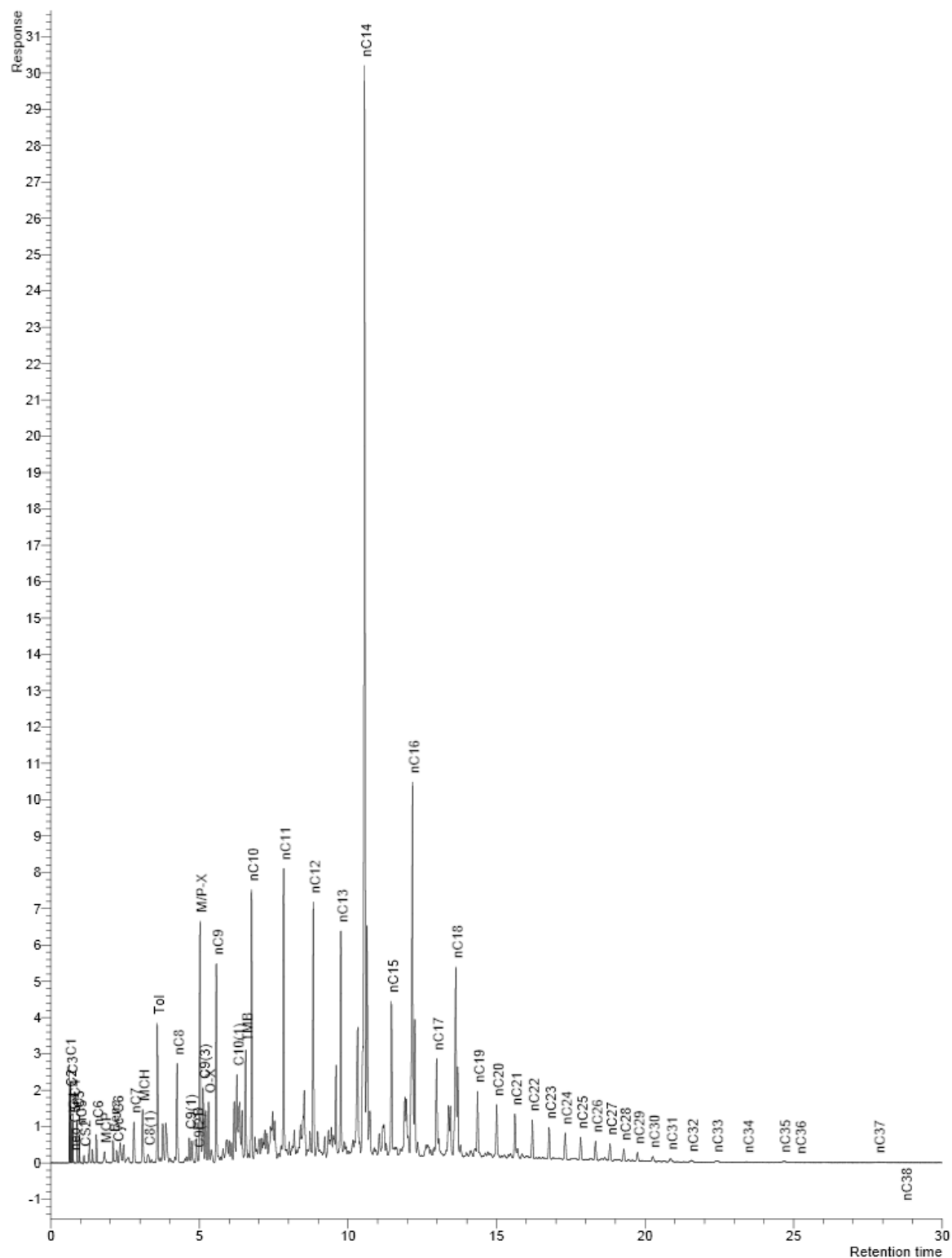
Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Fingerprint Profile of Flashed Condensate, Sample No .1.06, MPSR 2489
Depth: 5021.0m MDRT

Pharos-1 S/F (1,1)
 Acquired Thursday, 18 September 2014 9:46:46 PM

wb14092014,as_2.10042,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Gas Sample No. 1.08 , MPSR 3374 to C36+
 Depth: 5021.0m MDRT**

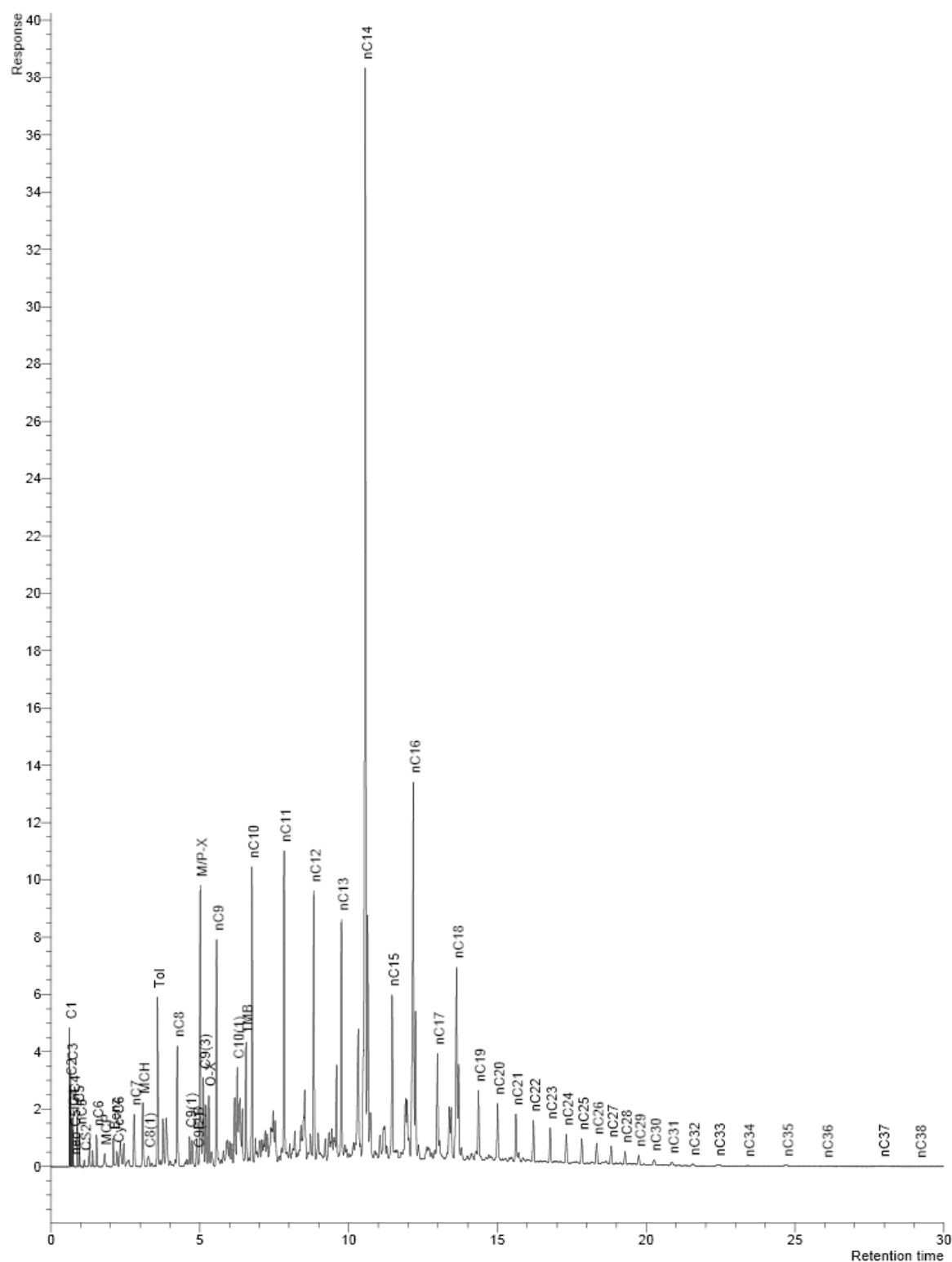
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.818	27.597	26.610
N ₂	Nitrogen	0.000	0.000	0.325	0.413	0.398
C ₁	Methane	1.470	0.140	78.011	56.796	54.768
C ₂	Ethane	0.527	0.094	4.832	6.594	6.361
C ₃	Propane	0.523	0.137	1.314	2.629	2.540
iC ₄	i-Butane	0.278	0.096	0.349	0.921	0.891
nC ₄	n-Butane	0.345	0.119	0.317	0.836	0.810
C ₅	neo-Pentane	0.014	0.006	0.005	0.017	0.017
iC ₅	i-Pentane	0.420	0.180	0.196	0.643	0.626
nC ₅	n-Pentane	0.294	0.126	0.109	0.356	0.348
C ₆	Hexanes	1.050	0.537	0.192	0.749	0.741
	Me-Cyclo-pentane	0.336	0.168	0.030	0.116	0.118
	Benzene	0.490	0.227	0.049	0.172	0.174
	Cyclo-hexane	0.314	0.157	0.054	0.207	0.205
C ₇	Heptanes	1.949	1.159	0.093	0.424	0.450
	Me-Cyclo-hexane	1.127	0.657	0.047	0.210	0.226
	Toluene	2.499	1.367	0.081	0.339	0.376
C ₈	Octanes	4.112	2.788	0.065	0.337	0.425
	Ethyl-benzene	0.357	0.225	0.005	0.022	0.029
	Meta/Para-xylene	4.080	2.571	0.032	0.156	0.242
	Ortho-xylene	0.944	0.595	0.005	0.023	0.043
C ₉	Nonanes	5.862	4.463	0.039	0.226	0.378
	Tri-Me-benzene	1.531	1.092	0.000	0.000	0.039
C ₁₀	Decanes	9.174	7.748	0.021	0.138	0.411
C ₁₁	Undecanes	8.552	7.462	0.009	0.062	0.327
C ₁₂	Dodecanes	7.096	6.782	0.002	0.016	0.258
C ₁₃	Tridecanes	6.204	6.445	0.000	0.001	0.232
C ₁₄	Tetradecanes	16.112	18.172	0.000	0.000	0.651
C ₁₅	Pentadecanes	4.123	5.041	0.000	0.000	0.181
C ₁₆	Hexadecanes	6.970	9.185	0.000	0.000	0.329
C ₁₇	Heptadecanes	2.201	3.096	0.000	0.000	0.111
C ₁₈	Octadecanes	4.037	6.015	0.000	0.000	0.215
C ₁₉	Nonadecanes	1.436	2.242	0.000	0.000	0.080
C ₂₀	Eicosanes	1.127	1.840	0.000	0.000	0.066
C ₂₁	Heneicosanes	0.842	1.455	0.000	0.000	0.052
C ₂₂	Docosanes	0.754	1.365	0.000	0.000	0.049
C ₂₃	Tricosanes	0.532	1.004	0.000	0.000	0.036
C ₂₄	Tetracosanes	0.451	0.887	0.000	0.000	0.032
C ₂₅	Pentacosanes	0.372	0.762	0.000	0.000	0.027
C ₂₆	Hexacosanes	0.297	0.632	0.000	0.000	0.023
C ₂₇	Heptacosanes	0.270	0.600	0.000	0.000	0.021
C ₂₈	Octacosanes	0.195	0.450	0.000	0.000	0.016
C ₂₉	Nonacosanes	0.161	0.385	0.000	0.000	0.014
C ₃₀	Triacontanes	0.129	0.319	0.000	0.000	0.011
C ₃₁	Hentriacontanes	0.104	0.265	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.077	0.202	0.000	0.000	0.007
C ₃₃	Trtriacontanes	0.072	0.195	0.000	0.000	0.007
C ₃₄	Tetratriacontanes	0.053	0.149	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.054	0.157	0.000	0.000	0.006
C ₃₆₊	Hexatriacontanes plus	0.081	0.241	0.000	0.000	0.009
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.08 , MPSR 3374 to C36+
Depth: 5021.0m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	95.078	0.533	2.449
Molecular Weight (g mol-1)	175	100.8	135
Density at 15.6°C (g cm-3)	0.8124	0.7729	0.7929
C₁₁₊			
Mole%	62.304	0.012	0.079
Molecular Weight (g mol-1)	204	150	202
Density at 15.6°C (g cm-3)	0.8307	0.7914	0.8295
C₂₀₊			
Mole%	5.573		0.000
Molecular Weight (g mol-1)	330		330
Density at 15.6°C (g cm-3)	0.8810		0.8810
C₃₆₊			
Mole %	0.081		0.000
Molecular Weight (g mol-1)	512		512
Density at 15.6°C (g cm-3)	0.9214		0.9214
Whole Sample Properties			
Average mole weight (g mol-1)	169 (Calculated)	22.0	22.8
Density (g cm-3 at 15.6°C)	0.8064 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7632	
GOR Ratio (scf/bbl) from flash at 19.4°C			130670.6
CGR Ratio (bbl/MMscf) from flash at 19.4°C			7.7
Gas Mole Fraction			0.9952
Liquid Mole Fraction			0.0048

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.08, MPSR 3374**
Depth: 5021.0m MDRTPharos-1 S/F (1,1)
Acquired Monday, 15 September 2014 7:55:24 PM

wb14092014,as_2_10006,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Gas Sample No. 1.09 , MPSR 5048 to C36+
 Depth: 5021.0m MDRT**

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.966	13.900	26.753
N ₂	Nitrogen	0.000	0.000	0.309	0.307	0.376
C ₁	Methane	1.574	0.150	77.709	77.351	54.271
C ₂	Ethane	0.515	0.092	4.854	4.834	6.357
C ₃	Propane	0.504	0.132	1.339	1.335	2.574
iC ₄	i-Butane	0.266	0.092	0.354	0.354	0.900
nC ₄	n-Butane	0.333	0.115	0.324	0.324	0.822
C ₅	neo-Pentane	0.009	0.004	0.005	0.005	0.015
iC ₅	i-Pentane	0.399	0.171	0.209	0.210	0.663
nC ₅	n-Pentane	0.280	0.120	0.116	0.117	0.368
C ₆	Hexanes	1.006	0.515	0.207	0.210	0.793
	Me-Cyclo-pentane	0.322	0.161	0.034	0.036	0.131
	Benzene	0.463	0.215	0.054	0.056	0.191
	Cyclo-hexane	0.310	0.155	0.061	0.063	0.230
C ₇	Heptanes	1.891	1.126	0.104	0.112	0.493
	Me-Cyclo-hexane	1.102	0.643	0.056	0.061	0.262
	Toluene	2.455	1.344	0.098	0.109	0.441
C ₈	Octanes	4.089	2.775	0.074	0.093	0.465
	Ethyl-benzene	0.357	0.225	0.006	0.007	0.034
	Meta/Para-xylene	4.106	2.590	0.039	0.059	0.273
	Ortho-xylene	0.942	0.594	0.007	0.011	0.052
C ₉	Nonanes	5.872	4.475	0.042	0.070	0.392
	Tri-Me-benzene	1.519	1.085	0.000	0.007	0.038
C ₁₀	Decanes	9.210	7.786	0.023	0.066	0.413
C ₁₁	Undecanes	8.556	7.473	0.008	0.048	0.308
C ₁₂	Dodecanes	7.167	6.856	0.001	0.035	0.243
C ₁₃	Tridecanes	6.124	6.368	0.001	0.030	0.226
C ₁₄	Tetradecanes	16.503	18.630	0.001	0.079	0.657
C ₁₅	Pentadecanes	4.071	4.983	0.000	0.019	0.173
C ₁₆	Hexadecanes	7.078	9.336	0.000	0.033	0.325
C ₁₇	Heptadecanes	2.141	3.015	0.000	0.010	0.105
C ₁₈	Octadecanes	4.099	6.113	0.000	0.019	0.213
C ₁₉	Nonadecanes	1.391	2.173	0.000	0.007	0.076
C ₂₀	Eicosanes	1.104	1.804	0.000	0.005	0.063
C ₂₁	Heneicosanes	0.794	1.373	0.000	0.004	0.048
C ₂₂	Docosanes	0.710	1.287	0.000	0.003	0.045
C ₂₃	Tricosanes	0.512	0.967	0.000	0.002	0.034
C ₂₄	Tetracosanes	0.434	0.854	0.000	0.002	0.030
C ₂₅	Pentacosanes	0.358	0.734	0.000	0.002	0.026
C ₂₆	Hexacosanes	0.286	0.609	0.000	0.001	0.021
C ₂₇	Heptacosanes	0.255	0.566	0.000	0.001	0.020
C ₂₈	Octacosanes	0.182	0.420	0.000	0.001	0.015
C ₂₉	Nonacosanes	0.152	0.364	0.000	0.001	0.013
C ₃₀	triacontanes	0.123	0.304	0.000	0.001	0.011
C ₃₁	Hentriacontanes	0.096	0.245	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.068	0.179	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.067	0.182	0.000	0.000	0.006
C ₃₄	Tetratriacontanes	0.051	0.143	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.055	0.158	0.000	0.000	0.005
C ₃₆₊	Hexatriacontanes plus	0.101	0.299	0.000	0.000	0.010
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Gas Sample No. 1.09 , MPSR 5048 to C36+
Depth: 5021.0m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	95.114	0.609	1.053
Molecular Weight (g mol-1)	175	100.5	132
Density at 15.6°C (g cm-3)	0.8122	0.7746	0.7920
C₁₁₊			
Mole%	62.477	0.010	0.303
Molecular Weight (g mol-1)	203	155	202
Density at 15.6°C (g cm-3)	0.8303	0.7957	0.8294
C₂₀₊			
Mole%	5.347		0.023
Molecular Weight (g mol-1)	330		330
Density at 15.6°C (g cm-3)	0.8811		0.8811
C₃₆₊			
Mole %	0.101		0.000
Molecular Weight (g mol-1)	517		517
Density at 15.6°C (g cm-3)	0.9222		0.9222
Whole Sample Properties			
Average mole weight (g mol-1)	168 (Calculated)	22.2	22.9
Density (g cm-3 at 15.6°C)	0.8062 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7680	
GOR Ratio (scf/bbl) from flash at 18.4°C			133728.0
CGR Ratio (bbl/MMscf) from flash at 18.4°C			7.5
Gas Mole Fraction			0.9953
Liquid Mole Fraction			0.0047

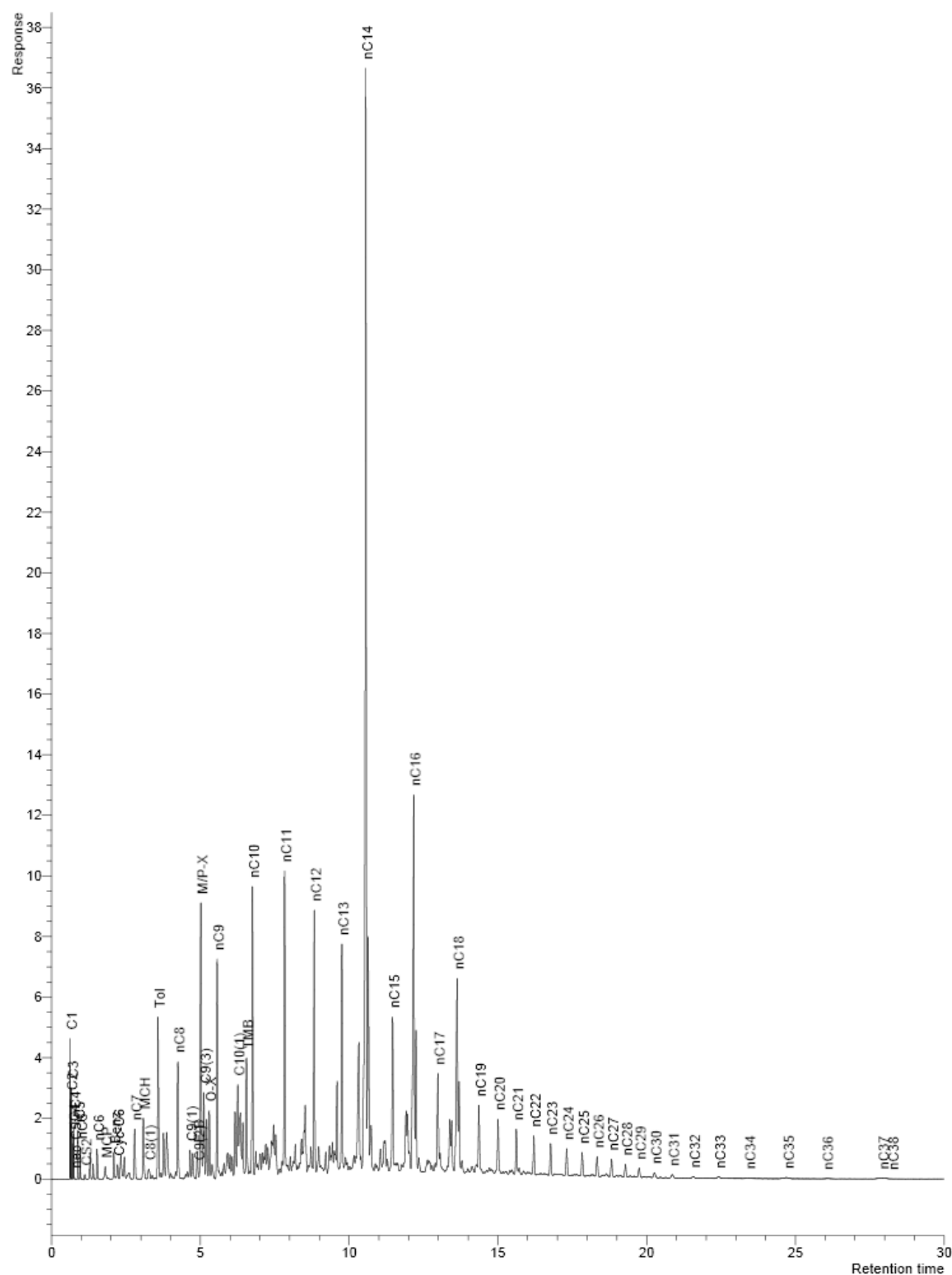
Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Fingerprint Profile of Flashed Condensate, Sample No .1.09, MPSR 5048
Depth: 5021.0m MDRT

Pharos-1 S/F (1,1)
 Acquired Friday, 12 September 2014 7:58:05 PM

wb07092014,as_2.10000,1,1,1



Section D - Impurities in Gas Data

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Trace Component Analysis**

Sample no.	Cylinder no.	Sample Depth m MDRT	Component			
			Helium He mole%	Hydrogen H2 mole%	Argon Ar mole%	Oxygen O2 mole%
1.01	2668	5000.2	0.008	0.001	0.001	0.003
1.05	5049	5021.0	0.010	0.002	0.002	0.004

Note:

1) 0.000 means < 0.0005 mol%

Section E - Synthetic Base Mud Composition

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Oil Based Drilling Mud to C36+
Sample No. 1.11, Mud Pit, 16/7/14

Component		Weight %
H ₂	Hydrogen	0.000
H ₂ S	Hydrogen Sulphide	0.000
CO ₂	Carbon Dioxide	0.000
N ₂	Nitrogen	0.000
C ₁	Methane	0.000
C ₂	Ethane	0.000
C ₃	Propane	0.000
iC ₄	i-Butane	0.000
nC ₄	n-Butane	0.000
C ₅	Neo-Pentane	0.000
iC ₅	i-Pentane	0.000
nC ₅	n-Pentane	0.000
C ₆	Hexanes	0.000
	M-C-Pentane	0.000
	Benzene	0.000
	Cyclohexane	0.010
C ₇	Heptanes	0.006
	M-C-Hexane	0.000
	Toluene	0.000
C ₈	Octanes	0.000
	E-Benzene	0.000
	M/P-Xylene	0.013
	O-Xylene	0.000
C ₉	Nonanes	0.033
	1,2,4-TMB	0.152
C ₁₀	Decanes	0.480
C ₁₁	Undecanes	0.075
C ₁₂	Dodecanes	0.753
C ₁₃	Tridecanes	0.276
C ₁₄	Tetradecanes	55.885
C ₁₅	Pentadecanes	1.427
C ₁₆	Hexadecanes	22.828
C ₁₇	Heptdecanes	0.908
C ₁₈	Octadecanes	14.440
C ₁₉	Nonadecanes	0.708
C ₂₀	Eicosanes	0.591
C ₂₁	Heneicosanes	0.099
C ₂₂	Docosanes	0.125
C ₂₃	Tricosanes	0.080
C ₂₄	Tetracosanes	0.140
C ₂₅	Pentacosanes	0.055
C ₂₆	Hexacosanes	0.040
C ₂₇	Heptacosanes	0.086
C ₂₈	Octacosanes	0.036
C ₂₉	Nonacosanes	0.038
C ₃₀	triacontanes	0.038
C ₃₁	Hentriacontanes	0.040
C ₃₂	Dotriacontanes	0.055
C ₃₃	Trtriacontanes	0.067
C ₃₄	Tetratriacontanes	0.080
C ₃₅	Pentatriacontanes	0.127
C ₃₆₊	Hexatriacontanes Plus	0.309
Totals :		100.000

Compositional Analysis of Oil Based Drilling Mud to C36+
Sample No. 1.11, Mud Pit, 16/7/14

Whole Sample Properties	Mole Weight (g.mol ⁻¹)	Density @ 15.6°C (g.cm ⁻³)
Measured Properties	227 (measured)	0.8132 (measured)

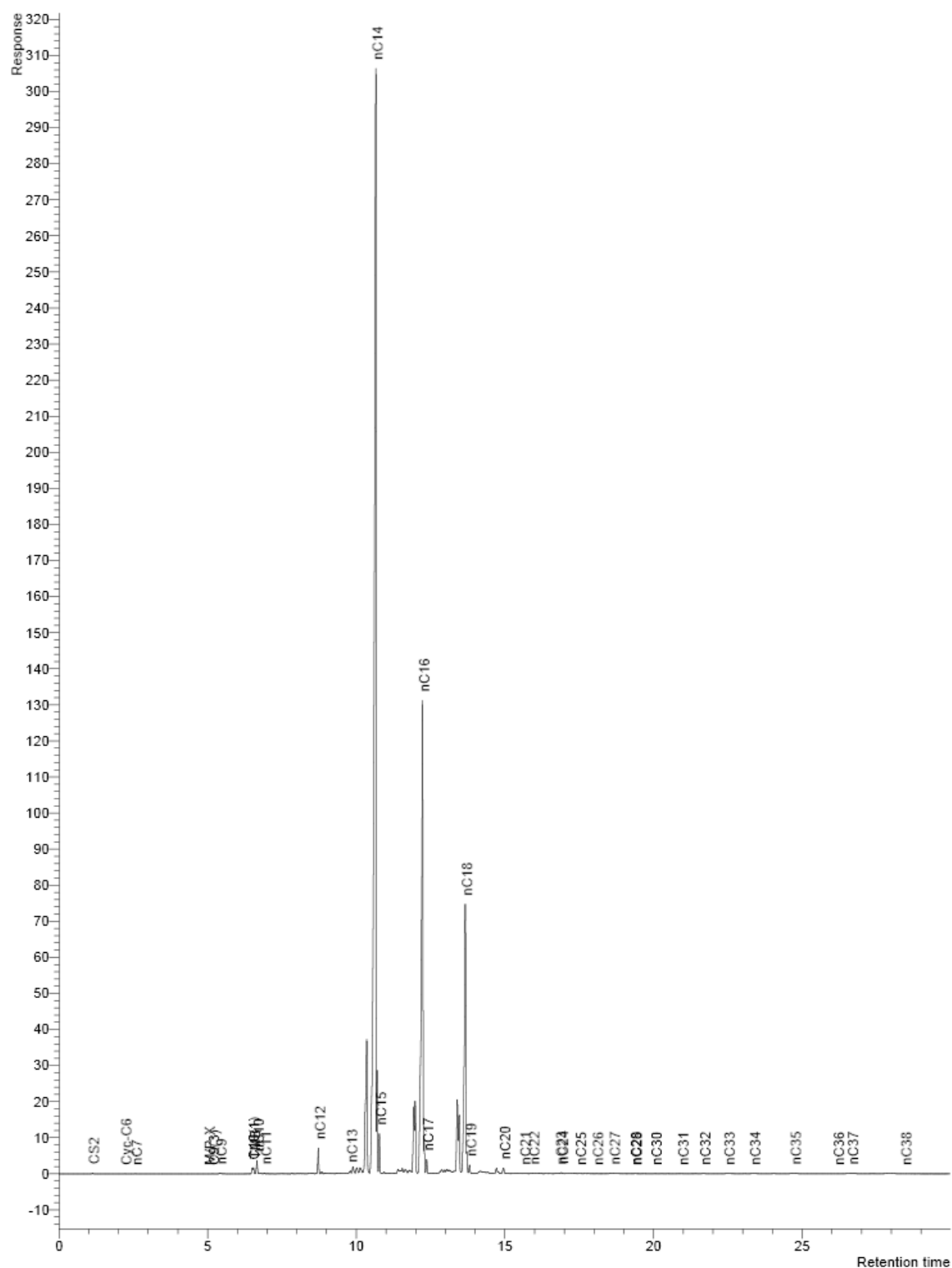
Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Chromatogram of Oil Based Drilling Mud
Sample No. 1.11, Mud Pit, 16/7/14

Pharos-1 Mud 1.11 (1,1)
 Acquired Wednesday, 3 September 2014 7:49:54 PM

wb31082014,as_2.9968,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Oil Based Drilling Mud to C36+
Sample No. 1.12, Mud Pit, 16/7/14

	Component	Weight %
H ₂	Hydrogen	0.000
H ₂ S	Hydrogen Sulphide	0.000
CO ₂	Carbon Dioxide	0.000
N ₂	Nitrogen	0.000
C ₁	Methane	0.000
C ₂	Ethane	0.000
C ₃	Propane	0.000
iC ₄	i-Butane	0.000
nC ₄	n-Butane	0.000
C ₅	Neo-Pentane	0.000
iC ₅	i-Pentane	0.000
nC ₅	n-Pentane	0.000
C ₆	Hexanes	0.000
	M-C-Pentane	0.000
	Benzene	0.000
	Cyclohexane	0.006
C ₇	Heptanes	0.002
	M-C-Hexane	0.000
	Toluene	0.001
C ₈	Octanes	0.000
	E-Benzene	0.000
	M/P-Xylene	0.001
	O-Xylene	0.000
C ₉	Nonanes	0.029
	1,2,4-TMB	0.147
C ₁₀	Decanes	0.419
C ₁₁	Undecanes	0.070
C ₁₂	Dodecanes	0.748
C ₁₃	Tridecanes	0.278
C ₁₄	Tetradecanes	56.076
C ₁₅	Pentadecanes	1.426
C ₁₆	Hexadecanes	22.798
C ₁₇	Heptdecanes	0.918
C ₁₈	Octadecanes	14.349
C ₁₉	Nonadecanes	0.727
C ₂₀	Eicosanes	0.587
C ₂₁	Heneicosanes	0.097
C ₂₂	Docosanes	0.126
C ₂₃	Tricosanes	0.077
C ₂₄	Tetracosanes	0.136
C ₂₅	Pentacosanes	0.053
C ₂₆	Hexacosanes	0.040
C ₂₇	Heptacosanes	0.084
C ₂₈	Octacosanes	0.036
C ₂₉	Nonacosanes	0.040
C ₃₀	Triacotanes	0.038
C ₃₁	Hentriacotanes	0.040
C ₃₂	Dotriacotanes	0.053
C ₃₃	Tritriacotanes	0.065
C ₃₄	Tetratriacotanes	0.079
C ₃₅	Pentatriacotanes	0.126
C ₃₆₊	Hexatriacotanes Plus	0.328
	Totals :	100.000

Compositional Analysis of Oil Based Drilling Mud to C36+
Sample No. 1.12, Mud Pit, 16/7/14

Whole Sample Properties	Mole Weight (g.mol ⁻¹)	Density @ 15.6°C (g.cm ⁻³)
Measured Properties	227 (measured)	0.8014 (measured)

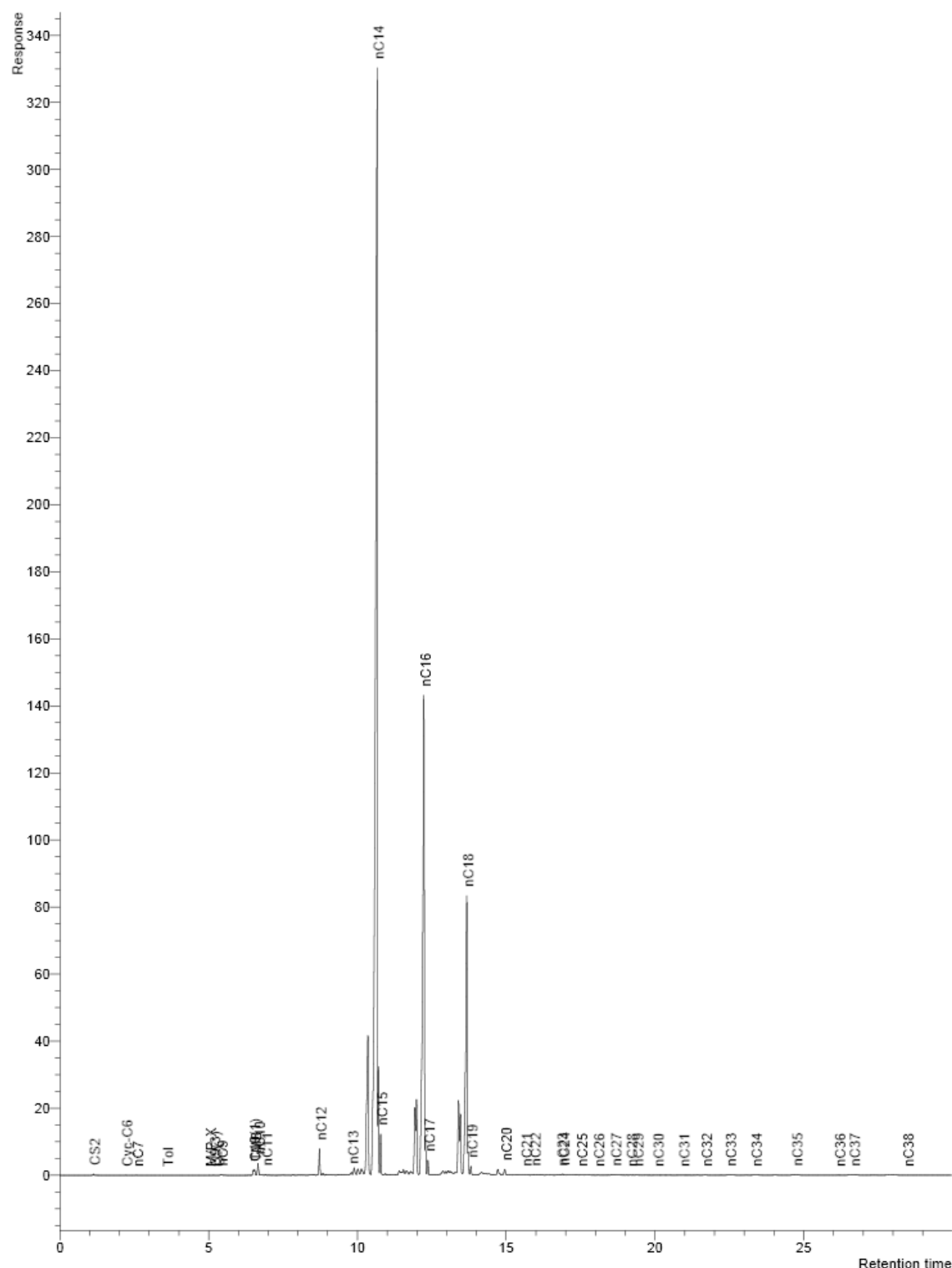
Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Chromatogram of Oil Based Drilling Mud
Sample No. 1.12, Mud Pit, 16/7/14

Pharos-1 Mud 1.12 (1,1)
 Acquired Wednesday, 3 September 2014 9:31:33 PM

wb31082014,as_2.9969,1,1,1



Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Section F - Calculated Uncontaminated Compositions - MDT gas

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.01, MPSR 2668 to C36+
Depth: 5000.2m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.802	26.855
N ₂	Nitrogen	0.320	0.397
C ₁	Methane	77.551	55.000
C ₂	Ethane	4.850	6.448
C ₃	Propane	1.345	2.622
iC ₄	i-Butane	0.354	0.910
nC ₄	n-Butane	0.323	0.831
C ₅	neo-Pentane	0.003	0.011
iC ₅	i-Pentane	0.209	0.666
nC ₅	n-Pentane	0.118	0.378
C ₆	Hexanes	0.209	0.797
	Me-Cyclo-pentane	0.034	0.127
	Benzene	0.055	0.188
	Cyclo-hexane	0.060	0.221
C ₇	Heptanes	0.110	0.485
	Me-Cyclo-hexane	0.056	0.245
	Toluene	0.109	0.445
C ₈	Octanes	0.080	0.405
	Ethyl-benzene	0.007	0.031
	Meta/Para-xylene	0.054	0.255
	Ortho-xylene	0.009	0.044
C ₉	Nonanes	0.068	0.384
	Tri-Me-benzene	0.006	0.033
C ₁₀	Decanes	0.062	0.392
C ₁₁	Undecanes	0.046	0.300
C ₁₂	Dodecanes	0.035	0.249
C ₁₃	Tridecanes	0.030	0.230
C ₁₄	Tetradecanes	0.021	0.179
C ₁₅	Pentadecanes	0.017	0.155
C ₁₆	Hexadecanes	0.012	0.120
C ₁₇	Heptadecanes	0.009	0.094
C ₁₈	Octadecanes	0.008	0.084
C ₁₉	Nonadecanes	0.006	0.070
C ₂₀	Eicosanes	0.004	0.054
C ₂₁	Heneicosanes	0.004	0.047
C ₂₂	Docosanes	0.003	0.041
C ₂₃	Tricosanes	0.002	0.033
C ₂₄	Tetracosanes	0.002	0.030
C ₂₅	Pentacosanes	0.002	0.025
C ₂₆	Hexacosanes	0.001	0.022
C ₂₇	Heptacosanes	0.001	0.019
C ₂₈	Octacosanes	0.001	0.016
C ₂₉	Nonacosanes	0.001	0.013
C ₃₀	Triacotanes	0.001	0.011
C ₃₁	Hentriacontanes	0.000	0.009
C ₃₂	Dotriacontanes	0.000	0.007
C ₃₃	Trtriacontanes	0.000	0.006
C ₃₄	Tetratriacontanes	0.000	0.005
C ₃₅	Pentatriacontanes	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.000	0.007
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

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Pharos-1 MDT

**Calculated Uncontaminated Composition of Sample 1.01, MPSR 2668 to C36+
Depth: 5000.2m MDRT**

Calculated Residue Properties		
C7+	Mole%	0.916
	Molecular Weight (g mol-1)	125
	Density at 15.6°C (g cm-3)	0.7869
C11+	Mole%	0.206
	Molecular Weight (g mol-1)	199
	Density at 15.6°C (g cm-3)	0.8282
C20+	Mole%	0.022
	Molecular Weight (g mol-1)	331
	Density at 15.6°C (g cm-3)	0.8813
C36+	Mole %	
	Molecular Weight (g mol-1)	514
	Density at 15.6°C (g cm-3)	0.9216
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	22.6
	Estimated SBM whole fluid contamination (wt%)	0.2
	Estimated SBM stock tank fluid contamination (wt%)	9.4

Note:

The mud composition from page E.1 was used to back out the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.02, MPSR 3351 to C36+
Depth: 5000.2m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.838	26.928
N ₂	Nitrogen	0.334	0.414
C ₁	Methane	77.543	55.004
C ₂	Ethane	4.849	6.447
C ₃	Propane	1.326	2.585
iC ₄	i-Butane	0.350	0.901
nC ₄	n-Butane	0.315	0.810
C ₅	neo-Pentane	0.004	0.014
iC ₅	i-Pentane	0.205	0.655
nC ₅	n-Pentane	0.114	0.364
C ₆	Hexanes	0.208	0.794
	Me-Cyclo-pentane	0.034	0.126
	Benzene	0.053	0.184
	Cyclo-hexane	0.059	0.220
C ₇	Heptanes	0.107	0.476
	Me-Cyclo-hexane	0.055	0.239
	Toluene	0.109	0.444
C ₈	Octanes	0.078	0.392
	Ethyl-benzene	0.007	0.031
	Meta/Para-xylene	0.052	0.246
	Ortho-xylene	0.010	0.045
C ₉	Nonanes	0.066	0.377
	Tri-Me-benzene	0.006	0.033
C ₁₀	Decanes	0.064	0.405
C ₁₁	Undecanes	0.049	0.318
C ₁₂	Dodecanes	0.036	0.259
C ₁₃	Tridecanes	0.030	0.231
C ₁₄	Tetradecanes	0.022	0.188
C ₁₅	Pentadecanes	0.018	0.160
C ₁₆	Hexadecanes	0.013	0.128
C ₁₇	Heptadecanes	0.009	0.095
C ₁₈	Octadecanes	0.008	0.089
C ₁₉	Nonadecanes	0.006	0.070
C ₂₀	Eicosanes	0.005	0.055
C ₂₁	Heneicosanes	0.004	0.046
C ₂₂	Docosanes	0.003	0.039
C ₂₃	Tricosanes	0.002	0.033
C ₂₄	Tetracosanes	0.002	0.029
C ₂₅	Pentacosanes	0.002	0.024
C ₂₆	Hexacosanes	0.001	0.021
C ₂₇	Heptacosanes	0.001	0.017
C ₂₈	Octacosanes	0.001	0.014
C ₂₉	Nonacosanes	0.001	0.012
C ₃₀	Triacotanes	0.001	0.010
C ₃₁	Hentriacotanes	0.000	0.008
C ₃₂	Dotriacotanes	0.000	0.006
C ₃₃	Tristriacotanes	0.000	0.005
C ₃₄	Tetratriacotanes	0.000	0.004
C ₃₅	Pentatriacotanes	0.000	0.003
C ₃₆₊	Hexatriacotanes plus	0.000	0.002
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.02, MPSR 3351 to C36+
Depth: 5000.2m MDRT

Calculated Residue Properties		
C7+	Mole%	0.914
	Molecular Weight (g mol-1)	126
	Density at 15.6°C (g cm-3)	0.7870
C11+	Mole%	0.214
	Molecular Weight (g mol-1)	197
	Density at 15.6°C (g cm-3)	0.8270
C20+	Mole%	0.023
	Molecular Weight (g mol-1)	327
	Density at 15.6°C (g cm-3)	0.8800
C36+	Mole %	
	Molecular Weight (g mol-1)	509
	Density at 15.6°C (g cm-3)	0.9207
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	22.6
	Estimated SBM whole fluid contamination (wt%)	0.2
	Estimated SBM stock tank fluid contamination (wt%)	7.7

Note:

The mud composition from page E.1 was used to back calculate the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.03, MPSR 2642 to C36+
Depth: 5000.2m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.596	26.631
N ₂	Nitrogen	0.366	0.457
C ₁	Methane	77.899	55.617
C ₂	Ethane	4.795	6.418
C ₃	Propane	1.316	2.583
iC ₄	i-Butane	0.345	0.893
nC ₄	n-Butane	0.312	0.807
C ₅	neo-Pentane	0.005	0.015
iC ₅	i-Pentane	0.197	0.631
nC ₅	n-Pentane	0.109	0.351
C ₆	Hexanes	0.197	0.754
	Me-Cyclo-pentane	0.032	0.119
	Benzene	0.050	0.175
	Cyclo-hexane	0.056	0.209
C ₇	Heptanes	0.101	0.453
	Me-Cyclo-hexane	0.053	0.230
	Toluene	0.103	0.422
C ₈	Octanes	0.073	0.372
	Ethyl-benzene	0.006	0.029
	Meta/Para-xylene	0.051	0.243
	Ortho-xylene	0.009	0.043
C ₉	Nonanes	0.064	0.363
	Tri-Me-benzene	0.006	0.033
C ₁₀	Decanes	0.061	0.388
C ₁₁	Undecanes	0.046	0.298
C ₁₂	Dodecanes	0.032	0.231
C ₁₃	Tridecanes	0.027	0.212
C ₁₄	Tetradecanes	0.020	0.168
C ₁₅	Pentadecanes	0.017	0.153
C ₁₆	Hexadecanes	0.012	0.122
C ₁₇	Heptadecanes	0.009	0.094
C ₁₈	Octadecanes	0.008	0.086
C ₁₉	Nonadecanes	0.006	0.069
C ₂₀	Eicosanes	0.004	0.053
C ₂₁	Heneicosanes	0.003	0.045
C ₂₂	Docosanes	0.003	0.040
C ₂₃	Tricosanes	0.002	0.032
C ₂₄	Tetracosanes	0.002	0.028
C ₂₅	Pentacosanes	0.002	0.024
C ₂₆	Hexacosanes	0.001	0.020
C ₂₇	Heptacosanes	0.001	0.018
C ₂₈	Octacosanes	0.001	0.014
C ₂₉	Nonacosanes	0.001	0.012
C ₃₀	triacontanes	0.001	0.010
C ₃₁	Hentriacontanes	0.000	0.009
C ₃₂	Dotriacontanes	0.000	0.007
C ₃₃	Trtriacontanes	0.000	0.006
C ₃₄	Tetratriacontanes	0.000	0.005
C ₃₅	Pentatriacontanes	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.000	0.004
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.03, MPSR 2642 to C36+
Depth: 5000.2m MDRT

Calculated Residue Properties		
C ₇₊	Mole%	0.863
	Molecular Weight (g mol-1)	126
	Density at 15.6°C (g cm-3)	0.7873
C ₁₁₊	Mole%	0.198
	Molecular Weight (g mol-1)	199
	Density at 15.6°C (g cm-3)	0.8282
C ₂₀₊	Mole%	0.021
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8808
C ₃₆₊	Mole %	
	Molecular Weight (g mol-1)	507
	Density at 15.6°C (g cm-3)	0.9206
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	22.5
	Estimated SBM whole fluid contamination (wt%)	0.4
	Estimated SBM stock tank fluid contamination (wt%)	14.2

Note:

The mud composition from page E.1 was used to back calculate the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.04, MPSR 5055 to C36+
Depth: 5000.2m MDRT**

Component		Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.925	27.140
N ₂	Nitrogen	0.294	0.365
C ₁	Methane	77.551	55.090
C ₂	Ethane	4.824	6.424
C ₃	Propane	1.326	2.589
iC ₄	i-Butane	0.350	0.902
nC ₄	n-Butane	0.317	0.817
C ₅	neo-Pentane	0.004	0.014
iC ₅	i-Pentane	0.209	0.669
nC ₅	n-Pentane	0.115	0.368
C ₆	Hexanes	0.207	0.791
	Me-Cyclo-pentane	0.034	0.126
	Benzene	0.053	0.183
	Cyclo-hexane	0.059	0.221
C ₇	Heptanes	0.106	0.472
	Me-Cyclo-hexane	0.055	0.238
	Toluene	0.102	0.414
C ₈	Octanes	0.084	0.425
	Ethyl-benzene	0.006	0.029
	Meta/Para-xylene	0.051	0.238
	Ortho-xylene	0.009	0.043
C ₉	Nonanes	0.062	0.353
	Tri-Me-benzene	0.006	0.032
C ₁₀	Decanes	0.059	0.369
C ₁₁	Undecanes	0.045	0.291
C ₁₂	Dodecanes	0.033	0.235
C ₁₃	Tridecanes	0.028	0.214
C ₁₄	Tetradecanes	0.020	0.169
C ₁₅	Pentadecanes	0.016	0.147
C ₁₆	Hexadecanes	0.011	0.111
C ₁₇	Heptadecanes	0.008	0.088
C ₁₈	Octadecanes	0.007	0.075
C ₁₉	Nonadecanes	0.005	0.063
C ₂₀	Eicosanes	0.004	0.050
C ₂₁	Heneicosanes	0.003	0.041
C ₂₂	Docosanes	0.003	0.036
C ₂₃	Tricosanes	0.002	0.029
C ₂₄	Tetracosanes	0.002	0.025
C ₂₅	Pentacosanes	0.001	0.022
C ₂₆	Hexacosanes	0.001	0.018
C ₂₇	Heptacosanes	0.001	0.016
C ₂₈	Octacosanes	0.001	0.012
C ₂₉	Nonacosanes	0.001	0.011
C ₃₀	Triacotanes	0.000	0.009
C ₃₁	Hentriacotanes	0.000	0.007
C ₃₂	Dotriacotanes	0.000	0.005
C ₃₃	Tritriacotanes	0.000	0.005
C ₃₄	Tetratriacotanes	0.000	0.004
C ₃₅	Pentatriacotanes	0.000	0.003
C ₃₆₊	Hexatriacotanes plus	0.000	0.002
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.04, MPSR 5055 to C36+
Depth: 5000.2m MDRT

Calculated Residue Properties		
C7+	Mole%	0.878
	Molecular Weight (g mol-1)	124
	Density at 15.6°C (g cm-3)	0.7850
C11+	Mole%	0.192
	Molecular Weight (g mol-1)	197
	Density at 15.6°C (g cm-3)	0.8267
C20+	Mole%	0.019
	Molecular Weight (g mol-1)	327
	Density at 15.6°C (g cm-3)	0.8800
C36+	Mole %	
	Molecular Weight (g mol-1)	507
	Density at 15.6°C (g cm-3)	0.9205
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	22.6
	Estimated SBM whole fluid contamination (wt%)	0.3
	Estimated SBM stock tank fluid contamination (wt%)	12.5

Note:

The mud composition from page E.1 was used to back out the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.05, MPSR 5049 to C36+
Depth: 5021.0m MDRT**

Component		Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.854	26.957
N ₂	Nitrogen	0.371	0.459
C ₁	Methane	77.510	54.978
C ₂	Ethane	4.844	6.440
C ₃	Propane	1.317	2.568
iC ₄	i-Butane	0.349	0.896
nC ₄	n-Butane	0.316	0.813
C ₅	neo-Pentane	0.003	0.009
iC ₅	i-Pentane	0.206	0.657
nC ₅	n-Pentane	0.114	0.364
C ₆	Hexanes	0.207	0.790
	Me-Cyclo-pentane	0.034	0.126
	Benzene	0.053	0.185
	Cyclo-hexane	0.059	0.220
C ₇	Heptanes	0.107	0.473
	Me-Cyclo-hexane	0.056	0.242
	Toluene	0.098	0.401
C ₈	Octanes	0.090	0.457
	Ethyl-benzene	0.006	0.029
	Meta/Para-xylene	0.053	0.247
	Ortho-xylene	0.009	0.042
C ₉	Nonanes	0.066	0.376
	Tri-Me-benzene	0.006	0.034
C ₁₀	Decanes	0.062	0.391
C ₁₁	Undecanes	0.048	0.313
C ₁₂	Dodecanes	0.035	0.251
C ₁₃	Tridecanes	0.029	0.226
C ₁₄	Tetradecanes	0.022	0.182
C ₁₅	Pentadecanes	0.017	0.156
C ₁₆	Hexadecanes	0.013	0.127
C ₁₇	Heptadecanes	0.009	0.095
C ₁₈	Octadecanes	0.008	0.086
C ₁₉	Nonadecanes	0.006	0.068
C ₂₀	Eicosanes	0.005	0.055
C ₂₁	Heneicosanes	0.004	0.048
C ₂₂	Docosanes	0.003	0.046
C ₂₃	Tricosanes	0.002	0.033
C ₂₄	Tetracosanes	0.002	0.028
C ₂₅	Pentacosanes	0.002	0.024
C ₂₆	Hexacosanes	0.001	0.020
C ₂₇	Heptacosanes	0.001	0.018
C ₂₈	Octacosanes	0.001	0.014
C ₂₉	Nonacosanes	0.001	0.012
C ₃₀	Triacotanes	0.001	0.010
C ₃₁	Hentriacontanes	0.000	0.008
C ₃₂	Dotriacontanes	0.000	0.006
C ₃₃	Trtriacontanes	0.000	0.005
C ₃₄	Tetratriacontanes	0.000	0.004
C ₃₅	Pentatriacontanes	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.000	0.007
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.05, MPSR 5049 to C36+
Depth: 5021.0m MDRT

Calculated Residue Properties		
C ₇₊	Mole%	0.909
	Molecular Weight (g mol-1)	126
	Density at 15.6°C (g cm-3)	0.7854
C ₁₁₊	Mole%	0.210
	Molecular Weight (g mol-1)	198
	Density at 15.6°C (g cm-3)	0.8277
C ₂₀₊	Mole%	0.023
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8805
C ₃₆₊	Mole %	
	Molecular Weight (g mol-1)	514
	Density at 15.6°C (g cm-3)	0.9217
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	22.6
	Estimated SBM whole fluid contamination (wt%)	0.9
	Estimated SBM stock tank fluid contamination (wt%)	26.9

Note:

The mud composition from page E.1 was used to back calculate the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.06, MPSR 2489 to C36+
Depth: 5021.0m MDRT**

Component		Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.826	26.868
N ₂	Nitrogen	0.320	0.396
C ₁	Methane	77.525	54.917
C ₂	Ethane	4.861	6.454
C ₃	Propane	1.329	2.588
iC ₄	i-Butane	0.352	0.903
nC ₄	n-Butane	0.318	0.817
C ₅	neo-Pentane	0.004	0.014
iC ₅	i-Pentane	0.205	0.655
nC ₅	n-Pentane	0.114	0.362
C ₆	Hexanes	0.208	0.791
	Me-Cyclo-pentane	0.034	0.126
	Benzene	0.054	0.185
	Cyclo-hexane	0.059	0.219
C ₇	Heptanes	0.108	0.476
	Me-Cyclo-hexane	0.055	0.238
	Toluene	0.110	0.446
C ₈	Octanes	0.078	0.393
	Ethyl-benzene	0.007	0.031
	Meta/Para-xylene	0.054	0.252
	Ortho-xylene	0.009	0.044
C ₉	Nonanes	0.069	0.389
	Tri-Me-benzene	0.007	0.035
C ₁₀	Decanes	0.068	0.426
C ₁₁	Undecanes	0.051	0.333
C ₁₂	Dodecanes	0.037	0.260
C ₁₃	Tridecanes	0.033	0.256
C ₁₄	Tetradecanes	0.025	0.210
C ₁₅	Pentadecanes	0.018	0.167
C ₁₆	Hexadecanes	0.014	0.138
C ₁₇	Heptadecanes	0.010	0.101
C ₁₈	Octadecanes	0.008	0.093
C ₁₉	Nonadecanes	0.006	0.074
C ₂₀	Eicosanes	0.005	0.058
C ₂₁	Heneicosanes	0.004	0.050
C ₂₂	Docosanes	0.004	0.047
C ₂₃	Tricosanes	0.002	0.034
C ₂₄	Tetracosanes	0.002	0.030
C ₂₅	Pentacosanes	0.002	0.025
C ₂₆	Hexacosanes	0.001	0.021
C ₂₇	Heptacosanes	0.001	0.019
C ₂₈	Octacosanes	0.001	0.014
C ₂₉	Nonacosanes	0.001	0.012
C ₃₀	Triacotanes	0.001	0.010
C ₃₁	Hentriacontanes	0.000	0.008
C ₃₂	Dotriacontanes	0.000	0.006
C ₃₃	Trtriacontanes	0.000	0.004
C ₃₄	Tetratriacontanes	0.000	0.003
C ₃₅	Pentatriacontanes	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.000	0.000
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.06, MPSR 2489 to C36+
Depth: 5021.0m MDRT

Calculated Residue Properties		
C ₇₊	Mole%	0.938
	Molecular Weight (g mol-1)	127
	Density at 15.6°C (g cm-3)	0.7875
C ₁₁₊	Mole%	0.226
	Molecular Weight (g mol-1)	197
	Density at 15.6°C (g cm-3)	0.8269
C ₂₀₊	Mole%	0.024
	Molecular Weight (g mol-1)	325
	Density at 15.6°C (g cm-3)	0.8792
C ₃₆₊	Mole %	
	Molecular Weight (g mol-1)	512
	Density at 15.6°C (g cm-3)	0.9214
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	22.6
	Estimated SBM whole fluid contamination (wt%)	0.8
	Estimated SBM stock tank fluid contamination (wt%)	23.8

Note:

The mud composition from page E.1 was used to back out the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.08, MPSR 3374 to C36+
Depth: 5021.0m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.764	26.832
N ₂	Nitrogen	0.323	0.401
C ₁	Methane	77.713	55.226
C ₂	Ethane	4.815	6.414
C ₃	Propane	1.311	2.561
iC ₄	i-Butane	0.349	0.898
nC ₄	n-Butane	0.317	0.817
C ₅	neo-Pentane	0.005	0.017
iC ₅	i-Pentane	0.198	0.631
nC ₅	n-Pentane	0.110	0.351
C ₆	Hexanes	0.196	0.747
	Me-Cyclo-pentane	0.032	0.119
	Benzene	0.051	0.175
	Cyclo-hexane	0.055	0.207
C ₇	Heptanes	0.102	0.454
	Me-Cyclo-hexane	0.052	0.228
	Toluene	0.093	0.379
C ₈	Octanes	0.085	0.429
	Ethyl-benzene	0.006	0.029
	Meta/Para-xylene	0.052	0.244
	Ortho-xylene	0.009	0.043
C ₉	Nonanes	0.067	0.381
	Tri-Me-benzene	0.007	0.038
C ₁₀	Decanes	0.065	0.410
C ₁₁	Undecanes	0.051	0.329
C ₁₂	Dodecanes	0.036	0.254
C ₁₃	Tridecanes	0.030	0.232
C ₁₄	Tetradecanes	0.023	0.190
C ₁₅	Pentadecanes	0.019	0.171
C ₁₆	Hexadecanes	0.014	0.141
C ₁₇	Heptadecanes	0.010	0.104
C ₁₈	Octadecanes	0.009	0.096
C ₁₉	Nonadecanes	0.006	0.075
C ₂₀	Eicosanes	0.005	0.062
C ₂₁	Heneicosanes	0.004	0.052
C ₂₂	Docosanes	0.004	0.048
C ₂₃	Tricosanes	0.003	0.036
C ₂₄	Tetracosanes	0.002	0.031
C ₂₅	Pentacosanes	0.002	0.027
C ₂₆	Hexacosanes	0.001	0.023
C ₂₇	Heptacosanes	0.001	0.020
C ₂₈	Octacosanes	0.001	0.016
C ₂₉	Nonacosanes	0.001	0.014
C ₃₀	triacontanes	0.001	0.011
C ₃₁	Hentriacontanes	0.000	0.009
C ₃₂	Dotriacontanes	0.000	0.007
C ₃₃	Trtriacontanes	0.000	0.006
C ₃₄	Tetratriacontanes	0.000	0.004
C ₃₅	Pentatriacontanes	0.000	0.005
C ₃₆₊	Hexatriacontanes plus	0.000	0.006
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

**Calculated Uncontaminated Composition of Sample 1.08, MPSR 3374 to C36+
Depth: 5021.0m MDRT**

Calculated Residue Properties

C₇₊	Mole%	0.899
	Molecular Weight (g mol-1)	128
	Density at 15.6°C (g cm-3)	0.7870
C₁₁₊	Mole%	0.223
	Molecular Weight (g mol-1)	200
	Density at 15.6°C (g cm-3)	0.8286
C₂₀₊	Mole%	0.025
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8806
C₃₆₊	Mole %	
	Molecular Weight (g mol-1)	494
	Density at 15.6°C (g cm-3)	0.9182

Calculated Whole Sample Properties

Average mole weight (g mol-1)	22.6
Estimated SBM whole fluid contamination (wt%)	0.8
Estimated SBM stock tank fluid contamination (wt%)	23.9

Note:

The mud composition from page E.1 was used to back calculate the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.09, MPSR 5048 to C36+
Depth: 5021.0m MDRT**

Component		Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	13.913	26.983
N ₂	Nitrogen	0.307	0.379
C ₁	Methane	77.423	54.736
C ₂	Ethane	4.838	6.412
C ₃	Propane	1.336	2.596
iC ₄	i-Butane	0.354	0.908
nC ₄	n-Butane	0.324	0.829
C ₅	neo-Pentane	0.005	0.015
iC ₅	i-Pentane	0.210	0.669
nC ₅	n-Pentane	0.117	0.371
C ₆	Hexanes	0.211	0.800
	Me-Cyclo-pentane	0.036	0.132
	Benzene	0.056	0.193
	Cyclo-hexane	0.063	0.232
C ₇	Heptanes	0.113	0.497
	Me-Cyclo-hexane	0.061	0.264
	Toluene	0.110	0.445
C ₈	Octanes	0.093	0.469
	Ethyl-benzene	0.007	0.034
	Meta/Para-xylene	0.059	0.275
	Ortho-xylene	0.011	0.052
C ₉	Nonanes	0.070	0.395
	Tri-Me-benzene	0.007	0.037
C ₁₀	Decanes	0.066	0.412
C ₁₁	Undecanes	0.048	0.310
C ₁₂	Dodecanes	0.034	0.239
C ₁₃	Tridecanes	0.029	0.226
C ₁₄	Tetradecanes	0.022	0.183
C ₁₅	Pentadecanes	0.018	0.162
C ₁₆	Hexadecanes	0.013	0.132
C ₁₇	Heptadecanes	0.009	0.098
C ₁₈	Octadecanes	0.008	0.091
C ₁₉	Nonadecanes	0.006	0.071
C ₂₀	Eicosanes	0.005	0.058
C ₂₁	Heneicosanes	0.004	0.048
C ₂₂	Docosanes	0.003	0.044
C ₂₃	Tricosanes	0.002	0.034
C ₂₄	Tetracosanes	0.002	0.029
C ₂₅	Pentacosanes	0.002	0.026
C ₂₆	Hexacosanes	0.001	0.021
C ₂₇	Heptacosanes	0.001	0.019
C ₂₈	Octacosanes	0.001	0.015
C ₂₉	Nonacosanes	0.001	0.013
C ₃₀	Triacotanes	0.001	0.011
C ₃₁	Hentriacontanes	0.000	0.009
C ₃₂	Dotriacontanes	0.000	0.006
C ₃₃	Trtriacontanes	0.000	0.005
C ₃₄	Tetratriacontanes	0.000	0.004
C ₃₅	Pentatriacontanes	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.000	0.007
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

**Calculated Uncontaminated Composition of Sample 1.09, MPSR 5048 to C36+
Depth: 5021.0m MDRT**

Calculated Residue Properties

C₇₊	Mole%	0.962
	Molecular Weight (g mol-1)	125
	Density at 15.6°C (g cm-3)	0.7860
C₁₁₊	Mole%	0.210
	Molecular Weight (g mol-1)	200
	Density at 15.6°C (g cm-3)	0.8285
C₂₀₊	Mole%	0.023
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8807
C₃₆₊	Mole %	
	Molecular Weight (g mol-1)	517
	Density at 15.6°C (g cm-3)	0.9222

Calculated Whole Sample Properties

Average mole weight (g mol-1)	22.7
Estimated SBM whole fluid contamination (wt%)	0.9
Estimated SBM stock tank fluid contamination (wt%)	24.9

Note:

The mud composition from page E.1 was used to back calculate the mud contamination.

The calculated uncontaminated compositions may contain errors due to a compositional mismatch between the mud measured for composition (page E.1) and the contamination present in the sample.

Section G - Compositional Analysis Data - SCAR and Slickline Gas

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.65, Cylinder 14952-IB to C
Depth: 4837.4m MDRT

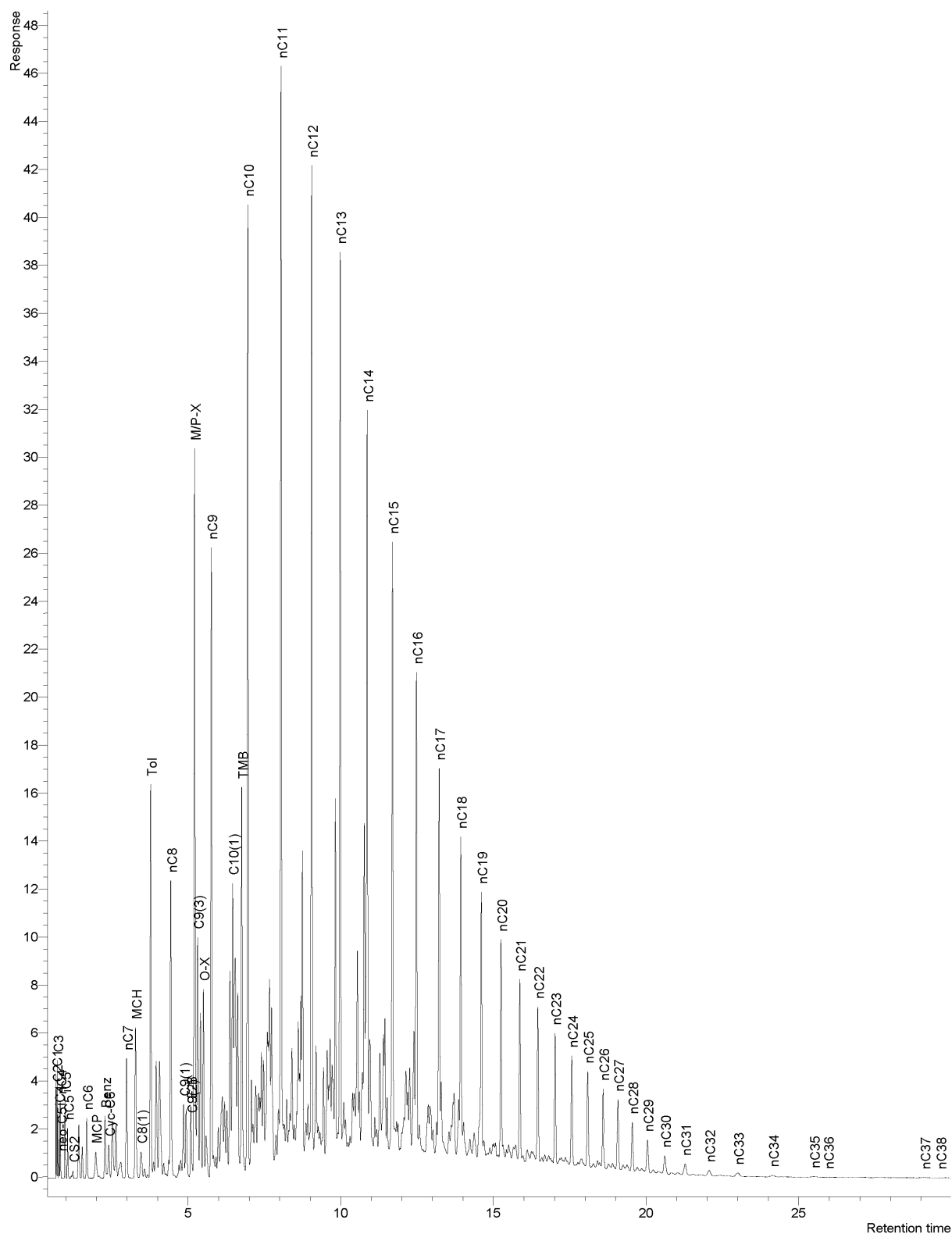
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.959	13.915	27.052
N ₂	Nitrogen	0.000	0.000	0.305	0.304	0.376
C ₁	Methane	0.422	0.040	77.724	77.480	54.909
G.1	Ethane	0.327	0.058	4.857	4.843	6.433
C ₃	Propane	0.384	0.100	1.335	1.332	2.595
iC ₄	i-Butane	0.219	0.075	0.353	0.353	0.906
nC ₄	n-Butane	0.251	0.086	0.319	0.318	0.817
C ₅	neo-Pentane	0.035	0.015	0.006	0.007	0.021
iC ₅	i-Pentane	0.361	0.154	0.212	0.212	0.676
nC ₅	n-Pentane	0.253	0.108	0.117	0.118	0.376
C ₆	Hexanes	0.926	0.471	0.204	0.206	0.783
	Me-Cyclo-pentane	0.306	0.152	0.033	0.034	0.126
	Benzene	0.451	0.208	0.053	0.054	0.187
	Cyclo-hexane	0.280	0.139	0.060	0.061	0.225
C ₇	Heptanes	1.802	1.066	0.102	0.108	0.477
	Me-Cyclo-hexane	1.043	0.605	0.053	0.056	0.243
	Toluene	2.308	1.256	0.095	0.102	0.415
C ₈	Octanes	3.991	2.692	0.073	0.085	0.431
	Ethyl-benzene	0.284	0.178	0.005	0.006	0.027
	Meta/Para-xylene	4.184	2.623	0.041	0.054	0.254
	Ortho-xylene	0.991	0.621	0.005	0.009	0.040
C ₉	Nonanes	6.348	4.808	0.047	0.067	0.377
	Tri-Me-benzene	1.829	1.298	0.000	0.006	0.030
C ₁₀	Decanes	10.975	9.221	0.029	0.063	0.396
C ₁₁	Undecanes	11.503	9.986	0.011	0.047	0.308
C ₁₂	Dodecanes	9.800	9.317	0.002	0.033	0.232
C ₁₃	Tridecanes	8.656	8.945	0.000	0.027	0.212
C ₁₄	Tetradecanes	7.656	8.590	0.000	0.024	0.202
C ₁₅	Pentadecanes	5.459	6.641	0.000	0.017	0.156
C ₁₆	Hexadecanes	4.333	5.680	0.000	0.014	0.133
C ₁₇	Heptadecanes	2.878	4.028	0.000	0.009	0.095
C ₁₈	Octadecanes	2.637	3.908	0.000	0.008	0.092
C ₁₉	Nonadecanes	1.860	2.889	0.000	0.006	0.068
C ₂₀	Eicosanes	1.408	2.287	0.000	0.004	0.054
C ₂₁	Heneicosanes	1.184	2.034	0.000	0.004	0.048
C ₂₂	Docosanes	0.931	1.677	0.000	0.003	0.039
C ₂₃	Tricosanes	0.738	1.385	0.000	0.002	0.033
C ₂₄	Tetracosanes	0.633	1.238	0.000	0.002	0.029
C ₂₅	Pentacosanes	0.502	1.022	0.000	0.002	0.024
C ₂₆	Hexacosanes	0.399	0.845	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.333	0.735	0.000	0.001	0.017
C ₂₈	Octacosanes	0.258	0.590	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.213	0.505	0.000	0.001	0.012
C ₃₀	Triacontanes	0.162	0.399	0.000	0.001	0.009
C ₃₁	Hentriacontanes	0.124	0.316	0.000	0.000	0.007
C ₃₂	Dotriacontanes	0.093	0.244	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.077	0.207	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.057	0.158	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.044	0.127	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.092	0.273	0.000	0.000	0.006
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.65, Cylinder 14952-IB to C3
Depth: 4837.4m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.822	0.609	0.912
Molecular Weight (g mol-1)	173	101.3	126
Density at 15.6°C (g cm-3)	0.8103	0.7738	0.7858
C₁₁₊			
Mole%	62.030	0.013	0.207
Molecular Weight (g mol-1)	202	150	199
Density at 15.6°C (g cm-3)	0.8299	0.7911	0.8279
C₂₀₊			
Mole%	7.248		0.022
Molecular Weight (g mol-1)	328		328
Density at 15.6°C (g cm-3)	0.8804		0.8804
C₃₆₊			
Mole %	0.092		0.000
Molecular Weight (g mol-1)	517		517
Density at 15.6°C (g cm-3)	0.9221		0.9221
Whole Sample Properties			
Average mole weight (g mol-1)	169 (Calculated)	22.2	22.6
Density (g cm-3 at 15.6°C)	0.8067 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7681	
GOR Ratio (scf/bbl) from flash at 20.3°C			200745.8
CGR Ratio (bbl/MMscf) from flash at 20.3°C			5.0
Gas Mole Fraction			0.9969
Liquid Mole Fraction			0.0031

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.65, Cylinder 14952**
Depth: 4837.4m MDRTPharos-1 S/F (1,1)
Acquired Friday, 24 October 2014 4:07:18 PM

wb19102014,as_2.10205,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.66, Cylinder 28985-IB to C
Depth: 4837.4m MDRT

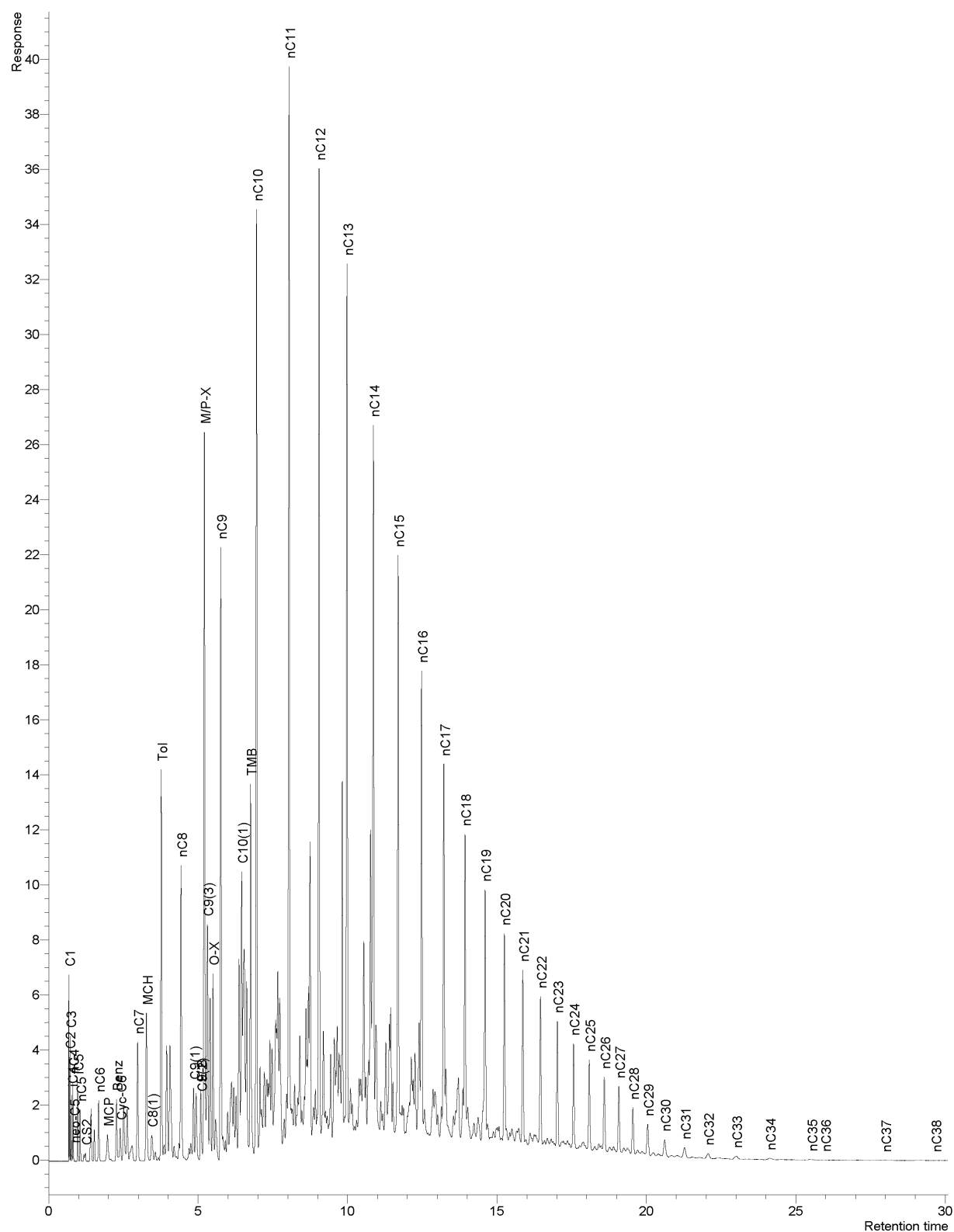
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.845	13.802	26.908
N ₂	Nitrogen	0.000	0.000	0.309	0.308	0.382
C ₁	Methane	0.842	0.080	77.829	77.592	55.139
G.1	Ethane	0.326	0.058	4.872	4.858	6.470
C ₃	Propane	0.348	0.091	1.344	1.341	2.618
iC ₄	i-Butane	0.195	0.067	0.356	0.356	0.915
nC ₄	n-Butane	0.238	0.082	0.317	0.317	0.816
C ₅	neo-Pentane	0.019	0.008	0.005	0.005	0.017
iC ₅	i-Pentane	0.330	0.141	0.210	0.210	0.671
nC ₅	n-Pentane	0.239	0.102	0.115	0.116	0.370
C ₆	Hexanes	0.903	0.461	0.205	0.207	0.789
	Me-Cyclo-pentane	0.305	0.152	0.033	0.034	0.127
	Benzene	0.445	0.206	0.053	0.054	0.187
	Cyclo-hexane	0.283	0.141	0.059	0.060	0.223
C ₇	Heptanes	1.791	1.063	0.102	0.107	0.474
	Me-Cyclo-hexane	1.042	0.606	0.052	0.055	0.239
	Toluene	2.316	1.264	0.093	0.100	0.408
C ₈	Octanes	4.013	2.715	0.074	0.086	0.435
	Ethyl-benzene	0.266	0.167	0.005	0.006	0.028
	Meta/Para-xylene	4.175	2.625	0.039	0.052	0.244
	Ortho-xylene	1.002	0.630	0.005	0.008	0.038
C ₉	Nonanes	6.339	4.815	0.042	0.062	0.349
	Tri-Me-benzene	1.647	1.172	0.000	0.005	0.027
C ₁₀	Decanes	11.124	9.373	0.024	0.058	0.365
C ₁₁	Undecanes	11.433	9.953	0.009	0.044	0.285
C ₁₂	Dodecanes	9.814	9.357	0.003	0.033	0.234
C ₁₃	Tridecanes	8.644	8.959	0.000	0.027	0.206
C ₁₄	Tetradecanes	7.582	8.532	0.000	0.023	0.193
C ₁₅	Pentadecanes	5.465	6.667	0.000	0.017	0.151
C ₁₆	Hexadecanes	4.285	5.633	0.000	0.013	0.128
C ₁₇	Heptadecanes	2.876	4.036	0.000	0.009	0.091
C ₁₈	Octadecanes	2.581	3.836	0.000	0.008	0.087
C ₁₉	Nonadecanes	1.875	2.921	0.000	0.006	0.066
C ₂₀	Eicosanes	1.396	2.274	0.000	0.004	0.051
C ₂₁	Heneicosanes	1.201	2.070	0.000	0.004	0.047
C ₂₂	Docosanes	0.934	1.687	0.000	0.003	0.038
C ₂₃	Tricosanes	0.746	1.405	0.000	0.002	0.032
C ₂₄	Tetracosanes	0.628	1.231	0.000	0.002	0.028
C ₂₅	Pentacosanes	0.508	1.038	0.000	0.002	0.023
C ₂₆	Hexacosanes	0.405	0.861	0.000	0.001	0.019
C ₂₇	Heptacosanes	0.332	0.736	0.000	0.001	0.017
C ₂₈	Octacosanes	0.258	0.593	0.000	0.001	0.013
C ₂₉	Nonacosanes	0.213	0.508	0.000	0.001	0.012
C ₃₀	Triacontanes	0.164	0.403	0.000	0.000	0.009
C ₃₁	Hentriacontanes	0.131	0.333	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.093	0.245	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.078	0.212	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.058	0.162	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.043	0.124	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.069	0.205	0.000	0.000	0.005
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.66, Cylinder 28985-IB to C3
Depth: 4837.4m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.560	0.593	0.888
Molecular Weight (g mol-1)	173	100.9	125
Density at 15.6°C (g cm-3)	0.8101	0.7738	0.7858
C₁₁₊			
Mole%	61.812	0.012	0.201
Molecular Weight (g mol-1)	202	151	199
Density at 15.6°C (g cm-3)	0.8299	0.7926	0.8280
C₂₀₊			
Mole%	7.257		0.021
Molecular Weight (g mol-1)	328		328
Density at 15.6°C (g cm-3)	0.8803		0.8803
C₃₆₊			
Mole %	0.069		0.000
Molecular Weight (g mol-1)	513		513
Density at 15.6°C (g cm-3)	0.9214		0.9214
Whole Sample Properties			
Average mole weight (g mol-1)	169 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8060 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7666	
GOR Ratio (scf/bbl) from flash at 20.1°C			208483.1
CGR Ratio (bbl/MMscf) from flash at 20.1°C			4.8
Gas Mole Fraction			0.9970
Liquid Mole Fraction			0.0030

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.66, Cylinder 28985**
Depth: 4837.4m MDRTPharos-1 S/F (1,1)
Acquired Friday, 17 October 2014 2:24:13 PM

wb12102014,as_2.10147,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of SCAR Gas Sample No. 1.67, Cylinder 5059-EA to C
Depth: 4837.4m MDRT**

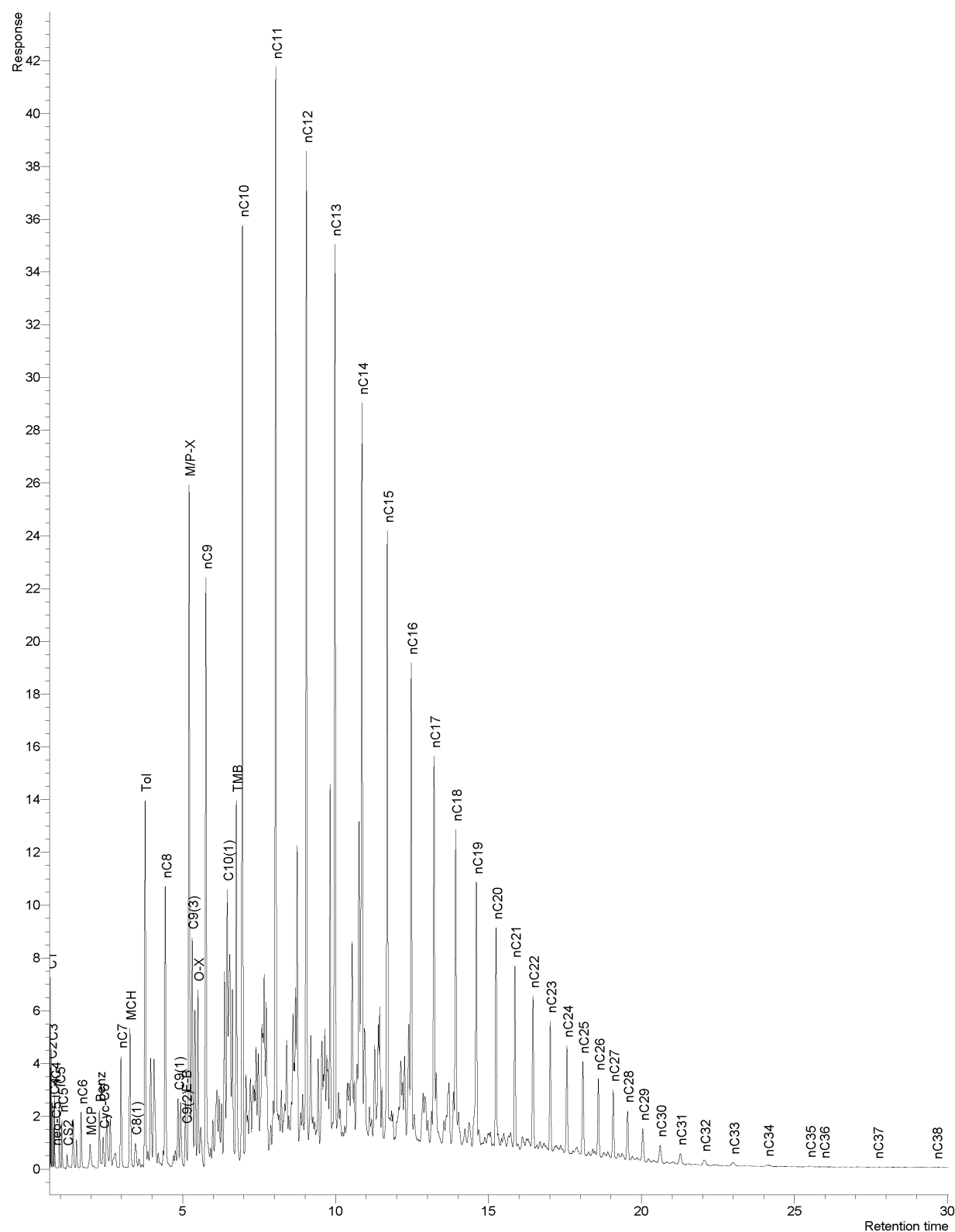
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.871	13.829	26.950
N ₂	Nitrogen	0.000	0.000	0.322	0.321	0.398
C ₁	Methane	0.819	0.077	77.797	77.564	55.099
G.1	Ethane	0.306	0.054	4.872	4.858	6.468
C ₃	Propane	0.325	0.084	1.348	1.345	2.626
iC ₄	i-Butane	0.179	0.061	0.358	0.358	0.921
nC ₄	n-Butane	0.202	0.069	0.322	0.322	0.828
C ₅	neo-Pentane	0.033	0.014	0.005	0.005	0.017
iC ₅	i-Pentane	0.305	0.129	0.208	0.209	0.666
nC ₅	n-Pentane	0.222	0.094	0.115	0.115	0.368
C ₆	Hexanes	0.853	0.431	0.202	0.204	0.778
	Me-Cyclo-pentane	0.284	0.140	0.032	0.033	0.121
	Benzene	0.419	0.192	0.052	0.053	0.182
	Cyclo-hexane	0.259	0.128	0.058	0.059	0.219
C ₇	Heptanes	1.705	1.002	0.097	0.101	0.450
	Me-Cyclo-hexane	0.994	0.572	0.052	0.054	0.237
	Toluene	2.195	1.186	0.087	0.093	0.381
C ₈	Octanes	3.858	2.584	0.074	0.085	0.431
	Ethyl-benzene	0.318	0.198	0.005	0.006	0.027
	Meta/Para-xylene	4.013	2.498	0.039	0.052	0.242
	Ortho-xylene	0.957	0.596	0.006	0.009	0.043
C ₉	Nonanes	6.175	4.644	0.043	0.061	0.349
	Tri-Me-benzene	1.791	1.262	0.000	0.005	0.029
C ₁₀	Decanes	10.831	9.035	0.025	0.058	0.366
C ₁₁	Undecanes	11.515	9.924	0.009	0.044	0.288
C ₁₂	Dodecanes	9.827	9.277	0.001	0.031	0.223
C ₁₃	Tridecanes	8.747	8.975	0.000	0.027	0.210
C ₁₄	Tetradecanes	7.732	8.614	0.000	0.024	0.199
C ₁₅	Pentadecanes	5.555	6.710	0.000	0.017	0.155
C ₁₆	Hexadecanes	4.386	5.709	0.000	0.013	0.132
C ₁₇	Heptadecanes	2.931	4.073	0.000	0.009	0.094
C ₁₈	Octadecanes	2.661	3.917	0.000	0.008	0.091
C ₁₉	Nonadecanes	1.908	2.942	0.000	0.006	0.068
C ₂₀	Eicosanes	1.444	2.329	0.000	0.004	0.054
C ₂₁	Heneicosanes	1.244	2.123	0.000	0.004	0.049
C ₂₂	Docosanes	1.028	1.838	0.000	0.003	0.043
C ₂₃	Tricosanes	0.804	1.499	0.000	0.002	0.035
C ₂₄	Tetracosanes	0.675	1.310	0.000	0.002	0.030
C ₂₅	Pentacosanes	0.532	1.076	0.000	0.002	0.025
C ₂₆	Hexacosanes	0.425	0.895	0.000	0.001	0.021
C ₂₇	Heptacosanes	0.364	0.799	0.000	0.001	0.018
C ₂₈	Octacosanes	0.277	0.631	0.000	0.001	0.015
C ₂₉	Nonacosanes	0.228	0.537	0.000	0.001	0.012
C ₃₀	Triacontanes	0.171	0.417	0.000	0.001	0.010
C ₃₁	Hentriacontanes	0.134	0.337	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.096	0.251	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.081	0.218	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.061	0.170	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.045	0.127	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.086	0.252	0.000	0.000	0.006
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.67, Cylinder 5059-EA to C36
Depth: 4837.4m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.756	0.580	0.870
Molecular Weight (g mol-1)	175	101.0	126
Density at 15.6°C (g cm-3)	0.8114	0.7734	0.7867
C₁₁₊			
Mole%	62.957	0.010	0.201
Molecular Weight (g mol-1)	203	149	200
Density at 15.6°C (g cm-3)	0.8305	0.7909	0.8289
C₂₀₊			
Mole%	7.695		0.022
Molecular Weight (g mol-1)	328		328
Density at 15.6°C (g cm-3)	0.8804		0.8804
C₃₆₊			
Mole %	0.086		0.000
Molecular Weight (g mol-1)	516		516
Density at 15.6°C (g cm-3)	0.9220		0.9220
Whole Sample Properties			
Average mole weight (g mol-1)	171 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8075 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7664	
GOR Ratio (scf/bbl) from flash at 19.9°C			204293.6
CGR Ratio (bbl/MMscf) from flash at 19.9°C			4.9
Gas Mole Fraction			0.9969
Liquid Mole Fraction			0.0031

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.67, Cylinder 5059-I**
Depth: 4837.4m MDRTPharos-1 S/F (1,1)
Acquired Wednesday, 22 October 2014 2:27:09 PM

wb19102014,as_2.10185,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.68, Cylinder 5142-EA to C
Depth: 4837.4m MDRT

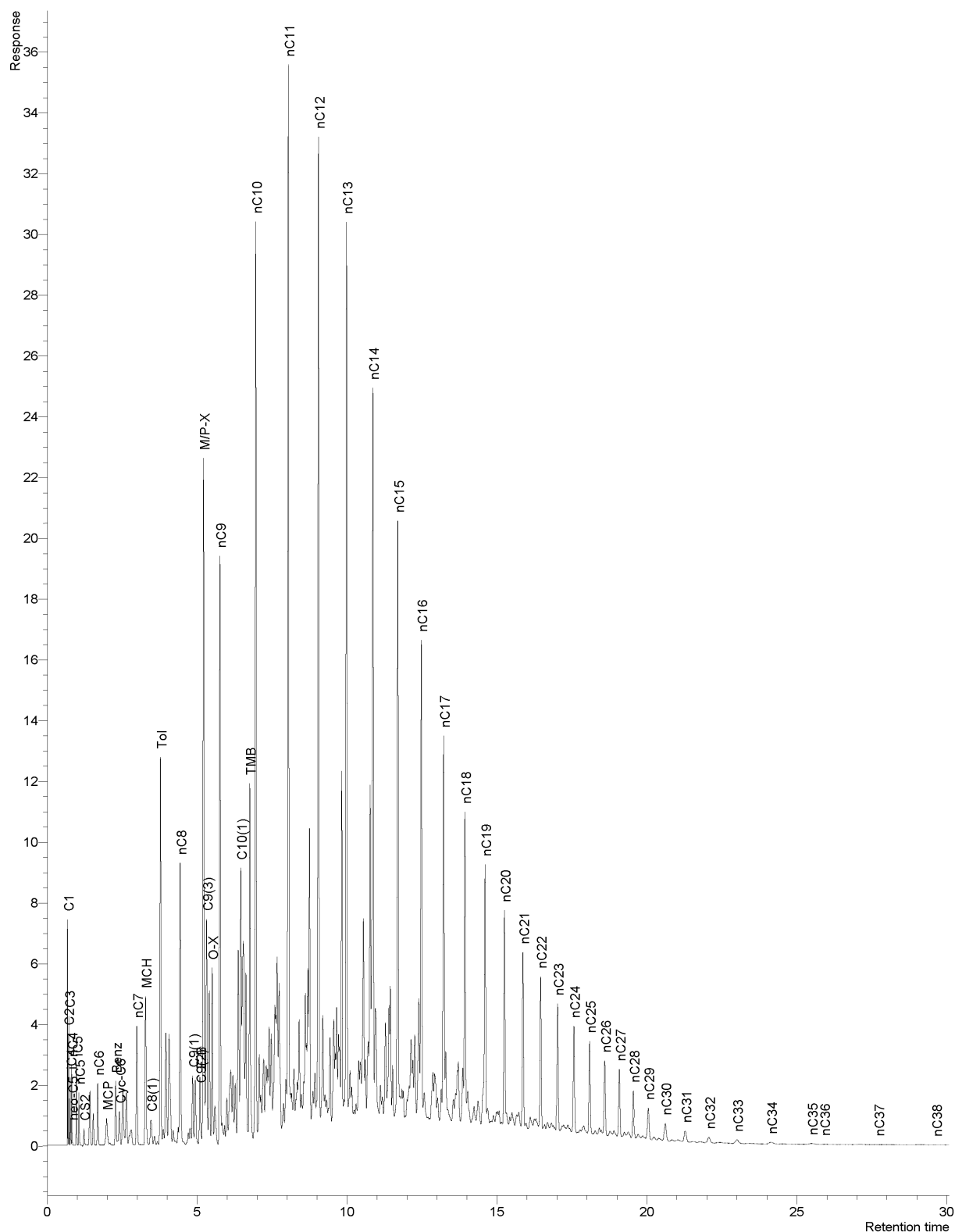
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.804	13.762	26.818
N ₂	Nitrogen	0.000	0.000	0.322	0.321	0.399
C ₁	Methane	1.025	0.097	77.842	77.606	55.126
G.1	Ethane	0.355	0.063	4.883	4.869	6.484
C ₃	Propane	0.369	0.096	1.344	1.341	2.618
iC ₄	i-Butane	0.207	0.071	0.355	0.354	0.912
nC ₄	n-Butane	0.236	0.081	0.318	0.318	0.818
C ₅	neo-Pentane	0.031	0.013	0.006	0.007	0.021
iC ₅	i-Pentane	0.345	0.147	0.209	0.210	0.670
nC ₅	n-Pentane	0.247	0.105	0.116	0.116	0.371
C ₆	Hexanes	0.925	0.470	0.203	0.205	0.783
	Me-Cyclo-pentane	0.306	0.152	0.033	0.033	0.125
	Benzene	0.445	0.205	0.052	0.054	0.185
	Cyclo-hexane	0.288	0.143	0.058	0.059	0.220
C ₇	Heptanes	1.792	1.059	0.099	0.105	0.464
	Me-Cyclo-hexane	1.034	0.599	0.052	0.055	0.238
	Toluene	2.269	1.233	0.090	0.096	0.393
C ₈	Octanes	3.863	2.603	0.075	0.087	0.439
	Ethyl-benzene	0.268	0.168	0.005	0.006	0.029
	Meta/Para-xylene	3.944	2.470	0.039	0.051	0.242
	Ortho-xylene	0.942	0.590	0.006	0.009	0.044
C ₉	Nonanes	6.021	4.555	0.044	0.062	0.354
	Tri-Me-benzene	1.770	1.255	0.000	0.005	0.029
C ₁₀	Decanes	10.525	8.832	0.029	0.061	0.383
C ₁₁	Undecanes	11.360	9.849	0.013	0.048	0.312
C ₁₂	Dodecanes	9.867	9.370	0.003	0.033	0.235
C ₁₃	Tridecanes	8.806	9.090	0.000	0.027	0.211
C ₁₄	Tetradecanes	7.834	8.779	0.000	0.024	0.202
C ₁₅	Pentadecanes	5.602	6.807	0.000	0.017	0.157
C ₁₆	Hexadecanes	4.424	5.793	0.000	0.014	0.133
C ₁₇	Heptadecanes	2.942	4.113	0.000	0.009	0.095
C ₁₈	Octadecanes	2.670	3.953	0.000	0.008	0.091
C ₁₉	Nonadecanes	1.906	2.957	0.000	0.006	0.068
C ₂₀	Eicosanes	1.434	2.326	0.000	0.004	0.054
C ₂₁	Heneicosanes	1.203	2.064	0.000	0.004	0.048
C ₂₂	Docosanes	0.951	1.711	0.000	0.003	0.039
C ₂₃	Tricosanes	0.758	1.421	0.000	0.002	0.033
C ₂₄	Tetracosanes	0.636	1.241	0.000	0.002	0.029
C ₂₅	Pentacosanes	0.511	1.039	0.000	0.002	0.024
C ₂₆	Hexacosanes	0.403	0.854	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.344	0.759	0.000	0.001	0.017
C ₂₈	Octacosanes	0.264	0.604	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.215	0.510	0.000	0.001	0.012
C ₃₀	Triacontanes	0.168	0.411	0.000	0.001	0.009
C ₃₁	Hentriacontanes	0.129	0.326	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.097	0.253	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.080	0.215	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.057	0.160	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.046	0.133	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.086	0.255	0.000	0.000	0.006
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.68, Cylinder 5142-EA to C36
Depth: 4837.4m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.260	0.598	0.891
Molecular Weight (g mol-1)	174	101.6	126
Density at 15.6°C (g cm-3)	0.8112	0.7665	0.7860
C₁₁₊			
Mole%	62.793	0.016	0.208
Molecular Weight (g mol-1)	203	150	198
Density at 15.6°C (g cm-3)	0.8301	0.7915	0.8278
C₂₀₊			
Mole%	7.382		0.022
Molecular Weight (g mol-1)	328		328
Density at 15.6°C (g cm-3)	0.8804		0.8804
C₃₆₊			
Mole %	0.086		0.000
Molecular Weight (g mol-1)	516		516
Density at 15.6°C (g cm-3)	0.9219		0.9219
Whole Sample Properties			
Average mole weight (g mol-1)	170 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8068 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7665	
GOR Ratio (scf/bbl) from flash at 21.6°C			204985.0
CGR Ratio (bbl/MMscf) from flash at 21.6°C			4.9
Gas Mole Fraction			0.9969
Liquid Mole Fraction			0.0031

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.68, Cylinder 5142-I**
Depth: 4837.4m MDRTPharos-1 S/F (1,1)
Acquired Wednesday, 22 October 2014 4:09:25 PM

wb19102014,as_2.10187,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.69, Cylinder 17242-IB to (
Depth: 4837.4m MDRT

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.856	13.812	26.881
N ₂	Nitrogen	0.000	0.000	0.308	0.307	0.380
C ₁	Methane	0.815	0.076	77.822	77.575	55.034
G.1	Ethane	0.326	0.057	4.869	4.854	6.454
C ₃	Propane	0.363	0.093	1.344	1.341	2.615
iC ₄	i-Butane	0.204	0.069	0.357	0.357	0.917
nC ₄	n-Butane	0.240	0.081	0.321	0.320	0.823
C ₅	neo-Pentane	0.024	0.010	0.004	0.004	0.014
iC ₅	i-Pentane	0.339	0.142	0.209	0.209	0.668
nC ₅	n-Pentane	0.241	0.101	0.115	0.116	0.369
C ₆	Hexanes	0.896	0.449	0.203	0.206	0.784
	Me-Cyclo-pentane	0.296	0.145	0.033	0.033	0.125
	Benzene	0.429	0.195	0.052	0.054	0.185
	Cyclo-hexane	0.278	0.136	0.059	0.059	0.221
C ₇	Heptanes	1.747	1.018	0.100	0.105	0.466
	Me-Cyclo-hexane	1.021	0.583	0.052	0.055	0.239
	Toluene	2.257	1.209	0.093	0.100	0.406
C ₈	Octanes	3.907	2.595	0.073	0.086	0.433
	Ethyl-benzene	0.269	0.166	0.005	0.006	0.029
	Meta/Para-xylene	4.079	2.518	0.039	0.052	0.244
	Ortho-xylene	0.966	0.596	0.006	0.009	0.042
C ₉	Nonanes	6.224	4.641	0.045	0.064	0.365
	Tri-Me-benzene	1.709	1.194	0.000	0.006	0.029
C ₁₀	Decanes	10.797	8.931	0.025	0.060	0.376
C ₁₁	Undecanes	11.232	9.599	0.009	0.045	0.293
C ₁₂	Dodecanes	9.567	8.955	0.001	0.032	0.225
C ₁₃	Tridecanes	8.461	8.609	0.000	0.027	0.212
C ₁₄	Tetradecanes	7.509	8.295	0.000	0.024	0.203
C ₁₅	Pentadecanes	5.295	6.342	0.000	0.017	0.155
C ₁₆	Hexadecanes	4.244	5.478	0.000	0.014	0.134
C ₁₇	Heptadecanes	2.808	3.869	0.000	0.009	0.095
C ₁₈	Octadecanes	2.588	3.776	0.000	0.008	0.093
C ₁₉	Nonadecanes	1.841	2.815	0.000	0.006	0.069
C ₂₀	Eicosanes	1.470	2.350	0.000	0.005	0.058
C ₂₁	Heneicosanes	1.409	2.383	0.000	0.005	0.058
C ₂₂	Docosanes	1.316	2.333	0.000	0.004	0.057
C ₂₃	Tricosanes	1.082	2.001	0.000	0.003	0.049
C ₂₄	Tetracosanes	0.836	1.609	0.000	0.003	0.039
C ₂₅	Pentacosanes	0.651	1.306	0.000	0.002	0.032
C ₂₆	Hexacosanes	0.531	1.108	0.000	0.002	0.027
C ₂₇	Heptacosanes	0.401	0.871	0.000	0.001	0.021
C ₂₈	Octacosanes	0.326	0.736	0.000	0.001	0.018
C ₂₉	Nonacosanes	0.252	0.590	0.000	0.001	0.014
C ₃₀	Triacontanes	0.192	0.465	0.000	0.001	0.011
C ₃₁	Hentriacontanes	0.146	0.366	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.108	0.278	0.000	0.000	0.007
C ₃₃	Trtriacontanes	0.089	0.238	0.000	0.000	0.006
C ₃₄	Tetratriacontanes	0.067	0.184	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.051	0.143	0.000	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.102	0.296	0.000	0.000	0.007
Totals :		100.001	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT

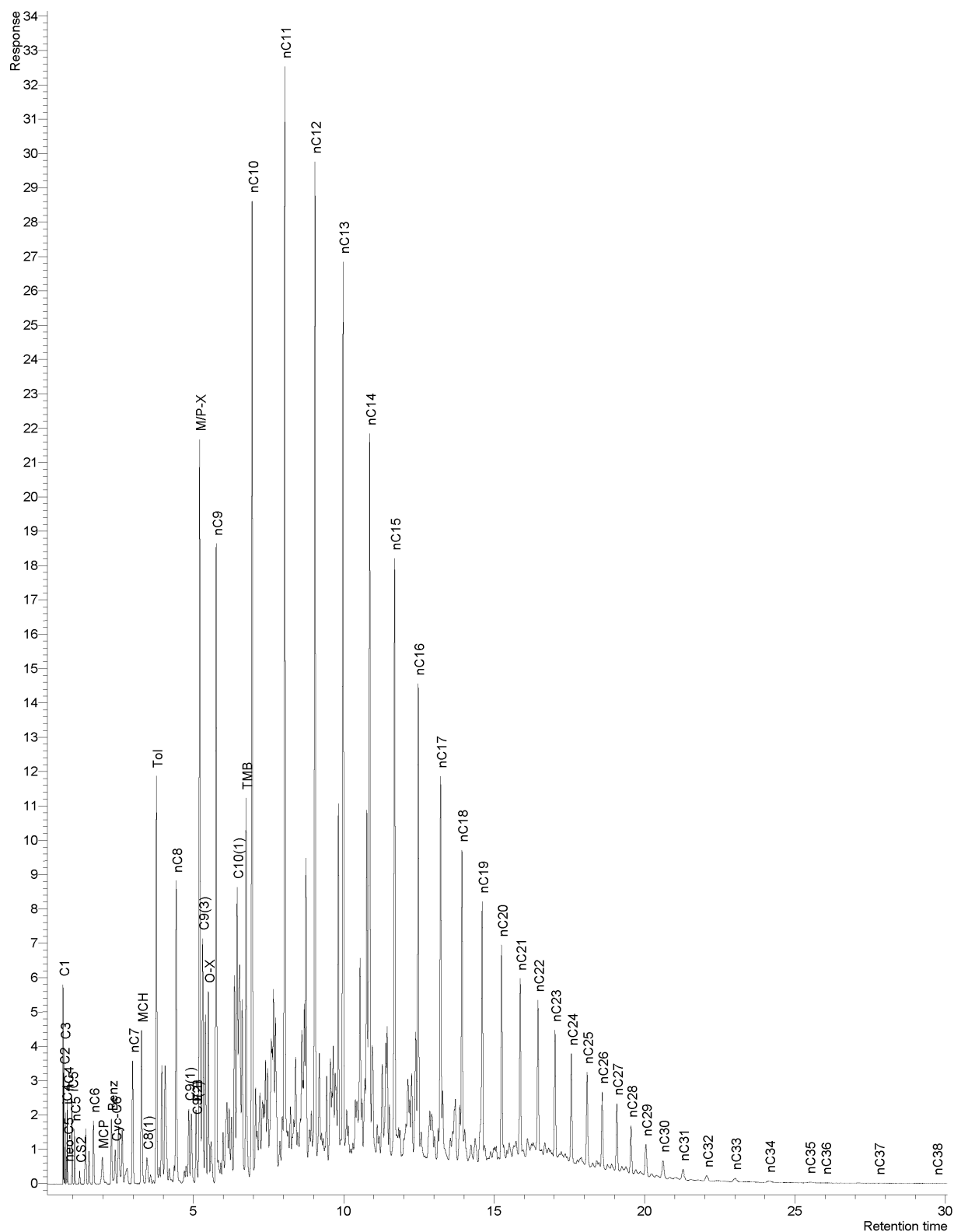
AFL 2014028

Compositional Analysis of SCAR Gas Sample No. 1.69, Cylinder 17242-IB to C3
Depth: 4837.4m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.553	0.592	0.899
Molecular Weight (g mol-1)	176	100.8	127
Density at 15.6°C (g cm-3)	0.8127	0.7736	0.7878
C₁₁₊			
Mole%	62.574	0.010	0.210
Molecular Weight (g mol-1)	206	148	203
Density at 15.6°C (g cm-3)	0.8323	0.7902	0.8309
C₂₀₊			
Mole%	9.029		0.028
Molecular Weight (g mol-1)	329		329
Density at 15.6°C (g cm-3)	0.8806		0.8806
C₃₆₊			
Mole %	0.102		0.000
Molecular Weight (g mol-1)	517		517
Density at 15.6°C (g cm-3)	0.9221		0.9221
Whole Sample Properties			
Average mole weight (g mol-1)	172 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8087 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7665	
GOR Ratio (scf/bbl) from flash at 19.9°C			192842.7
CGR Ratio (bbl/MMscf) from flash at 19.9°C			5.2
Gas Mole Fraction			0.9968
Liquid Mole Fraction			0.0032

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.69, Cylinder 17242**
Depth: 4837.4m MDRTPharos-1 S/F (1,1)
Acquired Thursday, 23 October 2014 2:25:51 PM

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Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.70, Cylinder 4990-EA to C
Depth: 4837.4m MDRT

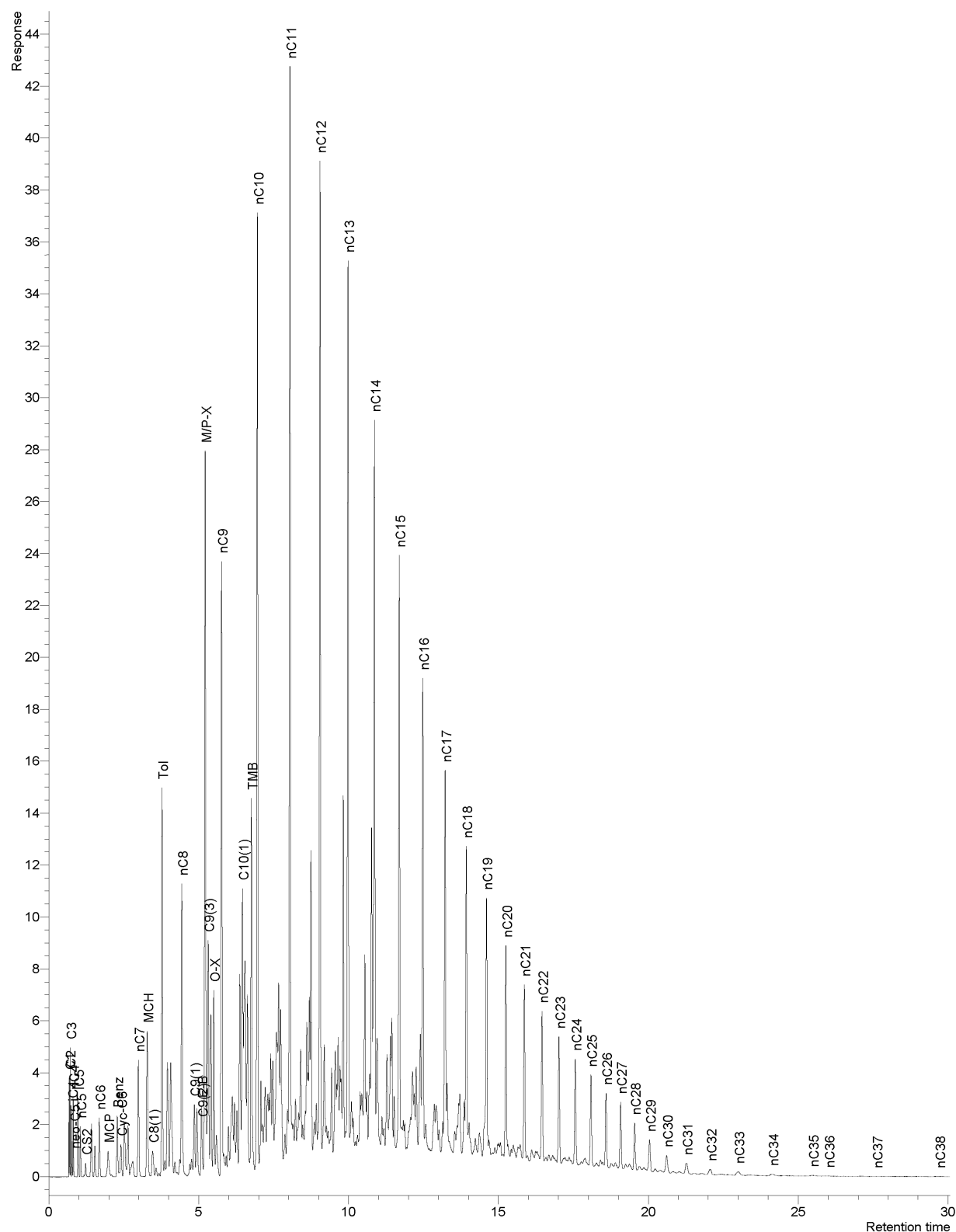
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.945	13.905	27.090
N ₂	Nitrogen	0.000	0.000	0.326	0.325	0.403
C ₁	Methane	0.475	0.045	77.730	77.511	55.043
G.1	Ethane	0.338	0.060	4.838	4.825	6.422
C ₃	Propane	0.395	0.103	1.333	1.331	2.598
iC ₄	i-Butane	0.221	0.076	0.354	0.354	0.910
nC ₄	n-Butane	0.247	0.085	0.322	0.321	0.827
C ₅	neo-Pentane	0.042	0.018	0.006	0.006	0.018
iC ₅	i-Pentane	0.364	0.155	0.213	0.213	0.681
nC ₅	n-Pentane	0.258	0.110	0.118	0.119	0.379
C ₆	Hexanes	0.927	0.472	0.209	0.211	0.807
	Me-Cyclo-pentane	0.308	0.153	0.034	0.035	0.130
	Benzene	0.444	0.205	0.056	0.057	0.196
	Cyclo-hexane	0.282	0.140	0.060	0.061	0.226
C ₇	Heptanes	1.775	1.051	0.100	0.105	0.467
	Me-Cyclo-hexane	1.029	0.597	0.053	0.056	0.244
	Toluene	2.294	1.249	0.097	0.103	0.421
C ₈	Octanes	3.966	2.677	0.075	0.086	0.433
	Ethyl-benzene	0.296	0.186	0.005	0.006	0.029
	Meta/Para-xylene	4.165	2.613	0.038	0.050	0.236
	Ortho-xylene	0.987	0.619	0.005	0.008	0.039
C ₉	Nonanes	6.341	4.806	0.045	0.063	0.355
	Tri-Me-benzene	1.546	1.098	0.000	0.004	0.023
C ₁₀	Decanes	11.170	9.391	0.026	0.058	0.364
C ₁₁	Undecanes	11.519	10.006	0.010	0.043	0.280
C ₁₂	Dodecanes	9.865	9.385	0.002	0.030	0.214
C ₁₃	Tridecanes	8.717	9.014	0.000	0.025	0.192
C ₁₄	Tetradecanes	7.690	8.634	0.000	0.022	0.184
C ₁₅	Pentadecanes	5.470	6.658	0.000	0.016	0.142
C ₁₆	Hexadecanes	4.359	5.718	0.000	0.012	0.122
C ₁₇	Heptadecanes	2.892	4.050	0.000	0.008	0.086
C ₁₈	Octadecanes	2.601	3.858	0.000	0.007	0.082
C ₁₉	Nonadecanes	1.873	2.911	0.000	0.005	0.062
C ₂₀	Eicosanes	1.407	2.286	0.000	0.004	0.049
C ₂₁	Heneicosanes	1.158	1.991	0.000	0.003	0.042
C ₂₂	Docosanes	0.916	1.650	0.000	0.003	0.035
C ₂₃	Tricosanes	0.724	1.361	0.000	0.002	0.029
C ₂₄	Tetracosanes	0.611	1.195	0.000	0.002	0.025
C ₂₅	Pentacosanes	0.492	1.002	0.000	0.001	0.021
C ₂₆	Hexacosanes	0.396	0.840	0.000	0.001	0.018
C ₂₇	Heptacosanes	0.330	0.730	0.000	0.001	0.016
C ₂₈	Octacosanes	0.249	0.572	0.000	0.001	0.012
C ₂₉	Nonacosanes	0.209	0.497	0.000	0.001	0.011
C ₃₀	Triacontanes	0.160	0.394	0.000	0.000	0.008
C ₃₁	Hentriacontanes	0.125	0.317	0.000	0.000	0.007
C ₃₂	Dotriacontanes	0.092	0.241	0.000	0.000	0.005
C ₃₃	Trtriacontanes	0.075	0.204	0.000	0.000	0.004
C ₃₄	Tetratriacontanes	0.060	0.167	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.045	0.129	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.095	0.281	0.000	0.000	0.006
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.70, Cylinder 4990-EA to C36
Depth: 4837.4m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.733	0.606	0.879
Molecular Weight (g mol-1)	173	100.8	124
Density at 15.6°C (g cm-3)	0.8100	0.7744	0.7852
C₁₁₊			
Mole%	62.130	0.012	0.187
Molecular Weight (g mol-1)	202	149	198
Density at 15.6°C (g cm-3)	0.8297	0.7909	0.8278
C₂₀₊			
Mole%	7.144		0.019
Molecular Weight (g mol-1)	328		328
Density at 15.6°C (g cm-3)	0.8805		0.8805
C₃₆₊			
Mole %	0.095		0.000
Molecular Weight (g mol-1)	518		518
Density at 15.6°C (g cm-3)	0.9223		0.9223
Whole Sample Properties			
Average mole weight (g mol-1)	169 (Calculated)	22.2	22.6
Density (g cm-3 at 15.6°C)	0.8062 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7679	
GOR Ratio (scf/bbl) from flash at 20.3°C			221220.1
CGR Ratio (bbl/MMscf) from flash at 20.3°C			4.5
Gas Mole Fraction			0.9971
Liquid Mole Fraction			0.0029

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.70, Cylinder 4990-I**
Depth: 4837.4m MDRTPharos-1 S/F (1,1)
Acquired Thursday, 23 October 2014 4:09:05 PM

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Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of SCAR Gas Sample No. 1.71, Cylinder 1473-EA to C
Depth: 4829.5m MDRT**

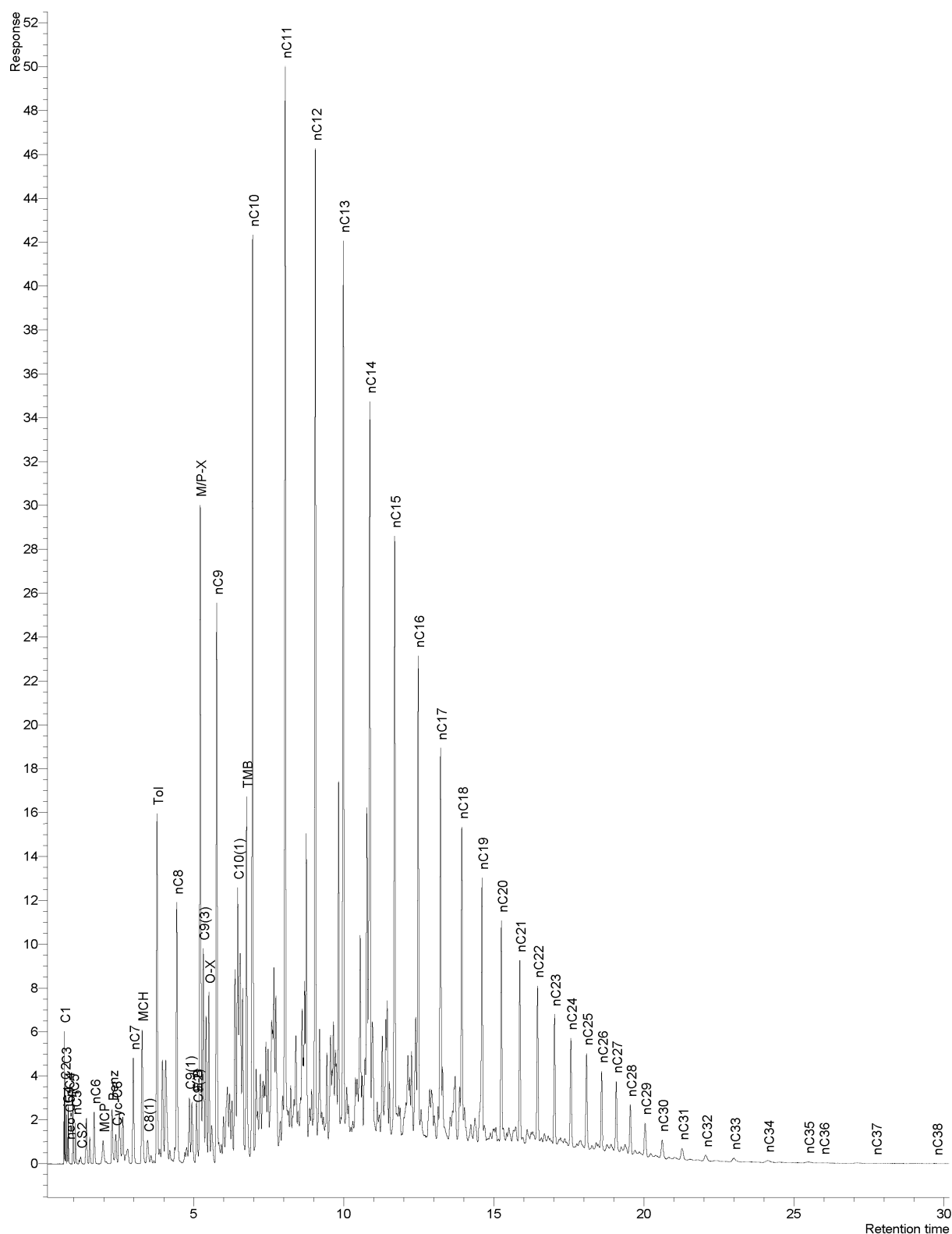
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.827	13.786	26.877
N ₂	Nitrogen	0.000	0.000	0.304	0.303	0.376
C ₁	Methane	0.602	0.056	77.844	77.619	55.163
G.1	Ethane	0.275	0.048	4.869	4.855	6.467
C ₃	Propane	0.301	0.077	1.348	1.345	2.628
iC ₄	i-Butane	0.169	0.057	0.356	0.356	0.915
nC ₄	n-Butane	0.184	0.062	0.320	0.320	0.823
C ₅	neo-Pentane	0.041	0.017	0.005	0.005	0.015
iC ₅	i-Pentane	0.299	0.125	0.208	0.208	0.666
nC ₅	n-Pentane	0.220	0.092	0.115	0.115	0.368
C ₆	Hexanes	0.843	0.421	0.202	0.204	0.777
	Me-Cyclo-pentane	0.289	0.141	0.033	0.034	0.125
	Benzene	0.426	0.193	0.052	0.053	0.184
	Cyclo-hexane	0.262	0.128	0.059	0.059	0.221
C ₇	Heptanes	1.700	0.987	0.101	0.105	0.467
	Me-Cyclo-hexane	0.993	0.565	0.052	0.055	0.237
	Toluene	2.212	1.181	0.091	0.097	0.395
C ₈	Octanes	3.798	2.514	0.074	0.085	0.431
	Ethyl-benzene	0.262	0.161	0.005	0.006	0.027
	Meta/Para-xylene	4.001	2.462	0.039	0.051	0.239
	Ortho-xylene	0.954	0.587	0.006	0.009	0.040
C ₉	Nonanes	6.014	4.470	0.045	0.063	0.356
	Tri-Me-benzene	1.763	1.228	0.000	0.005	0.027
C ₁₀	Decanes	10.652	8.783	0.028	0.059	0.369
C ₁₁	Undecanes	11.453	9.758	0.012	0.045	0.295
C ₁₂	Dodecanes	9.864	9.204	0.003	0.032	0.230
C ₁₃	Tridecanes	8.785	8.910	0.002	0.028	0.215
C ₁₄	Tetradecanes	7.773	8.559	0.000	0.023	0.192
C ₁₅	Pentadecanes	5.579	6.660	0.000	0.016	0.149
C ₁₆	Hexadecanes	4.377	5.631	0.000	0.013	0.126
C ₁₇	Heptadecanes	2.948	4.049	0.000	0.009	0.090
C ₁₈	Octadecanes	2.660	3.869	0.000	0.008	0.086
C ₁₉	Nonadecanes	1.906	2.905	0.000	0.006	0.065
C ₂₀	Eicosanes	1.468	2.339	0.000	0.004	0.052
C ₂₁	Heneicosanes	1.286	2.168	0.000	0.004	0.048
C ₂₂	Docosanes	1.092	1.930	0.000	0.003	0.043
C ₂₃	Tricosanes	0.897	1.654	0.000	0.003	0.037
C ₂₄	Tetracosanes	0.753	1.445	0.000	0.002	0.032
C ₂₅	Pentacosanes	0.592	1.184	0.000	0.002	0.026
C ₂₆	Hexacosanes	0.495	1.029	0.000	0.001	0.023
C ₂₇	Heptacosanes	0.423	0.917	0.000	0.001	0.020
C ₂₈	Octacosanes	0.317	0.712	0.000	0.001	0.016
C ₂₉	Nonacosanes	0.261	0.609	0.000	0.001	0.014
C ₃₀	Triacontanes	0.199	0.479	0.000	0.001	0.011
C ₃₁	Hentriacontanes	0.154	0.384	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.114	0.294	0.000	0.000	0.007
C ₃₃	Trtriacontanes	0.093	0.246	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.076	0.208	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.061	0.171	0.000	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.114	0.331	0.000	0.000	0.007
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.71, Cylinder 1473-EA to C36+
Depth: 4829.5m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	97.066	0.602	0.884
Molecular Weight (g mol-1)	176	101.8	126
Density at 15.6°C (g cm-3)	0.8127	0.7735	0.7863
C₁₁₊			
Mole%	63.740	0.017	0.203
Molecular Weight (g mol-1)	205	153	200
Density at 15.6°C (g cm-3)	0.8316	0.7945	0.8290
C₂₀₊			
Mole%	8.395		0.023
Molecular Weight (g mol-1)	331		331
Density at 15.6°C (g cm-3)	0.8813		0.8813
C₃₆₊			
Mole %	0.114		0.000
Molecular Weight (g mol-1)	516		516
Density at 15.6°C (g cm-3)	0.9220		0.9220
Whole Sample Properties			
Average mole weight (g mol-1)	173 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8093 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7667	
GOR Ratio (scf/bbl) from flash at 21.3°C			212401.3
CGR Ratio (bbl/MMscf) from flash at 21.3°C			4.7
Gas Mole Fraction			0.9971
Liquid Mole Fraction			0.0029

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.71, Cylinder 1473-I**
Depth: 4829.5m MDRTPharos-1 S/F (1,1)
Acquired Friday, 17 October 2014 5:51:29 PM

wb12102014,as_2.10155,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of SCAR Gas Sample No. 1.72, Cylinder 9083-MA to C
Depth: 4829.5m MDRT**

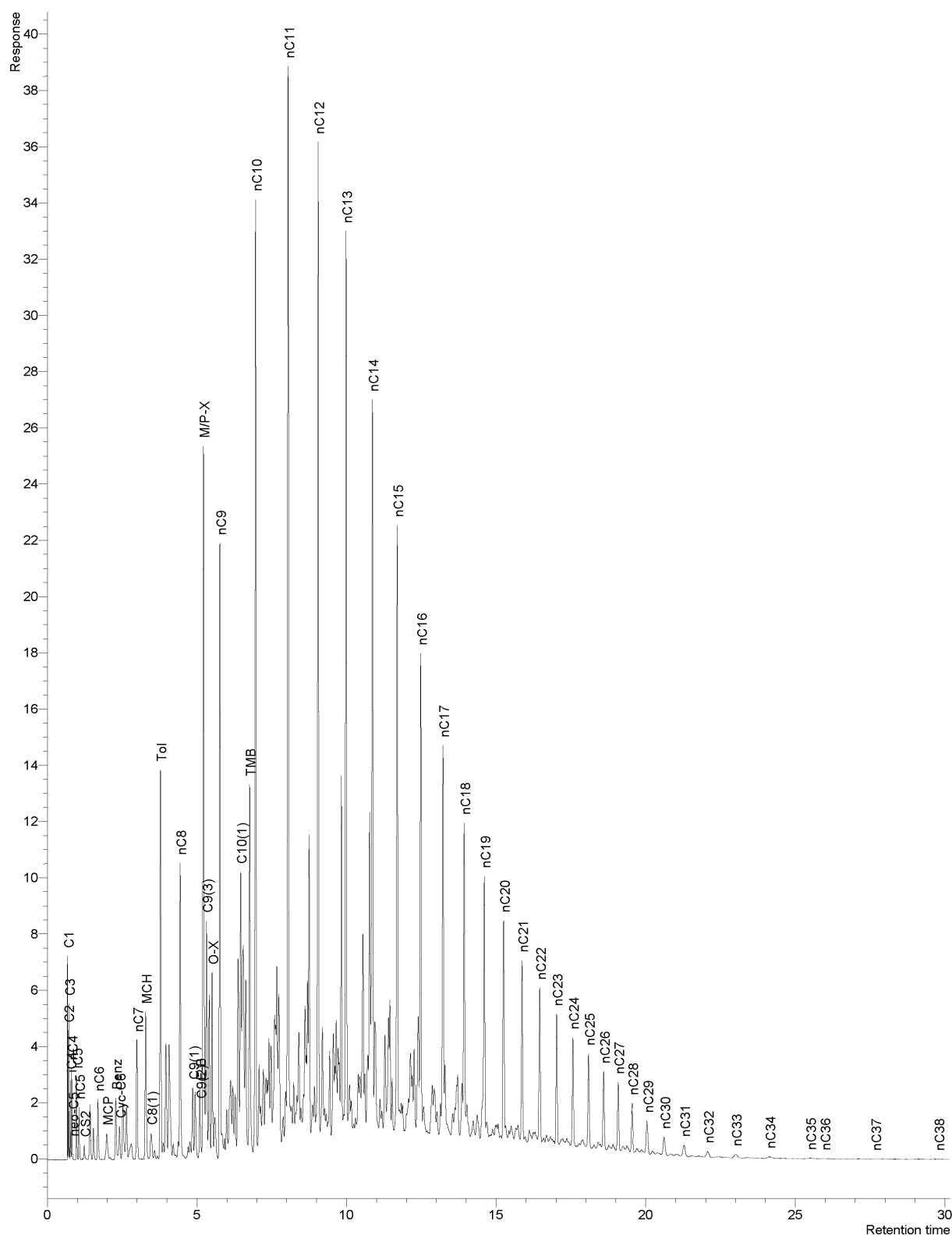
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.876	13.834	26.981
N ₂	Nitrogen	0.000	0.000	0.325	0.324	0.402
C ₁	Methane	0.854	0.081	77.800	77.570	55.149
G.1	Ethane	0.405	0.072	4.877	4.863	6.480
C ₃	Propane	0.437	0.114	1.343	1.340	2.619
iC ₄	i-Butane	0.233	0.080	0.356	0.356	0.917
nC ₄	n-Butane	0.265	0.091	0.319	0.319	0.821
C ₅	neo-Pentane	0.042	0.018	0.004	0.004	0.012
iC ₅	i-Pentane	0.368	0.157	0.205	0.205	0.656
nC ₅	n-Pentane	0.263	0.112	0.113	0.114	0.363
C ₆	Hexanes	0.937	0.477	0.200	0.202	0.773
	Me-Cyclo-pentane	0.312	0.155	0.032	0.033	0.123
	Benzene	0.453	0.209	0.052	0.053	0.184
	Cyclo-hexane	0.281	0.140	0.058	0.059	0.218
C ₇	Heptanes	1.812	1.073	0.099	0.105	0.464
	Me-Cyclo-hexane	1.037	0.602	0.051	0.054	0.235
	Toluene	2.294	1.249	0.093	0.100	0.406
C ₈	Octanes	3.973	2.682	0.069	0.081	0.410
	Ethyl-benzene	0.292	0.183	0.006	0.006	0.031
	Meta/Para-xylene	4.064	2.550	0.038	0.050	0.235
	Ortho-xylene	0.966	0.606	0.005	0.008	0.039
C ₉	Nonanes	6.224	4.718	0.045	0.063	0.359
	Tri-Me-benzene	1.758	1.249	0.000	0.005	0.028
C ₁₀	Decanes	10.797	9.079	0.025	0.057	0.359
C ₁₁	Undecanes	11.348	9.858	0.008	0.042	0.274
C ₁₂	Dodecanes	9.670	9.201	0.001	0.030	0.217
C ₁₃	Tridecanes	8.674	8.971	0.000	0.026	0.203
C ₁₄	Tetradecanes	7.590	8.523	0.000	0.023	0.192
C ₁₅	Pentadecanes	5.463	6.651	0.000	0.016	0.150
C ₁₆	Hexadecanes	4.306	5.650	0.000	0.013	0.127
C ₁₇	Heptadecanes	2.886	4.042	0.000	0.009	0.091
C ₁₈	Octadecanes	2.626	3.896	0.000	0.008	0.088
C ₁₉	Nonadecanes	1.883	2.927	0.000	0.006	0.066
C ₂₀	Eicosanes	1.416	2.301	0.000	0.004	0.052
C ₂₁	Heneicosanes	1.208	2.077	0.000	0.004	0.047
C ₂₂	Docosanes	0.970	1.748	0.000	0.003	0.039
C ₂₃	Tricosanes	0.766	1.439	0.000	0.002	0.032
C ₂₄	Tetracosanes	0.644	1.260	0.000	0.002	0.028
C ₂₅	Pentacosanes	0.512	1.043	0.000	0.002	0.023
C ₂₆	Hexacosanes	0.417	0.885	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.342	0.757	0.000	0.001	0.017
C ₂₈	Octacosanes	0.263	0.604	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.219	0.521	0.000	0.001	0.012
C ₃₀	Triacontanes	0.169	0.416	0.000	0.001	0.009
C ₃₁	Hentriacontanes	0.131	0.334	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.100	0.262	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.083	0.224	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.061	0.169	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.053	0.152	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.133	0.392	0.000	0.000	0.009
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Compositional Analysis of SCAR Gas Sample No. 1.72, Cylinder 9083-MA to C3**
Depth: 4829.5m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.196	0.582	0.869
Molecular Weight (g mol-1)	174	100.7	125
Density at 15.6°C (g cm-3)	0.8109	0.7740	0.7864
C₁₁₊			
Mole%	61.933	0.009	0.195
Molecular Weight (g mol-1)	203	149	200
Density at 15.6°C (g cm-3)	0.8305	0.7911	0.8290
C₂₀₊			
Mole%	7.487		0.022
Molecular Weight (g mol-1)	330		330
Density at 15.6°C (g cm-3)	0.8809		0.8809
C₃₆₊			
Mole %	0.133		0.000
Molecular Weight (g mol-1)	523		523
Density at 15.6°C (g cm-3)	0.9231		0.9231
Whole Sample Properties			
Average mole weight (g mol-1)	169 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8064 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7663	
GOR Ratio (scf/bbl) from flash at 20.9°C			209773.5
CGR Ratio (bbl/MMscf) from flash at 20.9°C			4.8
Gas Mole Fraction			0.9970
Liquid Mole Fraction			0.0030

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.72, Cylinder 9083-I**
Depth: 4829.5m MDRTPharos-1 S/F (1,1)
Acquired Monday, 20 October 2014 5:54:03 PM

wb19102014,as_2.10168,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.73, Cylinder 14809-QA to Depth: 4829.5m MDRT

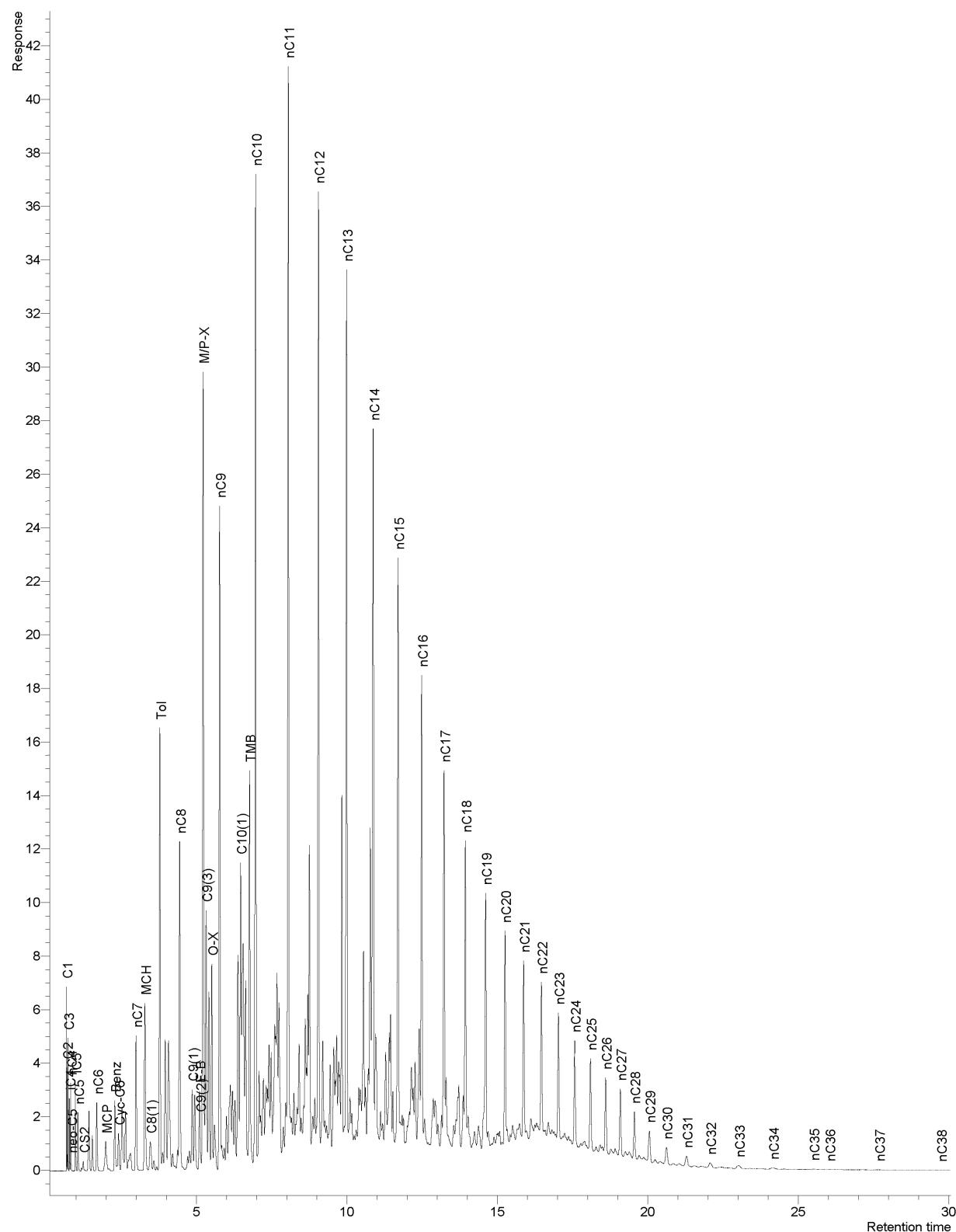
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.947	13.902	27.035
N ₂	Nitrogen	0.000	0.000	0.322	0.321	0.397
C ₁	Methane	0.781	0.073	77.761	77.513	54.948
G.1	Ethane	0.297	0.052	4.844	4.829	6.417
C ₃	Propane	0.343	0.088	1.335	1.332	2.595
iC ₄	i-Butane	0.198	0.067	0.354	0.353	0.908
nC ₄	n-Butane	0.242	0.082	0.317	0.316	0.813
C ₅	neo-Pentane	0.021	0.009	0.006	0.006	0.018
iC ₅	i-Pentane	0.345	0.145	0.211	0.212	0.676
nC ₅	n-Pentane	0.250	0.105	0.117	0.117	0.373
C ₆	Hexanes	0.956	0.480	0.204	0.206	0.786
	Me-Cyclo-pentane	0.322	0.158	0.033	0.034	0.125
	Benzene	0.468	0.213	0.053	0.054	0.188
	Cyclo-hexane	0.298	0.146	0.059	0.060	0.222
C ₇	Heptanes	1.904	1.111	0.100	0.106	0.467
	Me-Cyclo-hexane	1.100	0.629	0.052	0.055	0.240
	Toluene	2.433	1.306	0.087	0.095	0.386
C ₈	Octanes	4.207	2.799	0.075	0.089	0.447
	Ethyl-benzene	0.335	0.207	0.005	0.006	0.029
	Meta/Para-xylene	4.284	2.649	0.039	0.052	0.246
	Ortho-xylene	1.020	0.631	0.005	0.009	0.040
C ₉	Nonanes	6.491	4.849	0.043	0.064	0.364
	Tri-Me-benzene	1.808	1.266	0.000	0.006	0.031
C ₁₀	Decanes	10.842	8.985	0.022	0.057	0.359
C ₁₁	Undecanes	11.039	9.452	0.009	0.044	0.288
C ₁₂	Dodecanes	9.189	8.617	0.001	0.031	0.222
C ₁₃	Tridecanes	8.065	8.221	0.000	0.026	0.204
C ₁₄	Tetradecanes	7.136	7.897	0.000	0.023	0.195
C ₁₅	Pentadecanes	5.052	6.062	0.000	0.016	0.149
C ₁₆	Hexadecanes	4.013	5.189	0.000	0.013	0.128
C ₁₇	Heptadecanes	2.718	3.752	0.000	0.009	0.092
C ₁₈	Octadecanes	2.462	3.599	0.000	0.008	0.089
C ₁₉	Nonadecanes	1.802	2.760	0.000	0.006	0.068
C ₂₀	Eicosanes	1.453	2.328	0.000	0.005	0.057
C ₂₁	Heneicosanes	1.480	2.508	0.000	0.005	0.062
C ₂₂	Docosanes	1.428	2.537	0.000	0.005	0.063
C ₂₃	Tricosanes	1.195	2.213	0.000	0.004	0.055
C ₂₄	Tetracosanes	0.933	1.798	0.000	0.003	0.044
C ₂₅	Pentacosanes	0.699	1.404	0.000	0.002	0.035
C ₂₆	Hexacosanes	0.530	1.109	0.000	0.002	0.027
C ₂₇	Heptacosanes	0.441	0.960	0.000	0.001	0.024
C ₂₈	Octacosanes	0.331	0.747	0.000	0.001	0.018
C ₂₉	Nonacosanes	0.269	0.631	0.000	0.001	0.016
C ₃₀	Triacontanes	0.200	0.485	0.000	0.001	0.012
C ₃₁	Hentriacontanes	0.149	0.374	0.000	0.000	0.009
C ₃₂	Dotriacontanes	0.111	0.287	0.000	0.000	0.007
C ₃₃	Trtriacontanes	0.088	0.236	0.000	0.000	0.006
C ₃₄	Tetratriacontanes	0.069	0.189	0.000	0.000	0.005
C ₃₅	Pentatriacontanes	0.055	0.156	0.000	0.000	0.004
C ₃₆₊	Hexatriacontanes plus	0.151	0.439	0.000	0.000	0.011
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.73, Cylinder 14809-QA to C
Depth: 4829.5m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.567	0.583	0.893
Molecular Weight (g mol-1)	176	100.7	127
Density at 15.6°C (g cm-3)	0.8128	0.7727	0.7877
C₁₁₊			
Mole%	61.056	0.010	0.206
Molecular Weight (g mol-1)	208	149	205
Density at 15.6°C (g cm-3)	0.8334	0.7910	0.8319
C₂₀₊			
Mole%	9.582		0.030
Molecular Weight (g mol-1)	330		330
Density at 15.6°C (g cm-3)	0.8809		0.8809
C₃₆₊			
Mole %	0.151		0.000
Molecular Weight (g mol-1)	521		521
Density at 15.6°C (g cm-3)	0.9227		0.9227
Whole Sample Properties			
Average mole weight (g mol-1)	172 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8088 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7670	
GOR Ratio (scf/bbl) from flash at 19.3°C			191741.7
CGR Ratio (bbl/MMscf) from flash at 19.3°C			5.2
Gas Mole Fraction			0.9967
Liquid Mole Fraction			0.0033

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.73, Cylinder 14809**
Depth: 4829.5m MDRTPharos-1 S/F (1,1)
Acquired Monday, 27 October 2014 2:25:00 PM

wb26102014,as_2.10207,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.74, Cylinder 18974-QA to Depth: 4829.5m MDRT

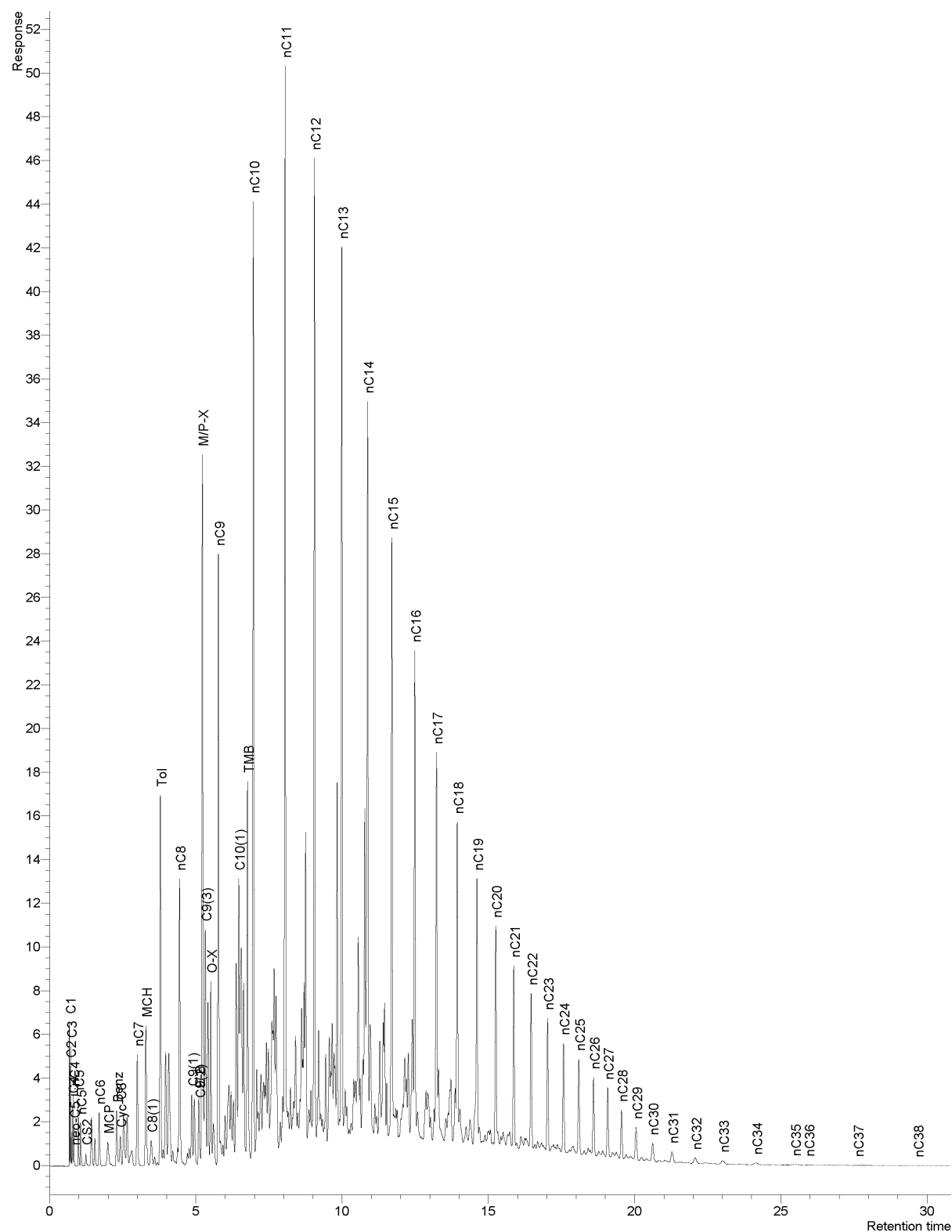
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.884	13.841	26.962
N ₂	Nitrogen	0.000	0.000	0.347	0.346	0.429
C ₁	Methane	0.636	0.060	77.735	77.501	55.032
G.1	Ethane	0.339	0.060	4.887	4.873	6.486
C ₃	Propane	0.366	0.095	1.356	1.353	2.640
iC ₄	i-Butane	0.193	0.066	0.360	0.359	0.924
nC ₄	n-Butane	0.225	0.077	0.323	0.323	0.830
C ₅	neo-Pentane	0.040	0.017	0.004	0.004	0.014
iC ₅	i-Pentane	0.325	0.138	0.207	0.208	0.664
nC ₅	n-Pentane	0.236	0.100	0.114	0.114	0.365
C ₆	Hexanes	0.880	0.446	0.201	0.203	0.776
	Me-Cyclo-pentane	0.297	0.147	0.032	0.033	0.124
	Benzene	0.442	0.203	0.051	0.052	0.180
	Cyclo-hexane	0.267	0.132	0.059	0.059	0.221
C ₇	Heptanes	1.760	1.037	0.099	0.104	0.461
	Me-Cyclo-hexane	1.020	0.589	0.052	0.055	0.238
	Toluene	2.266	1.228	0.096	0.102	0.417
C ₈	Octanes	3.958	2.659	0.063	0.075	0.379
	Ethyl-benzene	0.259	0.162	0.005	0.006	0.027
	Meta/Para-xylene	4.092	2.555	0.036	0.048	0.226
	Ortho-xylene	0.979	0.611	0.006	0.009	0.042
C ₉	Nonanes	6.233	4.701	0.042	0.061	0.346
	Tri-Me-benzene	1.677	1.185	0.000	0.005	0.027
C ₁₀	Decanes	10.964	9.174	0.025	0.059	0.370
C ₁₁	Undecanes	11.422	9.874	0.012	0.047	0.304
C ₁₂	Dodecanes	9.852	9.328	0.004	0.034	0.243
C ₁₃	Tridecanes	8.735	8.989	0.000	0.027	0.207
C ₁₄	Tetradecanes	7.707	8.611	0.000	0.024	0.198
C ₁₅	Pentadecanes	5.541	6.713	0.000	0.017	0.155
C ₁₆	Hexadecanes	4.351	5.680	0.000	0.013	0.131
C ₁₇	Heptadecanes	2.922	4.073	0.000	0.009	0.094
C ₁₈	Octadecanes	2.664	3.932	0.000	0.008	0.090
C ₁₉	Nonadecanes	1.899	2.937	0.000	0.006	0.068
C ₂₀	Eicosanes	1.431	2.315	0.000	0.004	0.053
C ₂₁	Heneicosanes	1.203	2.059	0.000	0.004	0.047
C ₂₂	Docosanes	0.957	1.716	0.000	0.003	0.039
C ₂₃	Tricosanes	0.774	1.447	0.000	0.002	0.033
C ₂₄	Tetracosanes	0.636	1.238	0.000	0.002	0.028
C ₂₅	Pentacosanes	0.513	1.040	0.000	0.002	0.024
C ₂₆	Hexacosanes	0.410	0.865	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.341	0.751	0.000	0.001	0.017
C ₂₈	Octacosanes	0.267	0.610	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.216	0.510	0.000	0.001	0.012
C ₃₀	Triacontanes	0.166	0.407	0.000	0.001	0.009
C ₃₁	Hentriacontanes	0.130	0.328	0.000	0.000	0.008
C ₃₂	Dotriacontanes	0.097	0.253	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.080	0.215	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.062	0.171	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.050	0.143	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.120	0.353	0.000	0.000	0.008
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of SCAR Gas Sample No. 1.74, Cylinder 18974-QA to C
Depth: 4829.5m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.760	0.582	0.875
Molecular Weight (g mol-1)	174	101	126
Density at 15.6°C (g cm-3)	0.8108	0.7756	0.7874
C₁₁₊			
Mole%	62.546	0.016	0.207
Molecular Weight (g mol-1)	203	151	199
Density at 15.6°C (g cm-3)	0.8303	0.7919	0.8280
C₂₀₊			
Mole%	7.453		0.022
Molecular Weight (g mol-1)	329		329
Density at 15.6°C (g cm-3)	0.8807		0.8807
C₃₆₊			
Mole %	0.120		0.000
Molecular Weight (g mol-1)	522		522
Density at 15.6°C (g cm-3)	0.9229		0.9229
Whole Sample Properties			
Average mole weight (g mol-1)	170 (Calculated)	22.1	22.6
Density (g cm-3 at 15.6°C)	0.8070 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7668	
GOR Ratio (scf/bbl) from flash at 21.0°C			205145.1
CGR Ratio (bbl/MMscf) from flash at 21.0°C			4.9
Gas Mole Fraction			0.9969
Liquid Mole Fraction			0.0031

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.74, Cylinder 18974**
Depth: 4829.5m MDRTPharos-1 S/F (1,1)
Acquired Monday, 20 October 2014 7:37:14 PM

wb19102014,as_2.10173,1,1,1



Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Slickline Gas Sample No. 1.77, Cylinder 18969-QA
Depth: 4986.5m MDRT

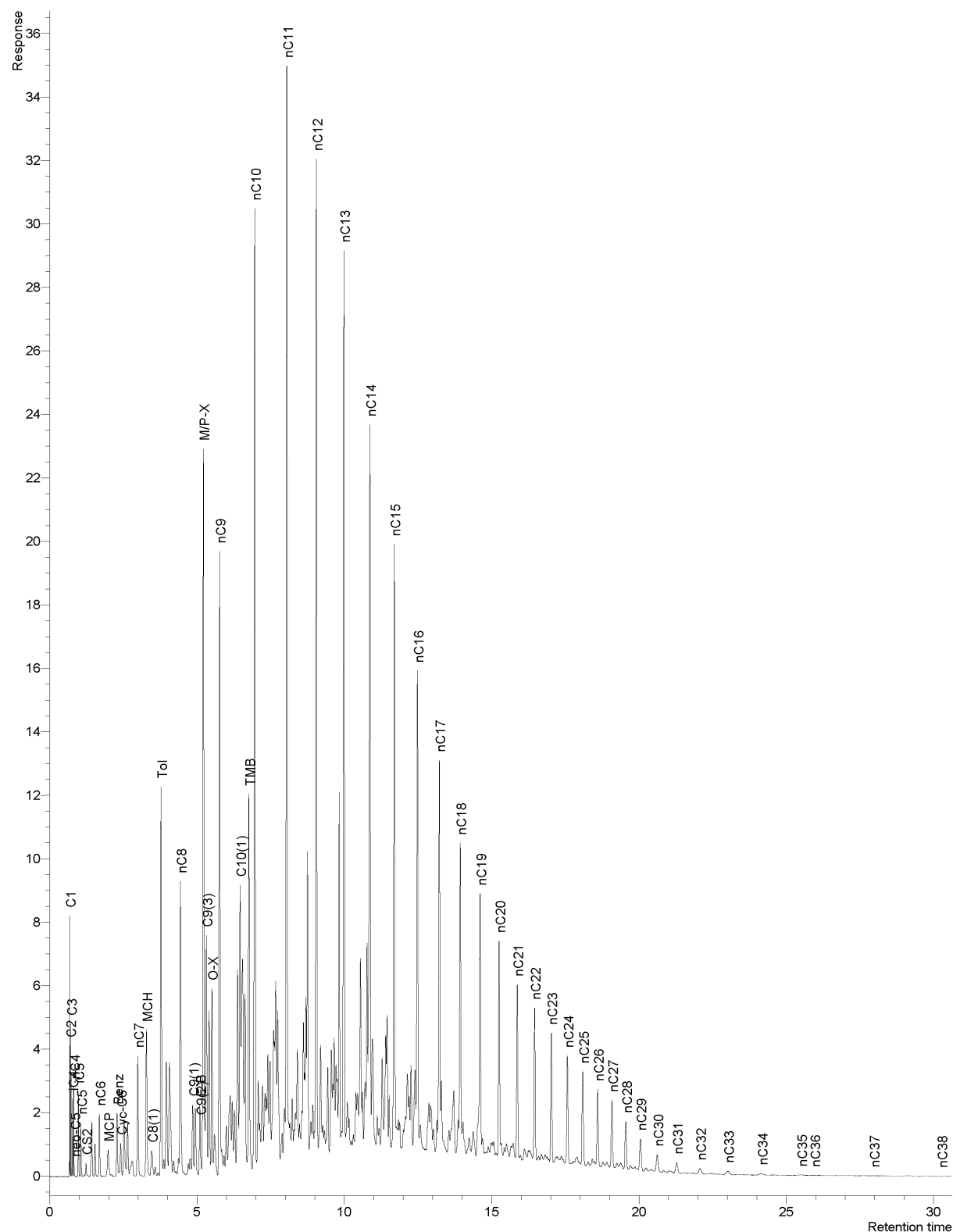
Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.941	13.898	27.041
N ₂	Nitrogen	0.000	0.000	0.309	0.308	0.381
C ₁	Methane	1.123	0.107	77.730	77.497	54.964
G.1	Ethane	0.381	0.068	4.866	4.852	6.450
C ₃	Propane	0.397	0.104	1.344	1.341	2.614
iC ₄	i-Butane	0.217	0.075	0.355	0.355	0.912
nC ₄	n-Butane	0.255	0.088	0.320	0.320	0.823
C ₅	neo-Pentane	0.023	0.010	0.003	0.003	0.011
iC ₅	i-Pentane	0.341	0.146	0.205	0.205	0.654
nC ₅	n-Pentane	0.243	0.104	0.113	0.114	0.363
C ₆	Hexanes	0.891	0.456	0.203	0.205	0.781
	Me-Cyclo-pentane	0.302	0.151	0.033	0.034	0.127
	Benzene	0.425	0.197	0.051	0.052	0.178
	Cyclo-hexane	0.280	0.140	0.060	0.060	0.225
C ₇	Heptanes	1.761	1.048	0.101	0.106	0.470
	Me-Cyclo-hexane	1.015	0.592	0.054	0.057	0.248
	Toluene	2.255	1.234	0.088	0.094	0.385
C ₈	Octanes	3.985	2.703	0.076	0.088	0.446
	Ethyl-benzene	0.290	0.183	0.005	0.006	0.029
	Meta/Para-xylene	4.116	2.595	0.044	0.057	0.267
	Ortho-xylene	0.985	0.621	0.006	0.009	0.042
C ₉	Nonanes	6.323	4.816	0.044	0.063	0.359
	Tri-Me-benzene	1.739	1.241	0.000	0.005	0.028
C ₁₀	Decanes	11.063	9.346	0.031	0.065	0.409
C ₁₁	Undecanes	11.522	10.058	0.013	0.048	0.312
C ₁₂	Dodecanes	9.831	9.399	0.005	0.034	0.245
C ₁₃	Tridecanes	8.712	9.054	0.000	0.027	0.205
C ₁₄	Tetradecanes	7.214	8.139	0.000	0.022	0.184
C ₁₅	Pentadecanes	5.472	6.694	0.000	0.017	0.152
C ₁₆	Hexadecanes	4.204	5.542	0.000	0.013	0.126
C ₁₇	Heptadecanes	2.902	4.084	0.000	0.009	0.093
C ₁₈	Octadecanes	2.556	3.809	0.000	0.008	0.086
C ₁₉	Nonadecanes	1.883	2.941	0.000	0.006	0.067
C ₂₀	Eicosanes	1.415	2.310	0.000	0.004	0.052
C ₂₁	Heneicosanes	1.182	2.042	0.000	0.004	0.046
C ₂₂	Docosanes	0.925	1.676	0.000	0.003	0.038
C ₂₃	Tricosanes	0.734	1.387	0.000	0.002	0.031
C ₂₄	Tetracosanes	0.624	1.227	0.000	0.002	0.028
C ₂₅	Pentacosanes	0.501	1.026	0.000	0.002	0.023
C ₂₆	Hexacosanes	0.409	0.872	0.000	0.001	0.020
C ₂₇	Heptacosanes	0.335	0.745	0.000	0.001	0.017
C ₂₈	Octacosanes	0.259	0.596	0.000	0.001	0.014
C ₂₉	Nonacosanes	0.214	0.510	0.000	0.001	0.012
C ₃₀	Triacontanes	0.168	0.415	0.000	0.001	0.009
C ₃₁	Hentriacontanes	0.128	0.326	0.000	0.000	0.007
C ₃₂	Dotriacontanes	0.098	0.259	0.000	0.000	0.006
C ₃₃	Trtriacontanes	0.079	0.216	0.000	0.000	0.005
C ₃₄	Tetratriacontanes	0.059	0.166	0.000	0.000	0.004
C ₃₅	Pentatriacontanes	0.051	0.146	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.113	0.336	0.000	0.000	0.008
Totals :		100.000	100.000	100.000	100.000	100.000

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Slickline Gas Sample No. 1.77, Cylinder 18969-QA t
Depth: 4986.5m MDRT

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.129	0.611	0.902
Molecular Weight (g mol-1)	173	102.0	125
Density at 15.6°C (g cm-3)	0.8103	0.7734	0.7850
C₁₁₊			
Mole%	61.590	0.018	0.206
Molecular Weight (g mol-1)	202	151	198
Density at 15.6°C (g cm-3)	0.8300	0.7921	0.8275
C₂₀₊			
Mole%	7.294		0.022
Molecular Weight (g mol-1)	329		329
Density at 15.6°C (g cm-3)	0.8807		0.8807
C₃₆₊			
Mole %	0.113		0.000
Molecular Weight (g mol-1)	519		519
Density at 15.6°C (g cm-3)	0.9224		0.9224
Whole Sample Properties			
Average mole weight (g mol-1)	168 (Calculated)	22.2	22.6
Density (g cm-3 at 15.6°C)	0.8056 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7681	
GOR Ratio (scf/bbl) from flash at 19.8°C			207740.0
CGR Ratio (bbl/MMscf) from flash at 19.8°C			4.8
Gas Mole Fraction			0.9969
Liquid Mole Fraction			0.0031

Conoco Phillips Australia
Pharos-1 MDT**AFL 2014028****Fingerprint Profile of Flashed Condensate, Sample No .1.77, Cylinder 18969**
Depth: 4986.5m MDRTPharos-1 S/F (1,1)
Acquired Monday, 20 October 2014 2:28:17 PM

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Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Compositional Analysis of Slickline Gas Sample No. 1.78, Cylinder 28960-IB
Depth: 4986.5m MDRT

Component		Flashed Liquid		Flashed Gas	Reservoir Fluid	
		Mole %	Weight %	Mole %	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000	0.000	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000	0.000	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000	13.966	13.925	27.119
N ₂	Nitrogen	0.000	0.000	0.314	0.313	0.388
C ₁	Methane	1.017	0.097	77.703	77.477	55.002
G.1	Ethane	0.425	0.076	4.870	4.857	6.462
C ₃	Propane	0.420	0.110	1.347	1.344	2.623
iC ₄	i-Butane	0.226	0.078	0.355	0.354	0.911
nC ₄	n-Butane	0.249	0.086	0.318	0.318	0.817
C ₅	neo-Pentane	0.035	0.015	0.005	0.005	0.017
iC ₅	i-Pentane	0.352	0.151	0.208	0.209	0.666
nC ₅	n-Pentane	0.247	0.106	0.115	0.115	0.368
C ₆	Hexanes	0.912	0.467	0.203	0.205	0.781
	Me-Cyclo-pentane	0.302	0.151	0.033	0.033	0.125
	Benzene	0.433	0.201	0.053	0.054	0.186
	Cyclo-hexane	0.286	0.143	0.059	0.060	0.222
C ₇	Heptanes	1.780	1.060	0.101	0.106	0.471
	Me-Cyclo-hexane	1.030	0.601	0.052	0.055	0.238
	Toluene	2.284	1.251	0.093	0.099	0.404
C ₈	Octanes	3.945	2.679	0.073	0.084	0.425
	Ethyl-benzene	0.311	0.196	0.006	0.007	0.031
	Meta/Para-xylene	4.090	2.581	0.040	0.052	0.243
	Ortho-xylene	0.978	0.617	0.006	0.009	0.043
C ₉	Nonanes	6.229	4.749	0.043	0.062	0.351
	Tri-Me-benzene	1.748	1.249	0.000	0.005	0.028
C ₁₀	Decanes	10.949	9.261	0.025	0.058	0.363
C ₁₁	Undecanes	11.482	10.034	0.010	0.044	0.285
C ₁₂	Dodecanes	9.890	9.466	0.002	0.031	0.223
C ₁₃	Tridecanes	8.753	9.106	0.000	0.026	0.203
C ₁₄	Tetradecanes	7.382	8.337	0.000	0.022	0.185
C ₁₅	Pentadecanes	5.498	6.732	0.000	0.016	0.149
C ₁₆	Hexadecanes	4.252	5.611	0.000	0.013	0.124
C ₁₇	Heptadecanes	2.890	4.072	0.000	0.009	0.090
C ₁₈	Octadecanes	2.572	3.837	0.000	0.008	0.085
C ₁₉	Nonadecanes	1.888	2.952	0.000	0.006	0.065
C ₂₀	Eicosanes	1.408	2.301	0.000	0.004	0.051
C ₂₁	Heneicosanes	1.175	2.033	0.000	0.003	0.045
C ₂₂	Docosanes	0.908	1.647	0.000	0.003	0.036
C ₂₃	Tricosanes	0.722	1.365	0.000	0.002	0.030
C ₂₄	Tetracosanes	0.613	1.206	0.000	0.002	0.027
C ₂₅	Pentacosanes	0.492	1.010	0.000	0.001	0.022
C ₂₆	Hexacosanes	0.398	0.849	0.000	0.001	0.019
C ₂₇	Heptacosanes	0.325	0.723	0.000	0.001	0.016
C ₂₈	Octacosanes	0.260	0.599	0.000	0.001	0.013
C ₂₉	Nonacosanes	0.207	0.495	0.000	0.001	0.011
C ₃₀	Triacontanes	0.159	0.394	0.000	0.000	0.009
C ₃₁	Hentriacontanes	0.125	0.319	0.000	0.000	0.007
C ₃₂	Dotriacontanes	0.091	0.241	0.000	0.000	0.005
C ₃₃	Trtriacontanes	0.073	0.198	0.000	0.000	0.004
C ₃₄	Tetratriacontanes	0.054	0.151	0.000	0.000	0.003
C ₃₅	Pentatriacontanes	0.043	0.124	0.000	0.000	0.003
C ₃₆₊	Hexatriacontanes plus	0.092	0.273	0.000	0.000	0.006
Totals :		100.000	100.000	100.000	100.000	100.000

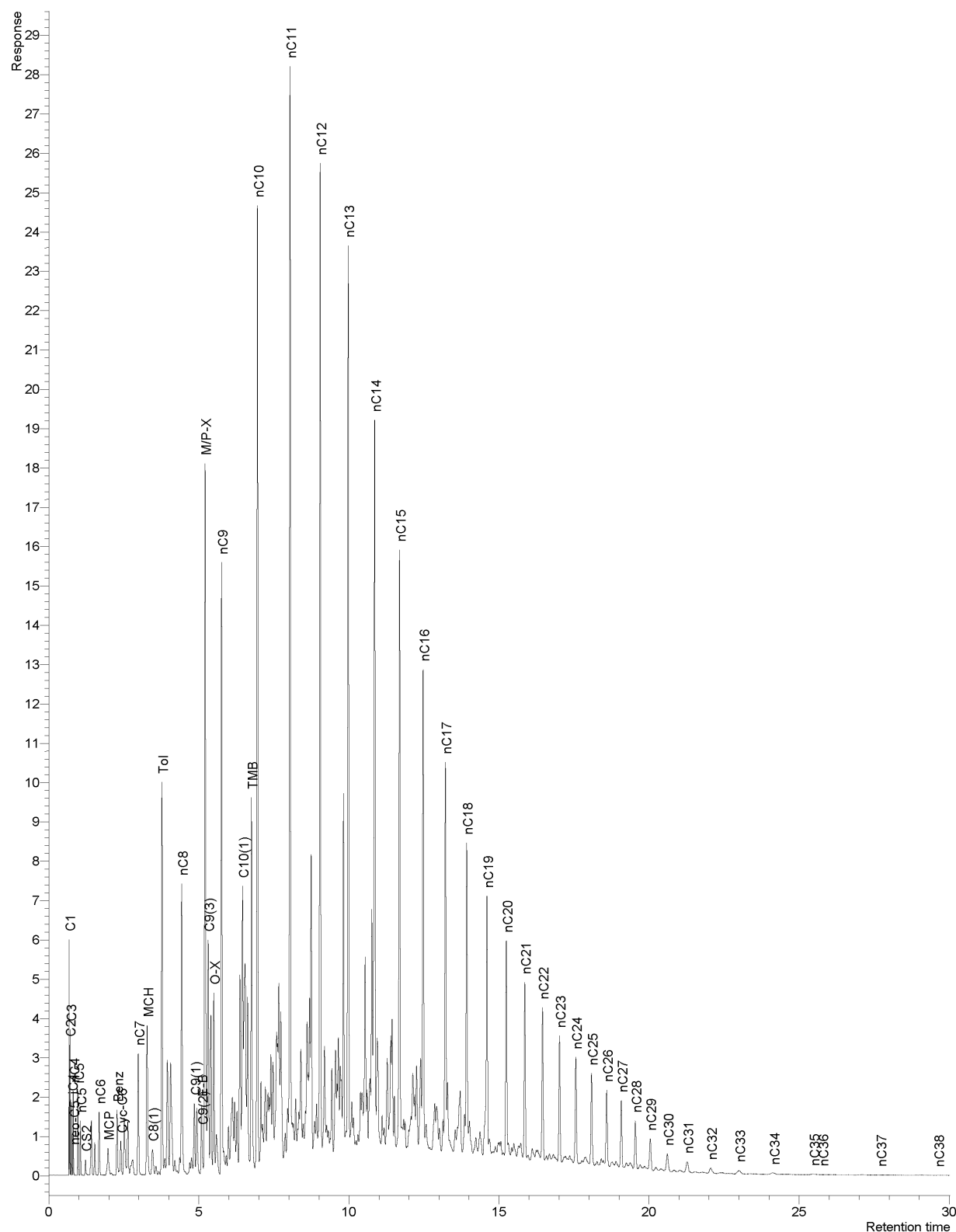
Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Compositional Analysis of Slickline Gas Sample No. 1.78, Cylinder 28960-IB to
 Depth: 4986.5m MDRT**

Calculated Properties	Flashed Liquid	Flashed Gas	Reservoir Fluid
C₇₊			
Mole%	96.117	0.596	0.878
Molecular Weight (g mol-1)	173	101.4	125
Density at 15.6°C (g cm-3)	0.8102		0.7857
C₁₁₊			
Mole%	61.752	0.012	0.194
Molecular Weight (g mol-1)	202	150	199
Density at 15.6°C (g cm-3)	0.8297		0.8279
C₂₀₊			
Mole%	7.145		0.019
Molecular Weight (g mol-1)	328		328
Density at 15.6°C (g cm-3)	0.8803		0.8803
C₃₆₊			
Mole %	0.092		0.000
Molecular Weight (g mol-1)	516		516
Density at 15.6°C (g cm-3)	0.9220		0.9220
Whole Sample Properties			
Average mole weight (g mol-1)	168 (Calculated)	22.2	22.6
Density (g cm-3 at 15.6°C)	0.8055 (Calculated)		
Real relative density (Air = 1.000 at 14.70 psia 15.6°C)		0.7677	
GOR Ratio (scf/bbl) from flash at 20.4°C			212834.9
CGR Ratio (bbl/MMscf) from flash at 20.4°C			4.7
Gas Mole Fraction			0.9970
Liquid Mole Fraction			0.0030

Fingerprint Profile of Flashed Condensate, Sample No .1.78, Cylinder 28960
Depth: 4986.5m MDRT

Pharos-1 S/F (1,1)
Acquired Wednesday, 22 October 2014 5:53:03 PM

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Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Section H - Calculated Uncontaminated Compositions - SCAR and Slickline ga

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.65, 14952-IB to C36+
Depth: 4837.4m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.430	0.041
C ₂	Ethane	0.333	0.059
C ₃	Propane	0.391	0.102
iC ₄	i-Butane	0.223	0.077
nC ₄	n-Butane	0.255	0.088
C ₅	neo-Pentane	0.036	0.015
iC ₅	i-Pentane	0.368	0.158
nC ₅	n-Pentane	0.258	0.111
C ₆	Hexanes	0.943	0.482
	Me-Cyclo-pentane	0.312	0.156
	Benzene	0.460	0.213
	Cyclo-hexane	0.285	0.142
C ₇	Heptanes	1.836	1.091
	Me-Cyclo-hexane	1.063	0.619
	Toluene	2.353	1.285
C ₈	Octanes	4.067	2.755
	Ethyl-benzene	0.289	0.182
	Meta/Para-xylene	4.263	2.684
	Ortho-xylene	1.009	0.635
C ₉	Nonanes	6.469	4.919
	Tri-Me-benzene	1.859	1.325
C ₁₀	Decanes	11.172	9.425
C ₁₁	Undecanes	11.725	10.217
C ₁₂	Dodecanes	9.969	9.517
C ₁₃	Tridecanes	8.815	9.147
C ₁₄	Tetradecanes	6.646	7.487
C ₁₅	Pentadecanes	5.537	6.763
C ₁₆	Hexadecanes	4.011	5.280
C ₁₇	Heptadecanes	2.918	4.101
C ₁₈	Octadecanes	2.461	3.662
C ₁₉	Nonadecanes	1.885	2.940
C ₂₀	Eicosanes	1.427	2.327
C ₂₁	Heneicosanes	1.205	2.079
C ₂₂	Docosanes	0.947	1.713
C ₂₃	Tricosanes	0.751	1.415
C ₂₄	Tetracosanes	0.644	1.264
C ₂₅	Pentacosanes	0.511	1.045
C ₂₆	Hexacosanes	0.406	0.864
C ₂₇	Heptacosanes	0.338	0.750
C ₂₈	Octacosanes	0.262	0.603
C ₂₉	Nonacosanes	0.216	0.516
C ₃₀	Triacotanes	0.165	0.407
C ₃₁	Hentriacotanes	0.126	0.322
C ₃₂	Dotriacotanes	0.094	0.248
C ₃₃	Tritriacotanes	0.077	0.210
C ₃₄	Tetratriacotanes	0.057	0.160
C ₃₅	Pentatriacotanes	0.044	0.127
C ₃₆₊	Hexatriacotanes plus	0.089	0.272
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia

AFL 2014028

Pharos-1 MDT

**Calculated Uncontaminated Composition of Sample 1.65, 14952-IB to C36+
Depth: 4837.4m MDRT**

Calculated Residue Properties		
C7+	Mole%	96.763
	Molecular Weight (g mol-1)	172
	Density at 15.6°C (g cm-3)	0.8099
C11+	Mole%	61.326
	Molecular Weight (g mol-1)	202
	Density at 15.6°C (g cm-3)	0.8299
C20+	Mole%	7.359
	Molecular Weight (g mol-1)	328
	Density at 15.6°C (g cm-3)	0.8804
C36+	Mole %	0.089
	Molecular Weight (g mol-1)	517
	Density at 15.6°C (g cm-3)	0.9221
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	168.7
	Estimated SBM whole fluid contamination (wt%)	<0.1
	Estimated SBM stock tank fluid contamination (wt%)	2

Note:

The Mole composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.66, 28985-IB to C36+
Depth: 4837.4m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.858	0.082
C ₂	Ethane	0.332	0.059
C ₃	Propane	0.355	0.093
iC ₄	i-Butane	0.198	0.069
nC ₄	n-Butane	0.243	0.084
C ₅	neo-Pentane	0.019	0.008
iC ₅	i-Pentane	0.336	0.144
nC ₅	n-Pentane	0.243	0.104
C ₆	Hexanes	0.921	0.472
	Me-Cyclo-pentane	0.311	0.156
	Benzene	0.454	0.211
	Cyclo-hexane	0.288	0.144
C ₇	Heptanes	1.826	1.088
	Me-Cyclo-hexane	1.062	0.620
	Toluene	2.361	1.294
C ₈	Octanes	4.091	2.779
	Ethyl-benzene	0.271	0.171
	Meta/Para-xylene	4.255	2.686
	Ortho-xylene	1.021	0.645
C ₉	Nonanes	6.460	4.928
	Tri-Me-benzene	1.673	1.196
C ₁₀	Decanes	11.324	9.582
C ₁₁	Undecanes	11.651	10.187
C ₁₂	Dodecanes	9.984	9.560
C ₁₃	Tridecanes	8.805	9.163
C ₁₄	Tetradecanes	6.564	7.417
C ₁₅	Pentadecanes	5.543	6.790
C ₁₆	Hexadecanes	3.960	5.228
C ₁₇	Heptadecanes	2.916	4.110
C ₁₈	Octadecanes	2.403	3.586
C ₁₉	Nonadecanes	1.901	2.973
C ₂₀	Eicosanes	1.415	2.314
C ₂₁	Heneicosanes	1.223	2.116
C ₂₂	Docosanes	0.950	1.724
C ₂₃	Tricosanes	0.759	1.436
C ₂₄	Tetracosanes	0.638	1.257
C ₂₅	Pentacosanes	0.517	1.061
C ₂₆	Hexacosanes	0.412	0.880
C ₂₇	Heptacosanes	0.338	0.751
C ₂₈	Octacosanes	0.263	0.606
C ₂₉	Nonacosanes	0.217	0.519
C ₃₀	Triacotanes	0.166	0.412
C ₃₁	Hentriacotanes	0.133	0.340
C ₃₂	Dotriacotanes	0.094	0.249
C ₃₃	Tristriacotanes	0.079	0.215
C ₃₄	Tettristriacotanes	0.058	0.164
C ₃₅	Pentatriacotanes	0.043	0.124
C ₃₆₊	Hexatriacotanes plus	0.066	0.203
	Totals :	100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia

AFL 2014028

Pharos-1 MDT

Calculated Uncontaminated Composition of Sample 1.66, 28985-IB to C36+
Depth: 4837.4m MDRT

Calculated Residue Properties		
C7+	Mole%	96.495
	Molecular Weight (g mol-1)	172
	Density at 15.6°C (g cm-3)	0.8096
C11+	Mole%	61.098
	Molecular Weight (g mol-1)	202
	Density at 15.6°C (g cm-3)	0.8298
C20+	Mole%	7.371
	Molecular Weight (g mol-1)	328
	Density at 15.6°C (g cm-3)	0.8802
C36+	Mole %	0.066
	Molecular Weight (g mol-1)	513
	Density at 15.6°C (g cm-3)	0.9214
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		168.1
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		2

Note:

The Mole composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.67, 5059-EA to C36+
Depth: 4837.4m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.830	0.078
C ₂	Ethane	0.310	0.055
C ₃	Propane	0.329	0.085
iC ₄	i-Butane	0.181	0.062
nC ₄	n-Butane	0.205	0.070
C ₅	neo-Pentane	0.034	0.014
iC ₅	i-Pentane	0.309	0.131
nC ₅	n-Pentane	0.225	0.096
C ₆	Hexanes	0.864	0.438
	Me-Cyclo-pentane	0.288	0.142
	Benzene	0.425	0.195
	Cyclo-hexane	0.263	0.130
C ₇	Heptanes	1.728	1.018
	Me-Cyclo-hexane	1.007	0.581
	Toluene	2.225	1.205
C ₈	Octanes	3.910	2.626
	Ethyl-benzene	0.322	0.201
	Meta/Para-xylene	4.066	2.538
	Ortho-xylene	0.970	0.606
C ₉	Nonanes	6.258	4.719
	Tri-Me-benzene	1.811	1.280
C ₁₀	Decanes	10.966	9.174
C ₁₁	Undecanes	11.668	10.087
C ₁₂	Dodecanes	9.946	9.416
C ₁₃	Tridecanes	8.860	9.116
C ₁₄	Tetradecanes	7.023	7.845
C ₁₅	Pentadecanes	5.611	6.796
C ₁₆	Hexadecanes	4.160	5.431
C ₁₇	Heptadecanes	2.960	4.124
C ₁₈	Octadecanes	2.538	3.746
C ₁₉	Nonadecanes	1.926	2.978
C ₂₀	Eicosanes	1.458	2.357
C ₂₁	Heneicosanes	1.260	2.156
C ₂₂	Docosanes	1.040	1.866
C ₂₃	Tricosanes	0.814	1.522
C ₂₄	Tetracosanes	0.683	1.329
C ₂₅	Pentacosanes	0.539	1.093
C ₂₆	Hexacosanes	0.431	0.909
C ₂₇	Heptacosanes	0.369	0.811
C ₂₈	Octacosanes	0.281	0.641
C ₂₉	Nonacosanes	0.231	0.545
C ₃₀	Triacotanes	0.173	0.423
C ₃₁	Hentriacotanes	0.135	0.342
C ₃₂	Dotriacotanes	0.097	0.254
C ₃₃	Tritriacotanes	0.082	0.220
C ₃₄	Tetratriacotanes	0.062	0.171
C ₃₅	Pentatriacotanes	0.044	0.127
C ₃₆₊	Hexatriacotanes plus	0.083	0.251
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia

AFL 2014028

Pharos-1 MDT

Calculated Uncontaminated Composition of Sample 1.67, 5059-EA to C36+
Depth: 4837.4m MDRT

Calculated Residue Properties		
C7+	Mole%	96.713
	Molecular Weight (g mol-1)	174
	Density at 15.6°C (g cm-3)	0.8110
C11+	Mole%	62.474
	Molecular Weight (g mol-1)	203
	Density at 15.6°C (g cm-3)	0.8305
C20+	Mole%	7.782
	Molecular Weight (g mol-1)	328
	Density at 15.6°C (g cm-3)	0.8804
C36+	Mole %	0.083
	Molecular Weight (g mol-1)	516
	Density at 15.6°C (g cm-3)	0.9220
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	170.1
	Estimated SBM whole fluid contamination (wt%)	<0.1
	Estimated SBM stock tank fluid contamination (wt%)	2

Note:

The Mole composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.68, 5142-EA to C36+
Depth: 4837.4m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	1.047	0.100
C ₂	Ethane	0.363	0.065
C ₃	Propane	0.377	0.099
iC ₄	i-Butane	0.212	0.073
nC ₄	n-Butane	0.241	0.083
C ₅	neo-Pentane	0.031	0.013
iC ₅	i-Pentane	0.353	0.151
nC ₅	n-Pentane	0.252	0.108
C ₆	Hexanes	0.945	0.482
	Me-Cyclo-pentane	0.313	0.156
	Benzene	0.455	0.210
	Cyclo-hexane	0.294	0.147
C ₇	Heptanes	1.830	1.087
	Me-Cyclo-hexane	1.057	0.615
	Toluene	2.318	1.266
C ₈	Octanes	3.947	2.672
	Ethyl-benzene	0.274	0.172
	Meta/Para-xylene	4.029	2.535
	Ortho-xylene	0.963	0.606
C ₉	Nonanes	6.150	4.674
	Tri-Me-benzene	1.803	1.284
C ₁₀	Decanes	10.737	9.053
C ₁₁	Undecanes	11.602	10.105
C ₁₂	Dodecanes	10.060	9.598
C ₁₃	Tridecanes	8.990	9.323
C ₁₄	Tetradecanes	6.692	7.534
C ₁₅	Pentadecanes	5.693	6.949
C ₁₆	Hexadecanes	4.062	5.343
C ₁₇	Heptadecanes	2.989	4.198
C ₁₈	Octadecanes	2.471	3.676
C ₁₉	Nonadecanes	1.936	3.016
C ₂₀	Eicosanes	1.456	2.372
C ₂₁	Heneicosanes	1.227	2.116
C ₂₂	Docosanes	0.970	1.753
C ₂₃	Tricosanes	0.773	1.456
C ₂₄	Tetracosanes	0.648	1.270
C ₂₅	Pentacosanes	0.521	1.065
C ₂₆	Hexacosanes	0.412	0.876
C ₂₇	Heptacosanes	0.351	0.777
C ₂₈	Octacosanes	0.269	0.619
C ₂₉	Nonacosanes	0.219	0.522
C ₃₀	Triacotanes	0.171	0.421
C ₃₁	Hentriacotanes	0.131	0.334
C ₃₂	Dotriacotanes	0.098	0.258
C ₃₃	Tristriacotanes	0.081	0.219
C ₃₄	Tetratriacotanes	0.058	0.162
C ₃₅	Pentatriacotanes	0.046	0.133
C ₃₆₊	Hexatriacotanes plus	0.083	0.254
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia

AFL 2014028

Pharos-1 MDT

**Calculated Uncontaminated Composition of Sample 1.68, 5142-EA to C36+
Depth: 4837.4m MDRT**

Calculated Residue Properties		
C ₇₊	Mole%	96.179
	Molecular Weight (g mol-1)	173
	Density at 15.6°C (g cm-3)	0.8107
C ₁₁₊	Mole%	62.009
	Molecular Weight (g mol-1)	202
	Density at 15.6°C (g cm-3)	0.8301
C ₂₀₊	Mole%	7.514
	Molecular Weight (g mol-1)	328
	Density at 15.6°C (g cm-3)	0.8804
C ₃₆₊	Mole %	0.083
	Molecular Weight (g mol-1)	516
	Density at 15.6°C (g cm-3)	0.9219
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		168.8
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		3

Note:

The Mole composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.69, 17242-IB to C36+
 Depth: 4837.4m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.832	0.078
C ₂	Ethane	0.333	0.058
C ₃	Propane	0.370	0.095
iC ₄	i-Butane	0.208	0.071
nC ₄	n-Butane	0.245	0.083
C ₅	neo-Pentane	0.024	0.010
iC ₅	i-Pentane	0.345	0.145
nC ₅	n-Pentane	0.246	0.103
C ₆	Hexanes	0.914	0.460
	Me-Cyclo-pentane	0.302	0.149
	Benzene	0.438	0.200
	Cyclo-hexane	0.283	0.139
C ₇	Heptanes	1.783	1.043
	Me-Cyclo-hexane	1.042	0.597
	Toluene	2.303	1.239
C ₈	Octanes	3.987	2.659
	Ethyl-benzene	0.274	0.170
	Meta/Para-xylene	4.162	2.580
	Ortho-xylene	0.985	0.611
C ₉	Nonanes	6.350	4.754
	Tri-Me-benzene	1.738	1.220
C ₁₀	Decanes	11.004	9.139
C ₁₁	Undecanes	11.463	9.833
C ₁₂	Dodecanes	9.743	9.157
C ₁₃	Tridecanes	8.628	8.814
C ₁₄	Tetradecanes	6.424	7.125
C ₁₅	Pentadecanes	5.375	6.463
C ₁₆	Hexadecanes	3.898	5.051
C ₁₇	Heptadecanes	2.849	3.942
C ₁₈	Octadecanes	2.398	3.514
C ₁₉	Nonadecanes	1.867	2.867
C ₂₀	Eicosanes	1.491	2.393
C ₂₁	Heneicosanes	1.436	2.439
C ₂₂	Docosanes	1.341	2.387
C ₂₃	Tricosanes	1.103	2.048
C ₂₄	Tetracosanes	0.851	1.645
C ₂₅	Pentacosanes	0.664	1.337
C ₂₆	Hexacosanes	0.541	1.134
C ₂₇	Heptacosanes	0.408	0.890
C ₂₈	Octacosanes	0.333	0.753
C ₂₉	Nonacosanes	0.257	0.604
C ₃₀	Triacotanes	0.196	0.476
C ₃₁	Hentriacotanes	0.149	0.374
C ₃₂	Dotriacotanes	0.109	0.283
C ₃₃	Tristriacotanes	0.091	0.242
C ₃₄	Tetratriacotanes	0.068	0.187
C ₃₅	Pentatriacotanes	0.051	0.143
C ₃₆₊	Hexatriacotanes plus	0.098	0.296
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

**Calculated Uncontaminated Composition of Sample 1.69, 17242-IB to C36+
Depth: 4837.4m MDRT**

Calculated Residue Properties

C₇₊	Mole%	96.483
	Molecular Weight (g mol-1)	176
	Density at 15.6°C (g cm-3)	0.8123
C₁₁₊	Mole%	61.832
	Molecular Weight (g mol-1)	206
	Density at 15.6°C (g cm-3)	0.8324
C₂₀₊	Mole%	9.187
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8805
C₃₆₊	Mole %	0.098
	Molecular Weight (g mol-1)	517
	Density at 15.6°C (g cm-3)	0.9221

Calculated Whole Sample Properties

Average mole weight (g mol-1)	171.3
Estimated SBM whole fluid contamination (wt%)	<0.1
Estimated SBM stock tank fluid contamination (wt%)	2

Note:

The calculated composition of the flashed condensate sample from 1.69 was used to back out the mud contamination.

The flashed condensate sample from 1.69 contains additional contamination in the C20 to C30 range. This represents approximately 5 to 10% by weight of the flashed fluid and could not be removed from the composition. The same contamination is also evident in sample 1.73.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.70, 4990-EA to C36+
Depth: 4837.4m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.486	0.046
C ₂	Ethane	0.345	0.062
C ₃	Propane	0.404	0.106
iC ₄	i-Butane	0.226	0.078
nC ₄	n-Butane	0.253	0.087
C ₅	neo-Pentane	0.043	0.019
iC ₅	i-Pentane	0.372	0.159
nC ₅	n-Pentane	0.264	0.113
C ₆	Hexanes	0.948	0.485
	Me-Cyclo-pentane	0.315	0.157
	Benzene	0.454	0.211
	Cyclo-hexane	0.287	0.144
C ₇	Heptanes	1.816	1.080
	Me-Cyclo-hexane	1.053	0.614
	Toluene	2.347	1.284
C ₈	Octanes	4.057	2.752
	Ethyl-benzene	0.303	0.191
	Meta/Para-xylene	4.260	2.686
	Ortho-xylene	1.009	0.636
C ₉	Nonanes	6.486	4.940
	Tri-Me-benzene	1.576	1.125
C ₁₀	Decanes	11.410	9.641
C ₁₁	Undecanes	11.782	10.282
C ₁₂	Dodecanes	10.069	9.627
C ₁₃	Tridecanes	8.909	9.259
C ₁₄	Tetradecanes	6.479	7.310
C ₁₅	Pentadecanes	5.562	6.805
C ₁₆	Hexadecanes	3.974	5.239
C ₁₇	Heptadecanes	2.940	4.138
C ₁₈	Octadecanes	2.389	3.562
C ₁₉	Nonadecanes	1.903	2.973
C ₂₀	Eicosanes	1.429	2.333
C ₂₁	Heneicosanes	1.183	2.044
C ₂₂	Docosanes	0.935	1.693
C ₂₃	Tricosanes	0.740	1.397
C ₂₄	Tetracosanes	0.623	1.225
C ₂₅	Pentacosanes	0.502	1.029
C ₂₆	Hexacosanes	0.405	0.862
C ₂₇	Heptacosanes	0.337	0.748
C ₂₈	Octacosanes	0.255	0.587
C ₂₉	Nonacosanes	0.214	0.510
C ₃₀	Triacotanes	0.164	0.404
C ₃₁	Hentriacotanes	0.127	0.325
C ₃₂	Dotriacotanes	0.093	0.246
C ₃₃	Tritriacotanes	0.076	0.208
C ₃₄	Tetratriacotanes	0.060	0.169
C ₃₅	Pentatriacotanes	0.045	0.129
C ₃₆₊	Hexatriacotanes plus	0.091	0.280
	Totals :	100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia

AFL 2014028

Pharos-1 MDT

Calculated Uncontaminated Composition of Sample 1.70, 4990-EA to C36+
Depth: 4837.4m MDRT

Calculated Residue Properties		
C7+	Mole%	96.659
	Molecular Weight (g mol-1)	172
	Density at 15.6°C (g cm-3)	0.8094
C11+	Mole%	61.286
	Molecular Weight (g mol-1)	202
	Density at 15.6°C (g cm-3)	0.8297
C20+	Mole%	7.279
	Molecular Weight (g mol-1)	328
	Density at 15.6°C (g cm-3)	0.8804
C36+	Mole %	0.091
	Molecular Weight (g mol-1)	518
	Density at 15.6°C (g cm-3)	0.9223
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		168.4
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		3

Note:

The MDT composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.71, 1473-EA to C36+
Depth: 4829.5m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.612	0.057
C ₂	Ethane	0.280	0.049
C ₃	Propane	0.306	0.078
iC ₄	i-Butane	0.172	0.058
nC ₄	n-Butane	0.187	0.063
C ₅	neo-Pentane	0.041	0.017
iC ₅	i-Pentane	0.304	0.127
nC ₅	n-Pentane	0.223	0.094
C ₆	Hexanes	0.856	0.429
	Me-Cyclo-pentane	0.294	0.144
	Benzene	0.433	0.197
	Cyclo-hexane	0.266	0.130
C ₇	Heptanes	1.726	1.005
	Me-Cyclo-hexane	1.008	0.575
	Toluene	2.246	1.203
C ₈	Octanes	3.856	2.560
	Ethyl-benzene	0.266	0.164
	Meta/Para-xylene	4.062	2.507
	Ortho-xylene	0.969	0.598
C ₉	Nonanes	6.105	4.551
	Tri-Me-benzene	1.786	1.248
C ₁₀	Decanes	10.804	8.935
C ₁₁	Undecanes	11.627	9.936
C ₁₂	Dodecanes	10.001	9.359
C ₁₃	Tridecanes	8.915	9.068
C ₁₄	Tetradecanes	6.966	7.693
C ₁₅	Pentadecanes	5.642	6.756
C ₁₆	Hexadecanes	4.120	5.316
C ₁₇	Heptadecanes	2.981	4.106
C ₁₈	Octadecanes	2.519	3.676
C ₁₉	Nonadecanes	1.927	2.945
C ₂₀	Eicosanes	1.483	2.371
C ₂₁	Heneicosanes	1.304	2.206
C ₂₂	Docosanes	1.107	1.963
C ₂₃	Tricosanes	0.910	1.683
C ₂₄	Tetracosanes	0.763	1.469
C ₂₅	Pentacosanes	0.601	1.205
C ₂₆	Hexacosanes	0.502	1.047
C ₂₇	Heptacosanes	0.429	0.932
C ₂₈	Octacosanes	0.321	0.724
C ₂₉	Nonacosanes	0.265	0.619
C ₃₀	Triacotanes	0.201	0.487
C ₃₁	Hentriacontanes	0.156	0.390
C ₃₂	Dotriacontanes	0.116	0.298
C ₃₃	Trtriacontanes	0.094	0.249
C ₃₄	Tetratriacontanes	0.077	0.210
C ₃₅	Pentatriacontanes	0.061	0.172
C ₃₆₊	Hexatriacontanes plus	0.110	0.331
	Totals :	100.000	100.000

Note: 0.000 means less than 0.0005.

Conoco Phillips Australia

AFL 2014028

Pharos-1 MDT

Calculated Uncontaminated Composition of Sample 1.71, 1473-EA to C36+
Depth: 4829.5m MDRT

Calculated Residue Properties		
C7+	Mole%	97.019
	Molecular Weight (g mol-1)	176
	Density at 15.6°C (g cm-3)	0.8124
C11+	Mole%	63.198
	Molecular Weight (g mol-1)	205
	Density at 15.6°C (g cm-3)	0.8316
C20+	Mole%	8.500
	Molecular Weight (g mol-1)	331
	Density at 15.6°C (g cm-3)	0.8813
C36+	Mole %	0.110
	Molecular Weight (g mol-1)	516
	Density at 15.6°C (g cm-3)	0.9220
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		172.0
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		2

Note:

The MDT composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.72, 9083-MA to C36+
 Depth: 4829.5m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.868	0.083
C ₂	Ethane	0.412	0.073
C ₃	Propane	0.444	0.116
iC ₄	i-Butane	0.237	0.082
nC ₄	n-Butane	0.269	0.093
C ₅	neo-Pentane	0.043	0.018
iC ₅	i-Pentane	0.374	0.160
nC ₅	n-Pentane	0.267	0.114
C ₆	Hexanes	0.951	0.486
	Me-Cyclo-pentane	0.317	0.158
	Benzene	0.460	0.213
	Cyclo-hexane	0.286	0.143
C ₇	Heptanes	1.841	1.094
	Me-Cyclo-hexane	1.054	0.614
	Toluene	2.330	1.273
C ₈	Octanes	4.036	2.734
	Ethyl-benzene	0.296	0.187
	Meta/Para-xylene	4.128	2.599
	Ortho-xylene	0.981	0.618
C ₉	Nonanes	6.322	4.809
	Tri-Me-benzene	1.782	1.270
C ₁₀	Decanes	10.958	9.246
C ₁₁	Undecanes	11.529	10.047
C ₁₂	Dodecanes	9.809	9.365
C ₁₃	Tridecanes	8.807	9.139
C ₁₄	Tetradecanes	6.750	7.606
C ₁₅	Pentadecanes	5.527	6.752
C ₁₆	Hexadecanes	4.039	5.317
C ₁₇	Heptadecanes	2.919	4.103
C ₁₈	Octadecanes	2.480	3.692
C ₁₉	Nonadecanes	1.904	2.970
C ₂₀	Eicosanes	1.431	2.334
C ₂₁	Heneicosanes	1.226	2.115
C ₂₂	Docosanes	0.984	1.779
C ₂₃	Tricosanes	0.777	1.465
C ₂₄	Tetracosanes	0.653	1.282
C ₂₅	Pentacosanes	0.519	1.062
C ₂₆	Hexacosanes	0.423	0.901
C ₂₇	Heptacosanes	0.347	0.770
C ₂₈	Octacosanes	0.267	0.615
C ₂₉	Nonacosanes	0.222	0.530
C ₃₀	Triacotanes	0.172	0.423
C ₃₁	Hentriacotanes	0.133	0.340
C ₃₂	Dotriacotanes	0.101	0.266
C ₃₃	Tritriacotanes	0.084	0.227
C ₃₄	Tetratriacotanes	0.061	0.171
C ₃₅	Pentatriacotanes	0.053	0.152
C ₃₆₊	Hexatriacotanes plus	0.127	0.394
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.72, 9083-MA to C36+
Depth: 4829.5m MDRT

Calculated Residue Properties		
C7+	Mole%	96.135
	Molecular Weight (g mol-1)	173
	Density at 15.6°C (g cm-3)	0.8105
C11+	Mole%	61.344
	Molecular Weight (g mol-1)	203
	Density at 15.6°C (g cm-3)	0.8305
C20+	Mole%	7.580
	Molecular Weight (g mol-1)	330
	Density at 15.6°C (g cm-3)	0.8809
C36+	Mole %	0.127
	Molecular Weight (g mol-1)	523
	Density at 15.6°C (g cm-3)	0.9231
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		168.6
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		2

Note:

The MDT composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.73, 14809-QA to C36+
Depth: 4829.5m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.792	0.074
C ₂	Ethane	0.301	0.053
C ₃	Propane	0.348	0.090
iC ₄	i-Butane	0.201	0.068
nC ₄	n-Butane	0.246	0.083
C ₅	neo-Pentane	0.022	0.009
iC ₅	i-Pentane	0.350	0.148
nC ₅	n-Pentane	0.253	0.107
C ₆	Hexanes	0.970	0.488
	Me-Cyclo-pentane	0.327	0.161
	Benzene	0.475	0.217
	Cyclo-hexane	0.302	0.148
C ₇	Heptanes	1.931	1.130
	Me-Cyclo-hexane	1.116	0.640
	Toluene	2.469	1.329
C ₈	Octanes	4.267	2.847
	Ethyl-benzene	0.340	0.211
	Meta/Para-xylene	4.345	2.695
	Ortho-xylene	1.035	0.642
C ₉	Nonanes	6.583	4.932
	Tri-Me-benzene	1.831	1.285
C ₁₀	Decanes	10.988	9.132
C ₁₁	Undecanes	11.197	9.614
C ₁₂	Dodecanes	9.307	8.753
C ₁₃	Tridecanes	8.177	8.358
C ₁₄	Tetradecanes	6.368	7.067
C ₁₅	Pentadecanes	5.105	6.142
C ₁₆	Hexadecanes	3.766	4.884
C ₁₇	Heptadecanes	2.746	3.801
C ₁₈	Octadecanes	2.327	3.412
C ₁₉	Nonadecanes	1.820	2.795
C ₂₀	Eicosanes	1.468	2.358
C ₂₁	Heneicosanes	1.500	2.550
C ₂₂	Docosanes	1.447	2.579
C ₂₃	Tricosanes	1.211	2.250
C ₂₄	Tetracosanes	0.945	1.827
C ₂₅	Pentacosanes	0.708	1.427
C ₂₆	Hexacosanes	0.538	1.127
C ₂₇	Heptacosanes	0.446	0.975
C ₂₈	Octacosanes	0.335	0.759
C ₂₉	Nonacosanes	0.273	0.641
C ₃₀	Triacotanes	0.203	0.493
C ₃₁	Hentriacontanes	0.151	0.380
C ₃₂	Dotriacontanes	0.112	0.291
C ₃₃	Trtriacontanes	0.089	0.239
C ₃₄	Tetratriacontanes	0.069	0.191
C ₃₅	Pentatriacontanes	0.055	0.157
C ₃₆₊	Hexatriacontanes plus	0.145	0.441
	Totals :	100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.73, 14809-QA to C36+
Depth: 4829.5m MDRT

Calculated Residue Properties		
C7+	Mole%	96.517
	Molecular Weight (g mol-1)	175
	Density at 15.6°C (g cm-3)	0.8125
C11+	Mole%	60.508
	Molecular Weight (g mol-1)	208
	Density at 15.6°C (g cm-3)	0.8335
C20+	Mole%	9.695
	Molecular Weight (g mol-1)	330
	Density at 15.6°C (g cm-3)	0.8809
C36+	Mole %	0.145
	Molecular Weight (g mol-1)	521
	Density at 15.6°C (g cm-3)	0.9227
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		171.2
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		2

Note:

The calculated composition of the flashed condensate sample from 1.73 contains additional contamination in the C20 to C30 range. This represents approximately 5 to 10% by weight of the flashed fluid and could not be removed from the composition. The same contamination is also evident in sample 1.69.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.74, 18974-QA to C36+
Depth: 4829.5m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	0.650	0.062
C ₂	Ethane	0.347	0.062
C ₃	Propane	0.374	0.097
iC ₄	i-Butane	0.197	0.068
nC ₄	n-Butane	0.230	0.079
C ₅	neo-Pentane	0.041	0.017
iC ₅	i-Pentane	0.332	0.142
nC ₅	n-Pentane	0.241	0.103
C ₆	Hexanes	0.899	0.458
	Me-Cyclo-pentane	0.303	0.151
	Benzene	0.452	0.208
	Cyclo-hexane	0.272	0.135
C ₇	Heptanes	1.798	1.064
	Me-Cyclo-hexane	1.042	0.604
	Toluene	2.315	1.260
C ₈	Octanes	4.044	2.729
	Ethyl-benzene	0.265	0.166
	Meta/Para-xylene	4.180	2.622
	Ortho-xylene	1.000	0.627
C ₉	Nonanes	6.367	4.824
	Tri-Me-benzene	1.707	1.212
C ₁₀	Decanes	11.187	9.403
C ₁₁	Undecanes	11.667	10.132
C ₁₂	Dodecanes	10.045	9.554
C ₁₃	Tridecanes	8.917	9.218
C ₁₄	Tetradecanes	6.564	7.367
C ₁₅	Pentadecanes	5.631	6.852
C ₁₆	Hexadecanes	3.987	5.229
C ₁₇	Heptadecanes	2.969	4.156
C ₁₈	Octadecanes	2.465	3.656
C ₁₉	Nonadecanes	1.928	2.996
C ₂₀	Eicosanes	1.453	2.360
C ₂₁	Heneicosanes	1.228	2.111
C ₂₂	Docosanes	0.976	1.758
C ₂₃	Tricosanes	0.789	1.483
C ₂₄	Tetracosanes	0.648	1.267
C ₂₅	Pentacosanes	0.523	1.066
C ₂₆	Hexacosanes	0.418	0.887
C ₂₇	Heptacosanes	0.348	0.768
C ₂₈	Octacosanes	0.273	0.625
C ₂₉	Nonacosanes	0.220	0.522
C ₃₀	Triacotanes	0.170	0.417
C ₃₁	Hentriacontanes	0.132	0.336
C ₃₂	Dotriacontanes	0.098	0.258
C ₃₃	Trtriacontanes	0.081	0.219
C ₃₄	Tetratriacontanes	0.062	0.173
C ₃₅	Pentatriacontanes	0.050	0.143
C ₃₆₊	Hexatriacontanes plus	0.115	0.354
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.74, 18974-QA to C36+
Depth: 4829.5m MDRT

Calculated Residue Properties		
C7+	Mole%	96.689
	Molecular Weight (g mol-1)	173
	Density at 15.6°C (g cm-3)	0.8103
C11+	Mole%	61.757
	Molecular Weight (g mol-1)	203
	Density at 15.6°C (g cm-3)	0.8302
C20+	Mole%	7.584
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8807
C36+	Mole %	0.115
	Molecular Weight (g mol-1)	522
	Density at 15.6°C (g cm-3)	0.9229
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	169.3
	Estimated SBM whole fluid contamination (wt%)	<0.1
	Estimated SBM stock tank fluid contamination (wt%)	3

Note:

The MDT composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.77, 18969-QA to C36+
Depth: 4986.5m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	1.141	0.109
C ₂	Ethane	0.387	0.069
C ₃	Propane	0.404	0.106
iC ₄	i-Butane	0.221	0.076
nC ₄	n-Butane	0.259	0.090
C ₅	neo-Pentane	0.024	0.010
iC ₅	i-Pentane	0.346	0.149
nC ₅	n-Pentane	0.247	0.106
C ₆	Hexanes	0.905	0.465
	Me-Cyclo-pentane	0.307	0.154
	Benzene	0.432	0.201
	Cyclo-hexane	0.284	0.143
C ₇	Heptanes	1.790	1.069
	Me-Cyclo-hexane	1.032	0.604
	Toluene	2.292	1.258
C ₈	Octanes	4.049	2.757
	Ethyl-benzene	0.295	0.187
	Meta/Para-xylene	4.182	2.646
	Ortho-xylene	1.001	0.633
C ₉	Nonanes	6.425	4.911
	Tri-Me-benzene	1.763	1.263
C ₁₀	Decanes	11.230	9.522
C ₁₁	Undecanes	11.706	10.256
C ₁₂	Dodecanes	9.975	9.570
C ₁₃	Tridecanes	8.848	9.228
C ₁₄	Tetradecanes	6.352	7.192
C ₁₅	Pentadecanes	5.538	6.798
C ₁₆	Hexadecanes	3.930	5.199
C ₁₇	Heptadecanes	2.936	4.147
C ₁₈	Octadecanes	2.405	3.598
C ₁₉	Nonadecanes	1.905	2.985
C ₂₀	Eicosanes	1.430	2.344
C ₂₁	Heneicosanes	1.200	2.081
C ₂₂	Docosanes	0.939	1.707
C ₂₃	Tricosanes	0.746	1.413
C ₂₄	Tetracosanes	0.633	1.249
C ₂₅	Pentacosanes	0.508	1.045
C ₂₆	Hexacosanes	0.415	0.889
C ₂₇	Heptacosanes	0.340	0.758
C ₂₈	Octacosanes	0.263	0.607
C ₂₉	Nonacosanes	0.217	0.519
C ₃₀	Triacotanes	0.170	0.422
C ₃₁	Hentriacotanes	0.129	0.332
C ₃₂	Dotriacotanes	0.099	0.263
C ₃₃	Tritriacotanes	0.080	0.219
C ₃₄	Tetratriacotanes	0.060	0.168
C ₃₅	Pentatriacotanes	0.051	0.146
C ₃₆₊	Hexatriacotanes plus	0.109	0.337
Totals :		100.000	100.000

Note: 0.000 means less than 0.0005.

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AFL 2014028

Pharos-1 MDT

Calculated Uncontaminated Composition of Sample 1.77, 18969-QA to C36+
Depth: 4986.5m MDRT

Calculated Residue Properties		
C7+	Mole%	96.066
	Molecular Weight (g mol-1)	173
	Density at 15.6°C (g cm-3)	0.8098
C11+	Mole%	60.984
	Molecular Weight (g mol-1)	202
	Density at 15.6°C (g cm-3)	0.8300
C20+	Mole%	7.389
	Molecular Weight (g mol-1)	329
	Density at 15.6°C (g cm-3)	0.8807
C36+	Mole %	0.109
	Molecular Weight (g mol-1)	519
	Density at 15.6°C (g cm-3)	0.9224
Calculated Whole Sample Properties		
Average mole weight (g mol-1)		167.8
Estimated SBM whole fluid contamination (wt%)		<0.1
Estimated SBM stock tank fluid contamination (wt%)		2

Note:

The MDT composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
**Calculated Uncontaminated Composition of Sample 1.78, 28960-QA to C36+
Depth: 4986.5m MDRT**

	Component	Mole %	Weight %
H ₂	Hydrogen	0.000	0.000
H ₂ S	Hydrogen sulphide	0.000	0.000
CO ₂	Carbon dioxide	0.000	0.000
N ₂	Nitrogen	0.000	0.000
C ₁	Methane	1.032	0.099
C ₂	Ethane	0.431	0.077
C ₃	Propane	0.426	0.112
iC ₄	i-Butane	0.229	0.079
nC ₄	n-Butane	0.253	0.088
C ₅	neo-Pentane	0.035	0.015
iC ₅	i-Pentane	0.357	0.154
nC ₅	n-Pentane	0.251	0.108
C ₆	Hexanes	0.925	0.475
	Me-Cyclo-pentane	0.306	0.154
	Benzene	0.439	0.205
	Cyclo-hexane	0.290	0.145
C ₇	Heptanes	1.805	1.079
	Me-Cyclo-hexane	1.045	0.612
	Toluene	2.317	1.273
C ₈	Octanes	4.003	2.727
	Ethyl-benzene	0.315	0.199
	Meta/Para-xylene	4.149	2.627
	Ortho-xylene	0.992	0.628
C ₉	Nonanes	6.318	4.833
	Tri-Me-benzene	1.770	1.268
C ₁₀	Decanes	11.099	9.417
C ₁₁	Undecanes	11.647	10.213
C ₁₂	Dodecanes	10.020	9.621
C ₁₃	Tridecanes	8.876	9.263
C ₁₄	Tetradecanes	6.612	7.492
C ₁₅	Pentadecanes	5.557	6.826
C ₁₆	Hexadecanes	4.007	5.305
C ₁₇	Heptadecanes	2.921	4.128
C ₁₈	Octadecanes	2.437	3.649
C ₁₉	Nonadecanes	1.908	2.992
C ₂₀	Eicosanes	1.422	2.331
C ₂₁	Heneicosanes	1.191	2.067
C ₂₂	Docosanes	0.920	1.674
C ₂₃	Tricosanes	0.732	1.388
C ₂₄	Tetracosanes	0.621	1.225
C ₂₅	Pentacosanes	0.499	1.027
C ₂₆	Hexacosanes	0.403	0.863
C ₂₇	Heptacosanes	0.329	0.734
C ₂₈	Octacosanes	0.263	0.609
C ₂₉	Nonacosanes	0.210	0.503
C ₃₀	Triacotanes	0.161	0.400
C ₃₁	Hentriacotanes	0.126	0.324
C ₃₂	Dotriacotanes	0.092	0.244
C ₃₃	Tristriacotanes	0.073	0.200
C ₃₄	Tetratriacotanes	0.054	0.152
C ₃₅	Pentatriacotanes	0.043	0.124
C ₃₆₊	Hexatriacotanes plus	0.089	0.272
	Totals :	100.000	100.000

Note: 0.000 means less than 0.0005.

Calculated Uncontaminated Composition of Sample 1.78, 28960-QA to C36+
Depth: 4986.5m MDRT

Calculated Residue Properties		
C7+	Mole%	96.061
	Molecular Weight (g mol-1)	172
	Density at 15.6°C (g cm-3)	0.8098
C11+	Mole%	61.213
	Molecular Weight (g mol-1)	202
	Density at 15.6°C (g cm-3)	0.8297
C20+	Mole%	7.228
	Molecular Weight (g mol-1)	328
	Density at 15.6°C (g cm-3)	0.8803
C36+	Mole %	0.089
	Molecular Weight (g mol-1)	516
	Density at 15.6°C (g cm-3)	0.9220
Calculated Whole Sample Properties		
	Average mole weight (g mol-1)	167.7
	Estimated SBM whole fluid contamination (wt%)	<0.1
	Estimated SBM stock tank fluid contamination (wt%)	2

Note:

The MDT composition from page 5.1 was used to back calculate the mud contamination

Conoco Phillips Australia
Pharos-1

AFL 2014028

Section I - Trace Component Analysis

Trace Component Analysis

Sample no.	Cylinder no.	Sample Depth m MDRT	Component			
			Helium He mole%	Hydrogen H2 mole%	Argon Ar mole%	Oxygen O2 mole%
1.65	14952-IB	4837.4	0.012	0.004	0.002	0.010
1.73	14809-QA	4829.5	0.011	0.002	0.003	0.012

Note:
1) 0.000 means < 0.0005 mol%

Sulphur Species Analysis

Sample no.	Chamber no.	Sample Depth m MD	Component				
			Hydrogen Sulphide H ₂ S	Carbonyl Sulphide COS	Methyl Mercaptan CH ₃ -SH	Ethyl Mercaptan C ₂ H ₅ -SH	Propyl Mercaptan C ₃ H ₇ -SH
			ppm (mol)	ppm (mol)	ppm (mol)	ppm (mol)	ppm (mol)
1.65	14952-IB	4837.4	6.5	0.0	0.0	0.0	0.0
1.73	14809-QA	4829.5	7.5	0.0	0.0	0.0	0.0

Note:
1) 0.0 means < 0.05 ppm (mol)

Conoco Phillips Australia
Pharos-1

AFL 2014028

Mercury Content of Gas Sample No. 1.65, Cylinder 14952-IB
Depth: 4837.4m MDRT

Mercury in Gas - Condensate trap and heated Amasil traps

Sample No.	Chamber No.	Depth (m MDRT)	Sampling Date	Time	Analysis Date	Mercury $\mu\text{g}/\text{m}^3$ at 15.6°C	Average $(\mu\text{g}/\text{m}^3 \text{ at } 15.6^\circ\text{C})$
1.65	14952-IB	4837.4	03-Aug-14	10:14	21-Oct-14	1.5	
						2.2	
						1.7	
						2.7	
						5.2	
						2.3	
						5.6	3.0

Mercury in Flashed Condensate

Sample No.	Chamber No.	Depth (m MD)	Sampling Date	Time	Analysis Date	Mercury (PPB wt or ng/g)	Average (PPB wt or ng/g)
1.65	14952-IB	4837.4	03-Aug-14	10:14	27-Oct-14	10.1	
						13.4	11.8

Mercury in Total Fluid

CGR	Mercury in Flashed Gas g/m^3 at 15.6°C	Mercury in Condensate (ng/g)	Mercury in Flashed Condensate per m^3 of Flashed Gas (μg)	Total Mercury from 1 m^3 of flashed gas at 15.6°C plus Corresponding Condensate (μg)
5.0	3.0	11.8	0.3	3.3

Note: Evaluation Only. Created with Aspose.Cells for Python via .NET. Copyright 2003 - 2025 Aspose Pty Ltd.
See page A.1 for analysis methods and further information.

Mercury Content of Gas Sample No. 1.73, Cylinder 14809-QA
Depth: 4829.5m MDRT

Mercury in Gas - Condensate trap and heated Amasil traps							
Sample No.	Chamber No.	Depth (m TVD)	Sampling Date	Time	Analysis Date	Mercury $\mu\text{g}/\text{m}^3$ at 15.6°C	Average $(\mu\text{g}/\text{m}^3 \text{ at } 15.6^\circ\text{C})$
1.73	14809-QA	4829.5	03-Aug-14	10:14	23-Oct-14	0.2	
						0.3	
						0.5	
						0.3	
						0.5	0.4

Mercury in Flashed Condensate							
Sample No.	Chamber No.	Depth (m MD)	Sampling Date	Time	Analysis Date	Mercury (PPB wt or ng/g)	Average (PPB wt or ng/g)
1.73	14809-QA	4829.50	03-Aug-14	10:14	27-Oct-14	11.1	
						12.4	11.8

Mercury in Total Fluid					Total Mercury from 1 m ³ of flashed gas at 15.6°C plus Corresponding Condensate (μg)
CGR	Mercury in Flashed Gas $\mu\text{g}/\text{m}^3$ at 15.6°C	Mercury in Condensate (ng/g)	Mercury in Flashed Condensate per m ³ of Flashed Gas (μg)		
5.2	0.4	11.8	0.3		0.6

Note: See page A.1 for analysis methods and further information.

Section J - Constant Composition Expansion Data - MDT Gas

Conoco Phillips Australia
Pharos-1
AFL 2014028
Constant Composition Expansion at 165.0°C
SCAR mix (9803-MA & 14974-QA); depth: 4829.5m MDBRT

Pressure (psig)	Relative Volume (1)	Retrograde Liquid Volume % (2) (3)		Density (g cm-3)	Deviation Factor Z	Calculated Gas Viscosity (cP) (4)
9995	0.3444			0.3156	1.355	0.0425
9495	0.3536			0.3074	1.322	0.0410
8995	0.3638			0.2988	1.288	0.0395
8495	0.3751			0.2898	1.254	0.0380
7995	0.3879			0.2803	1.221	0.0364
7495	0.4025			0.2701	1.188	0.0349
7266	Reservoir pressure	0.4100		0.2651	1.173	0.0342
6995		0.4196		0.2591	1.156	0.0333
6495		0.4398		0.2472	1.125	0.0317
5995		0.4640		0.2343	1.096	0.0301
5495		0.4934		0.2203	1.069	0.0285
4995		0.5300		0.2051	1.044	0.0269
4495		0.5761		0.1887	1.021	0.0253
3995		0.6358		0.1710	1.002	0.0237
3495		0.7149		0.1520	0.986	0.0222
2995		0.8236		0.1320	0.974	0.0207
2795		0.8789		0.1237	0.971	0.0202
2595		0.9436		0.1152	0.968	0.0196
2495		0.9803		0.1109	0.967	0.0194
2445	Dew point pressure	1.0000	0.00	0.00	0.1087	0.967
2395		1.0205	0.00	0.00		
2295		1.0643	0.01	0.01		
2195		1.1123	0.01	0.01		
1995		1.2233	0.02	0.02		
1795		1.3597	0.03	0.02		
1595		1.5310	0.04	0.02		
1395		1.7521	0.04	0.02		
1194		2.0493	0.04	0.02		
894		2.7410	0.04	0.01		
694		3.5301	0.04	0.01		

(1) Relative Volume = V / V_{sat} ie. volume at indicated pressure per volume at dew point pressure.

(2) Retrograde liquid volume as a percentage of the volume at dew point pressure.

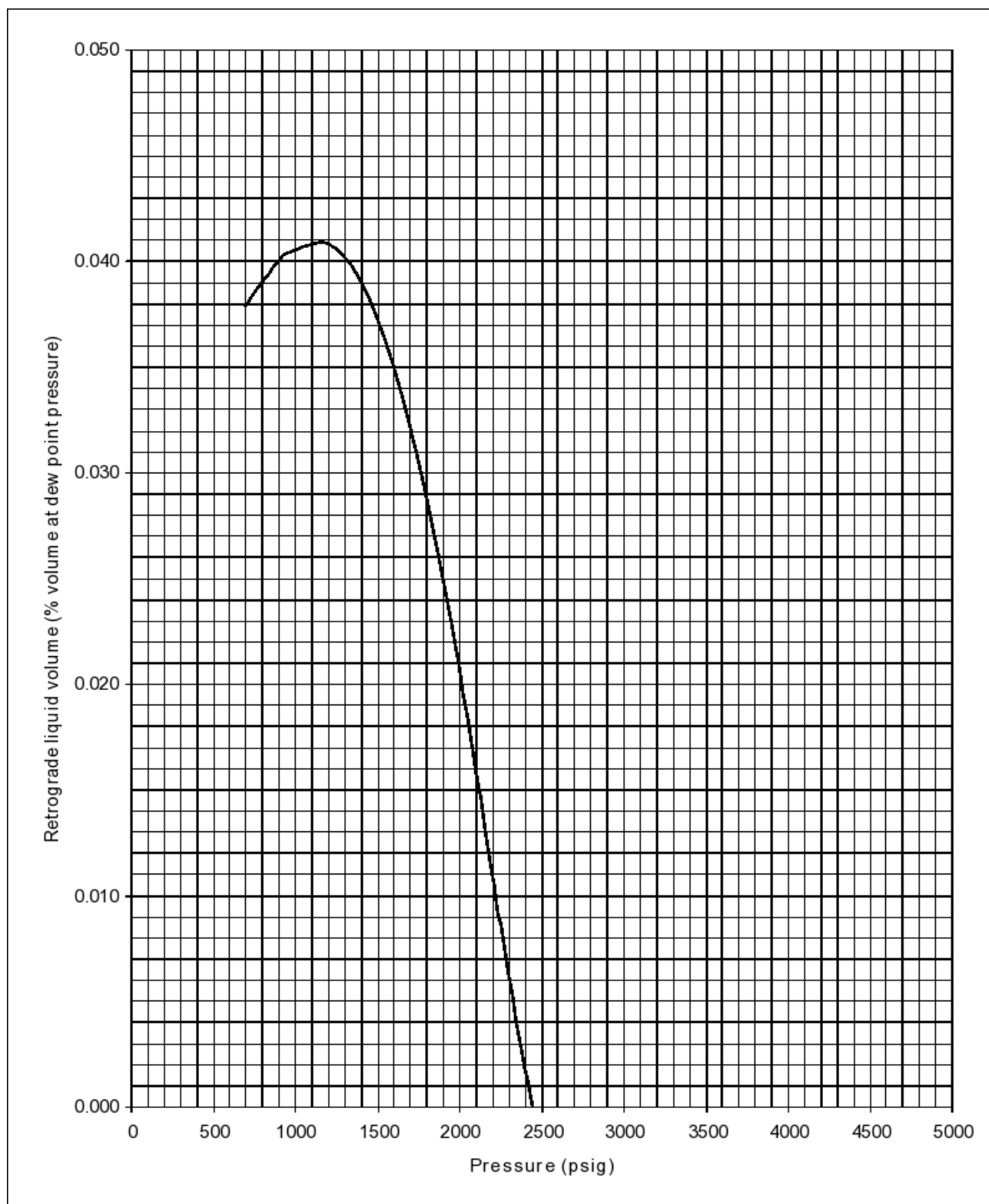
(3) Retrograde liquid volume as a percentage of the volume at indicated pressure.

(4) Calculated using the method of Lee, Gonzales and Eakin, JPT, Aug 1966.

Conoco Phillips Australia
Pharos-1

AFL 2014028

Retrograde Liquid Curve During Constant Composition Expansion at 165°C
SCAR mix (9803-MA & 14974-QA); depth: 4829.5m MDBRT



Conoco Phillips Australia
Pharos-1
AFL 2014028
Constant Composition Expansion at 165.0°C
Slickline (18969-QA); depth: 4986.0m MDBRT

Pressure (psig)	Relative Volume (1)	Retrograde Liquid Volume % (2)	(3)	Density (g cm-3)	Deviation Factor Z	Calculated Gas Viscosity (cP) (4)
9995	0.3440			0.3173	1.351	0.0428
9495	0.3535			0.3087	1.319	0.0412
8995	0.3639			0.2999	1.286	0.0396
8495	0.3753			0.2907	1.253	0.0381
7995	0.3881			0.2811	1.220	0.0366
7495	0.4027			0.2709	1.187	0.0350
7266	Reservoir pressure	0.4102		0.2660	1.172	0.0343
6995		0.4197		0.2600	1.154	0.0335
6495		0.4396		0.2482	1.123	0.0319
5995		0.4635		0.2354	1.093	0.0303
5495		0.4926		0.2215	1.065	0.0286
4995		0.5288		0.2063	1.039	0.0270
4495		0.5747		0.1899	1.017	0.0254
3995		0.6343		0.1720	0.998	0.0238
3495		0.7136		0.1529	0.983	0.0222
2995		0.8228		0.1326	0.972	0.0208
2795		0.8785		0.1242	0.968	0.0202
2595		0.9436		0.1156	0.966	0.0197
2495		0.9803		0.1113	0.965	0.0194
2445	Dew point pressure	1.0000	0.00	0.1091	0.965	0.0193
2395		1.0205	0.00	0.00		
2295		1.0644	0.01	0.01		
2195		1.1124	0.01	0.01		
1995		1.2234	0.02	0.02		
1795		1.3599	0.03	0.02		
1595		1.5313	0.03	0.02		
1395		1.7526	0.04	0.02		
1194		2.0501	0.04	0.02		
894		2.7422	0.04	0.01		
694		3.5319	0.04	0.01		

(1) Relative Volume = V / V_{sat} ie. volume at indicated pressure per volume at dew point pressure.

(2) Retrograde liquid volume as a percentage of the volume at dew point pressure.

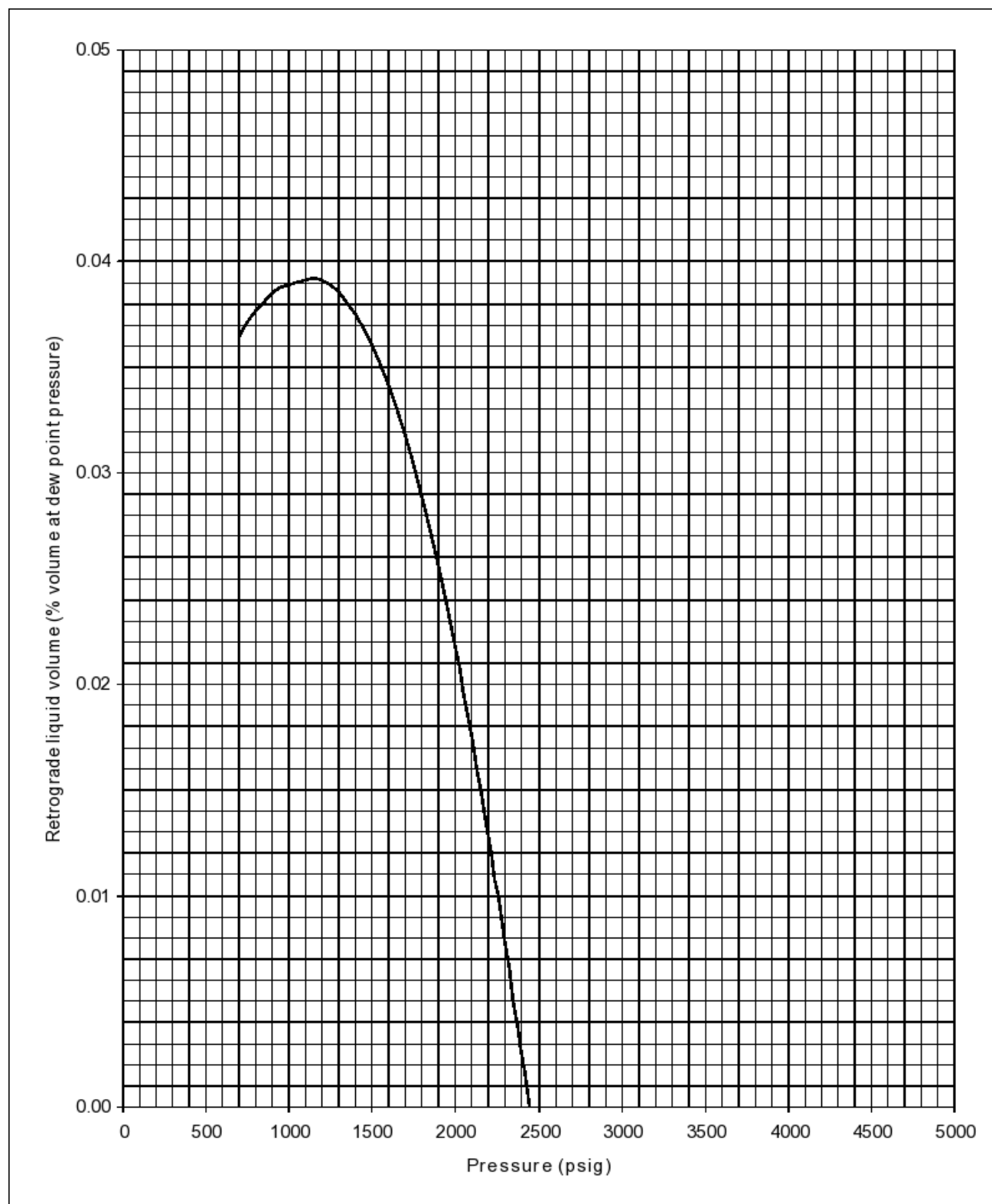
(3) Retrograde liquid volume as a percentage of the volume at indicated pressure.

(4) Calculated using the method of Lee, Gonzales and Eakin, JPT, Aug 1966.

Conoco Phillips Australia
Pharos-1

AFL 2014028

Retrograde Liquid Curve During Constant Composition Expansion at 165°C
Slickline (18969-QA); depth: 4986.0m MDBRT



Section K- Constant Volume Depletion Data

Conoco Phillips Australia
Pharos-1**AFL 2014028****Summary of Wellstream Data From Constant Volume Depletion at 165.0°C**
SCAR mix (9803-MA & 14974-QA); depth: 4829.5m MDBRT

Reservoir Pressure (psig)	Deviation Factor Z Equilibrium Gas	Factor Z 2 Phase	Density (g cm-3) (1)	Produced Wellstream (cumulative mole%)	Retrograde Liquid Volume% (2)	Calculated Gas Viscosity (cP) (3)
2445	0.967	0.967	0.1087	0.00	0.000	0.019
2095	0.966	0.966	0.0925	14.20	0.015	0.018
1795	0.967	0.967	0.0792	26.46	0.023	0.018
1495	0.970	0.970	0.0659	38.82	0.028	0.017
1195	0.974	0.973	0.0525	51.16	0.031	0.017
894	0.979	0.979	0.0392	63.51	0.030	0.016
694	0.984	0.983	0.0304	71.66	0.027	0.016

Note:

(1) Density at indicated pressure and temperature.

(2) Liquid volume as a percentage of the sample volume at dew point pressure.

(3) Calculated by the method of Lee, Gonzales and Eakin, JPT, Aug 1966, pp 997-1000.

Conoco Phillips Australia
Pharos-1
AFL 2014028
**Compositions of Produced Wellstreams From Constant Volume Depletion at 16
 SCAR mix (9803-MA & 14974-QA); depth: 4829.5m MDBRT**

Reservoir pressure (psig) 2445* 2445** 2095 1795 1495 1195 894 694 694***		Component (mole%)								
H ₂	Hydrogen	0.00	0.00	0.000	0.00	0.00	0.00	0.00	0.00	0.00
H ₂ S	Hydrogen Sulphide	0.00	0.00	0.000	0.00	0.00	0.00	0.00	0.00	0.00
CO ₂	Carbon Dioxide	13.83	13.89	13.838	13.84	13.88	13.90	13.92	13.91	0.00
N ₂	Nitrogen	0.32	0.32	0.316	0.32	0.32	0.32	0.32	0.32	0.00
C ₁	Methane	77.57	77.80	77.787	77.79	77.78	77.83	77.84	77.88	0.58
C ₂	Ethane	4.86	4.88	4.862	4.88	4.88	4.89	4.88	4.89	0.18
C ₃	Propane	1.34	1.28	1.289	1.30	1.29	1.29	1.28	1.28	0.11
iC ₄	i-Butane	0.36	0.30	0.312	0.31	0.31	0.30	0.30	0.30	0.05
nC ₄	n-Butane	0.32	0.27	0.278	0.27	0.28	0.27	0.27	0.27	0.06
C ₅	Neo-Pentane	0.00	0.01	0.004	0.01	0.01	0.01	0.00	0.01	0.00
iC ₅	i-Pentane	0.21	0.18	0.186	0.18	0.19	0.18	0.18	0.18	0.10
nC ₅	n-Pentane	0.11	0.10	0.102	0.10	0.10	0.10	0.10	0.10	0.08
C ₆	Hexanes	0.20	0.19	0.196	0.19	0.19	0.19	0.18	0.18	0.59
	M-C-Pentane	0.03	0.03	0.035	0.04	0.03	0.03	0.03	0.03	0.20
	Benzene	0.05	0.05	0.053	0.05	0.05	0.05	0.05	0.04	0.25
	Cyclohexane	0.06	0.05	0.058	0.06	0.05	0.05	0.05	0.05	0.17
C ₇	Heptanes	0.11	0.09	0.102	0.10	0.10	0.09	0.09	0.08	1.22
	M-C-Hexane	0.05	0.04	0.049	0.05	0.05	0.04	0.04	0.04	0.77
	Toluene	0.10	0.09	0.099	0.09	0.09	0.08	0.09	0.08	1.73
C ₈	Octanes	0.08	0.08	0.077	0.08	0.08	0.08	0.08	0.08	3.39
	E-Benzene	0.01	0.00	0.004	0.00	0.00	0.00	0.00	0.00	0.33
	M/P-Xylene	0.05	0.05	0.053	0.05	0.05	0.05	0.04	0.04	3.52
	O-Xylene	0.01	0.01	0.008	0.01	0.01	0.01	0.01	0.01	0.86
C ₉	Nonanes	0.06	0.05	0.061	0.06	0.05	0.05	0.05	0.05	5.90
	1,2,4-TMB	0.01	0.01	0.006	0.01	0.01	0.01	0.01	0.00	1.77
C ₁₀	Decanes	0.06	0.05	0.052	0.05	0.05	0.05	0.05	0.05	10.54
C ₁₁	Undecanes	0.04	0.04	0.038	0.04	0.04	0.04	0.04	0.04	11.72
C ₁₂₊	Dodecanes plus	0.15	0.14	0.135	0.14	0.14	0.13	0.12	0.12	55.88
Totals :		100.00	100.00	###	100.00	100.00	100.00	100.00	100.00	###

Calculated Wellstream Properties
C₁₂₊

Molecular Weight (g mol-	215	-	179	182	186	192	192	189	210
Density at 15.6°C (g cm-	0.8369	-	0.8140	0.8163	0.8192	0.8229	0.8227	0.8208	0.8340

Whole Sample

Molecular Weight (g mol-	22.6	-	22.4	22.4	22.4	22.3	22.3	22.3	173
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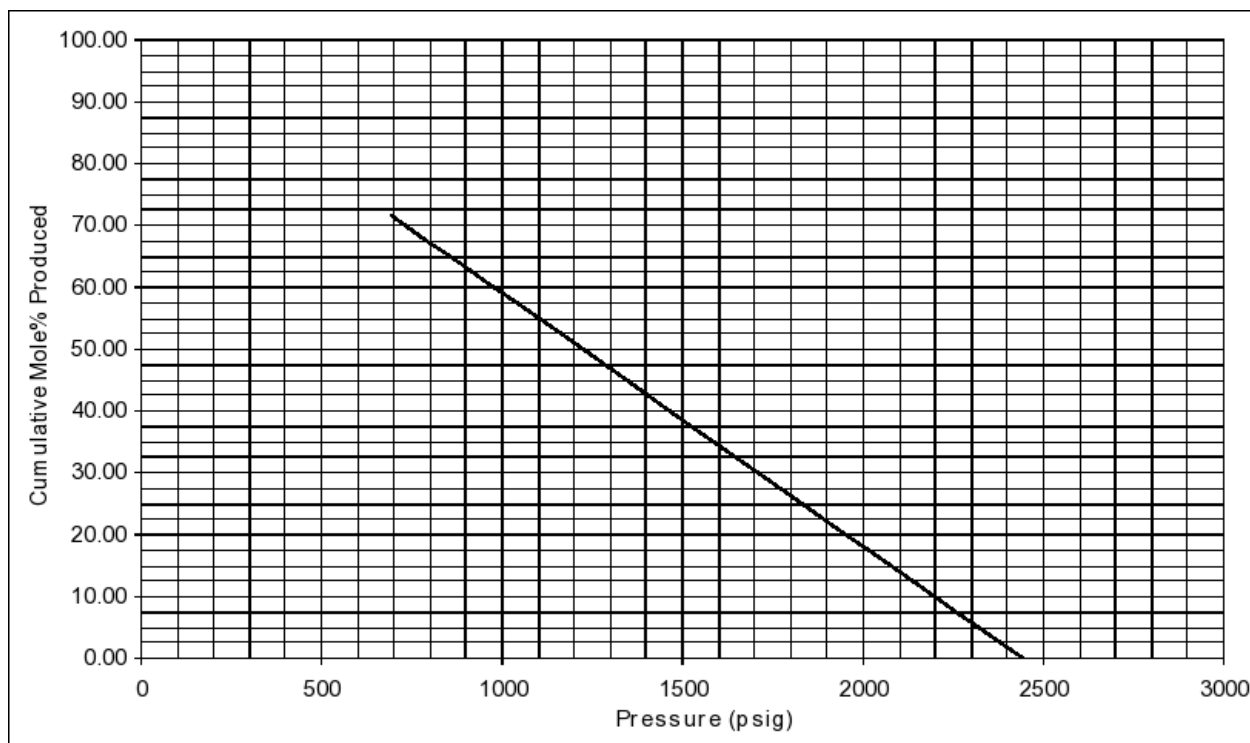
- 1) * Measured reservoir fluid composition.
 2) ** Recovered material balance composition
 3) *** Insufficient liquid at 700psig to perform an equilibrium liquid composition. Hence the sample was flashed to atmospheric conditions and the residual liquid composition determined.

Conoco Phillips Australia Pharos-1

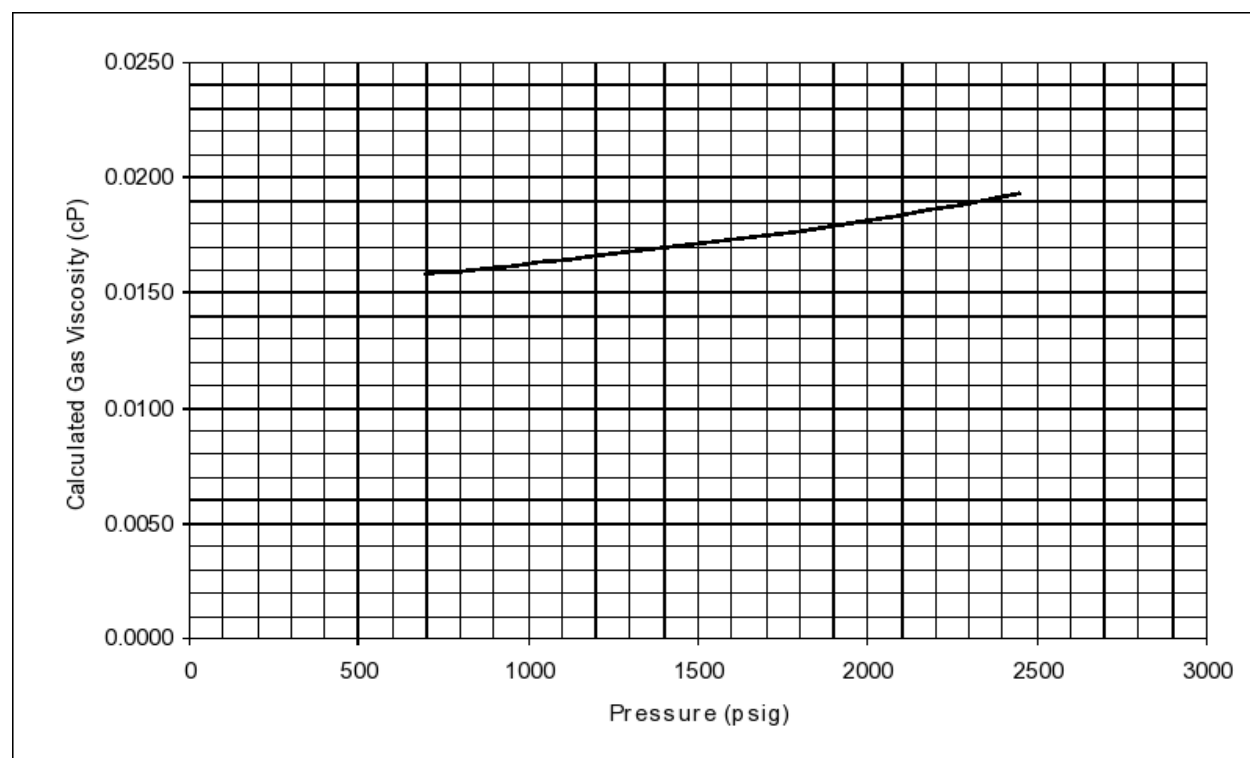
AFL 2014028

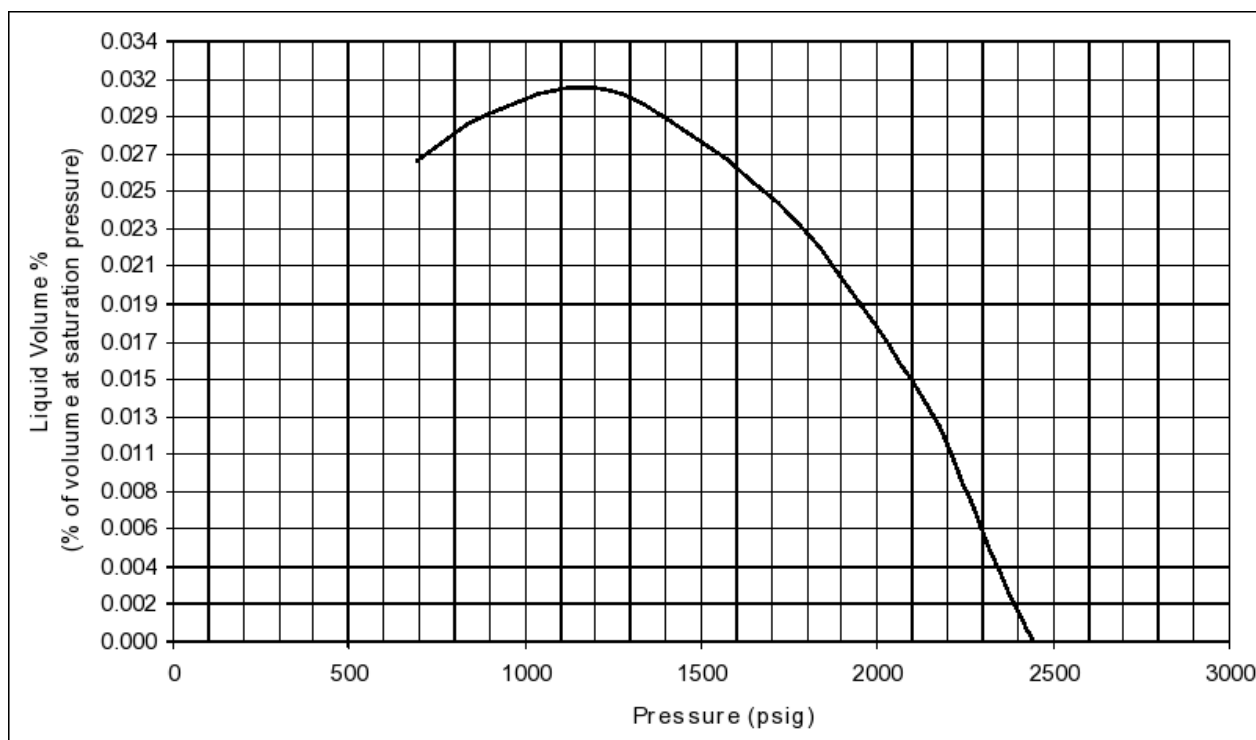
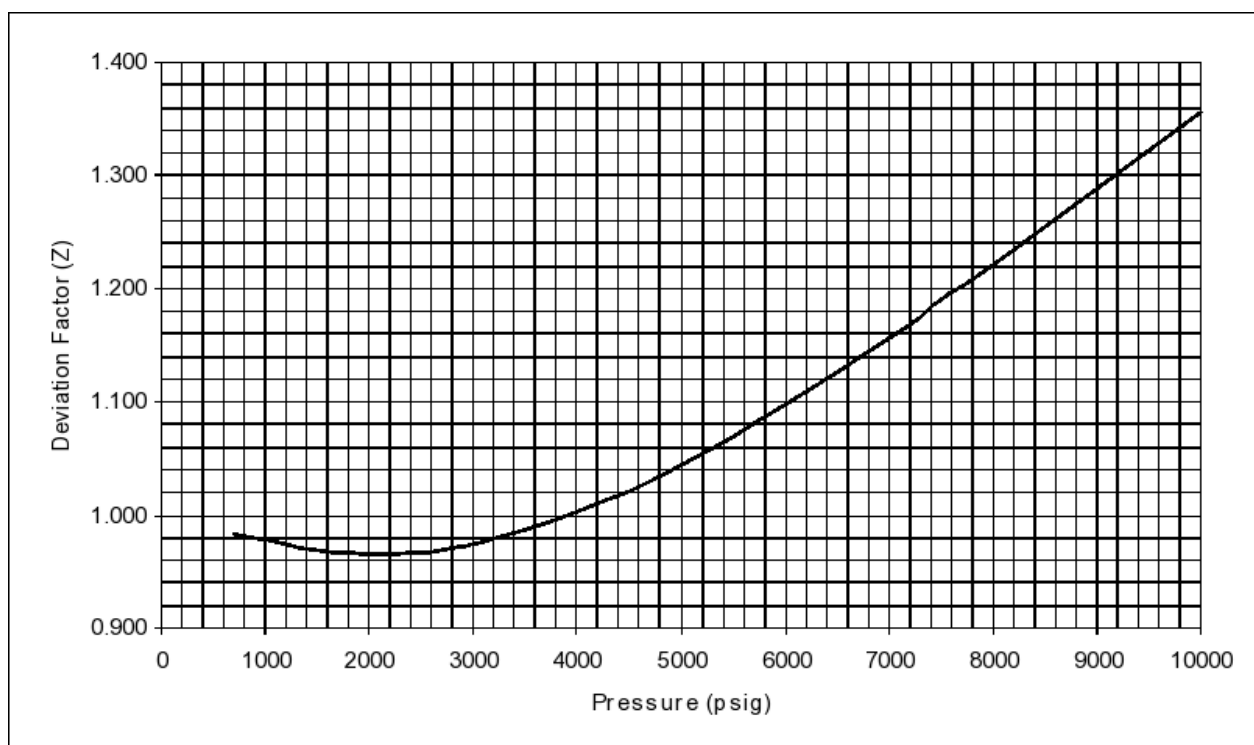
Graphs of Constant Volume Depletion Data at 165.0°C SCAR mix (9803-MA & 14974-QA); depth: 4829.5m MDBRT

Pressure v Cumulative Mole% Produced



Pressure v Calculated Gas Viscosity



**Graphs of CVD and CCE Data at 165.0°C
SCAR mix (9803-MA & 14974-QA); depth: 4829.5m MDBRT****Pressure v Liquid Volume%****Pressure v Deviation Factor Z**

Conoco Phillips Australia
Pharos-1**AFL 2014028****Summary of Wellstream Data From Constant Volume Depletion at 165.0°C**
Slickline (18969-QA); depth: 4986.0m MDBRT

Reservoir Pressure (psig)	Deviation Factor Z Equilibrium Gas	Factor Z 2 Phase	Density (g cm-3) (1)	Produced Wellstream (cumulative mole%)	Retrograde Liquid Volume% (2)	Calculated Gas Viscosity (cP) (3)
2445	0.965	0.965	0.1091	0.00	0.000	0.019
2095	0.964	0.964	0.0928	14.16	0.016	0.018
1795	0.966	0.966	0.0793	26.46	0.026	0.018
1495	0.969	0.968	0.066	38.81	0.032	0.017
1195	0.972	0.972	0.0526	51.15	0.037	0.017
894	0.978	0.977	0.0393	63.51	0.037	0.016
694	0.982	0.981	0.0305	71.67	0.035	0.016

Note:

(1) Density at indicated pressure and temperature.

(2) Liquid volume as a percentage of the sample volume at dew point pressure.

(3) Calculated by the method of Lee, Gonzales and Eakin, JPT, Aug 1966, pp 997-1000.

Conoco Phillips Australia
Pharos-1
AFL 2014028
Compositions of Produced Wellstreams From Constant Volume Depletion at 16 Slickline (18969-QA); depth: 4986.0m MDBRT

Reservoir pressure (psig) 2445* 2445** 2095 1795 1495 1195 894 694 694***		Component (mole%)								
H ₂	Hydrogen	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
H ₂ S	Hydrogen Sulphide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CO ₂	Carbon Dioxide	13.90	13.95	13.92	13.95	13.92	13.95	13.97	13.99	0.00
N ₂	Nitrogen	0.31	0.30	0.30	0.30	0.30	0.29	0.30	0.31	0.00
C ₁	Methane	77.50	77.76	77.80	77.78	77.76	77.83	77.77	77.78	0.10
C ₂	Ethane	4.85	4.86	4.83	4.87	4.87	4.87	4.89	4.87	0.03
C ₃	Propane	1.34	1.28	1.26	1.29	1.28	1.28	1.29	1.28	0.03
iC ₄	i-Butane	0.36	0.31	0.30	0.31	0.32	0.30	0.31	0.31	0.02
nC ₄	n-Butane	0.32	0.27	0.27	0.27	0.28	0.26	0.28	0.28	0.04
C ₅	Neo-Pentane	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.00
iC ₅	i-Pentane	0.21	0.18	0.18	0.18	0.19	0.17	0.18	0.18	0.15
nC ₅	n-Pentane	0.11	0.10	0.10	0.10	0.11	0.09	0.10	0.10	0.16
C ₆	Hexanes	0.21	0.19	0.19	0.19	0.20	0.19	0.19	0.19	2.10
	M-C-Pentane	0.03	0.03	0.04	0.03	0.03	0.03	0.03	0.03	0.82
	Benzene	0.05	0.04	0.05	0.05	0.05	0.04	0.04	0.04	1.08
	Cyclohexane	0.06	0.05	0.06	0.06	0.06	0.05	0.05	0.05	0.76
C ₇	Heptanes	0.11	0.10	0.10	0.10	0.10	0.09	0.09	0.09	4.86
	M-C-Hexane	0.06	0.05	0.06	0.05	0.05	0.05	0.05	0.04	2.68
	Toluene	0.09	0.08	0.09	0.08	0.08	0.08	0.08	0.08	5.29
C ₈	Octanes	0.09	0.08	0.09	0.08	0.08	0.08	0.07	0.07	8.00
	E-Benzene	0.01	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.58
	M/P-Xylene	0.06	0.05	0.06	0.05	0.05	0.05	0.04	0.04	5.53
	O-Xylene	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1.24
C ₉	Nonanes	0.06	0.06	0.06	0.05	0.05	0.06	0.05	0.05	7.87
	1,2,4-TMB	0.01	0.01	0.01	0.01	0.01	0.00	0.00	0.00	1.62
C ₁₀	Decanes	0.07	0.06	0.06	0.05	0.05	0.06	0.05	0.05	10.08
C ₁₁	Undecanes	0.05	0.04	0.04	0.04	0.04	0.04	0.04	0.04	9.09
C ₁₂₊	Dodecanes plus	0.16	0.14	0.14	0.13	0.13	0.12	0.12	0.12	37.87
Totals :		100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

Calculated Wellstream Properties
C₁₂₊

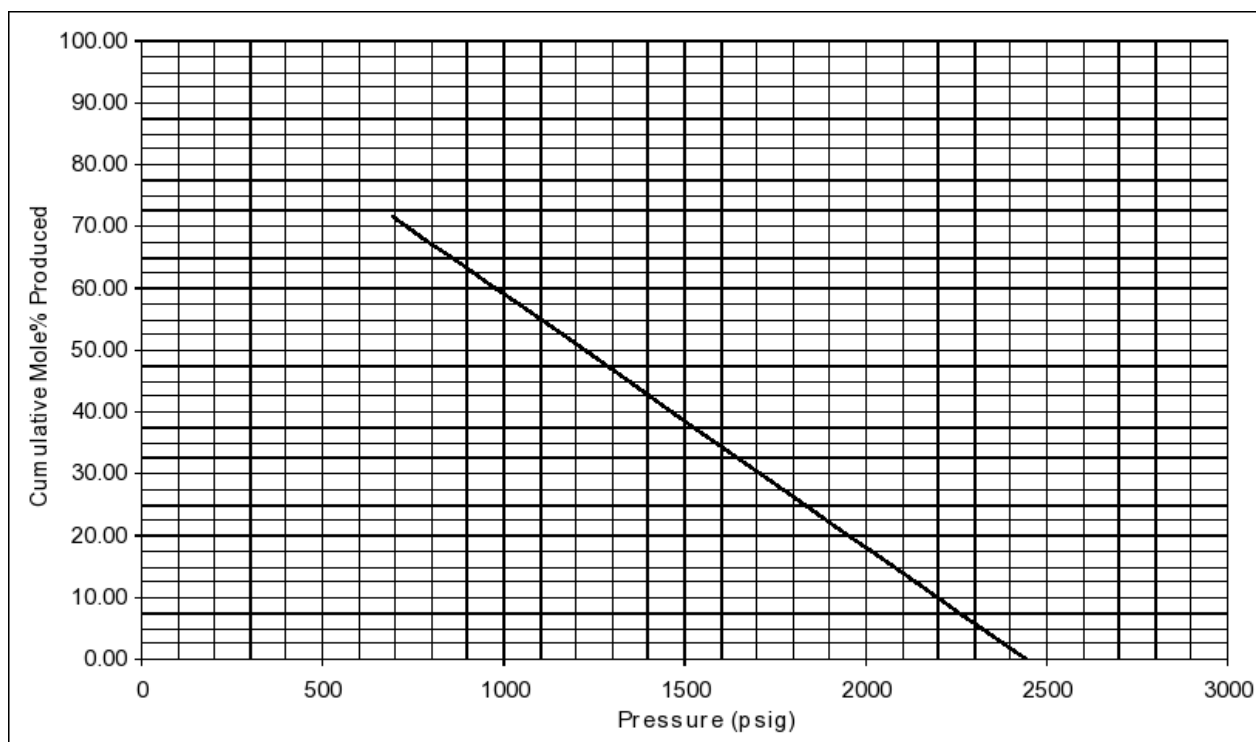
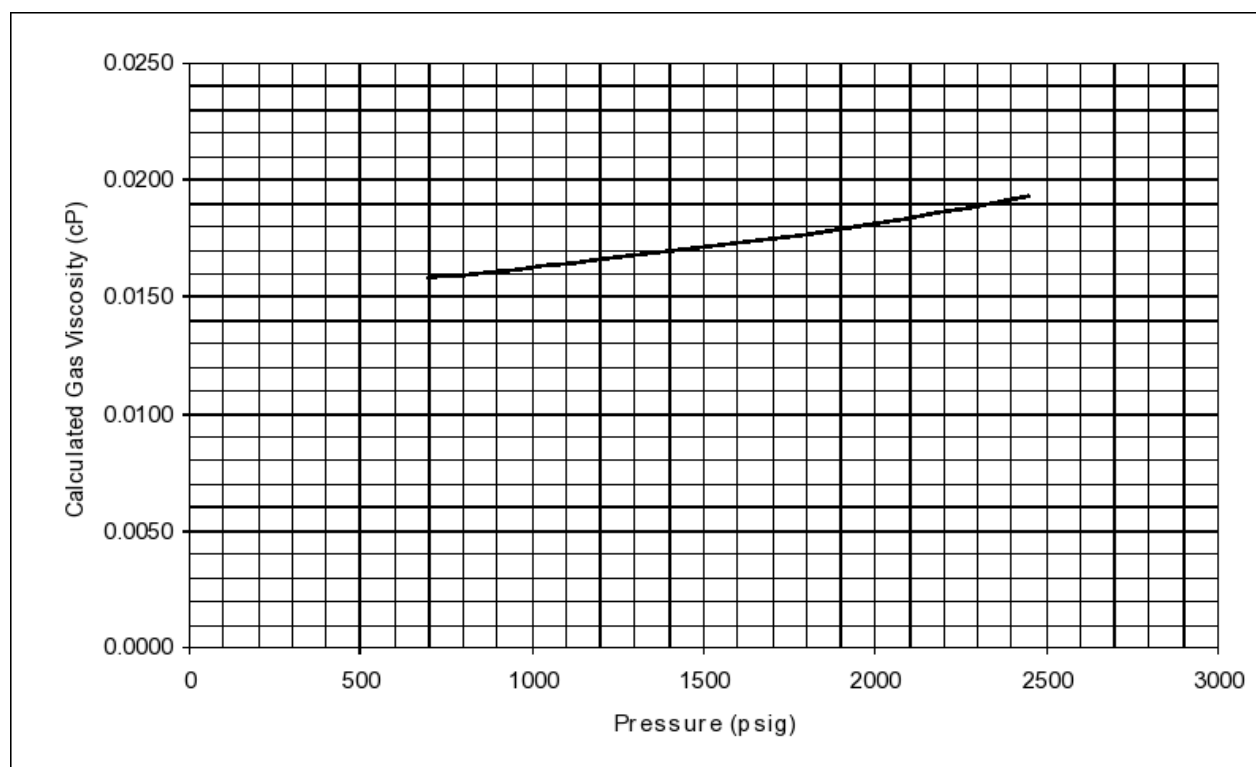
Molecular Weight (g mol ⁻¹)	213	-	178	179	188	183	186	192	209
Density at 15.6°C (g cm ⁻³)	0.836	-	0.8138	0.8143	0.8201	0.8169	0.8194	0.8228	0.8333

Whole Sample

Molecular Weight (g mol ⁻¹)	22.6	-	22.4	22.3	22.4	22.3	22.3	22.3	152
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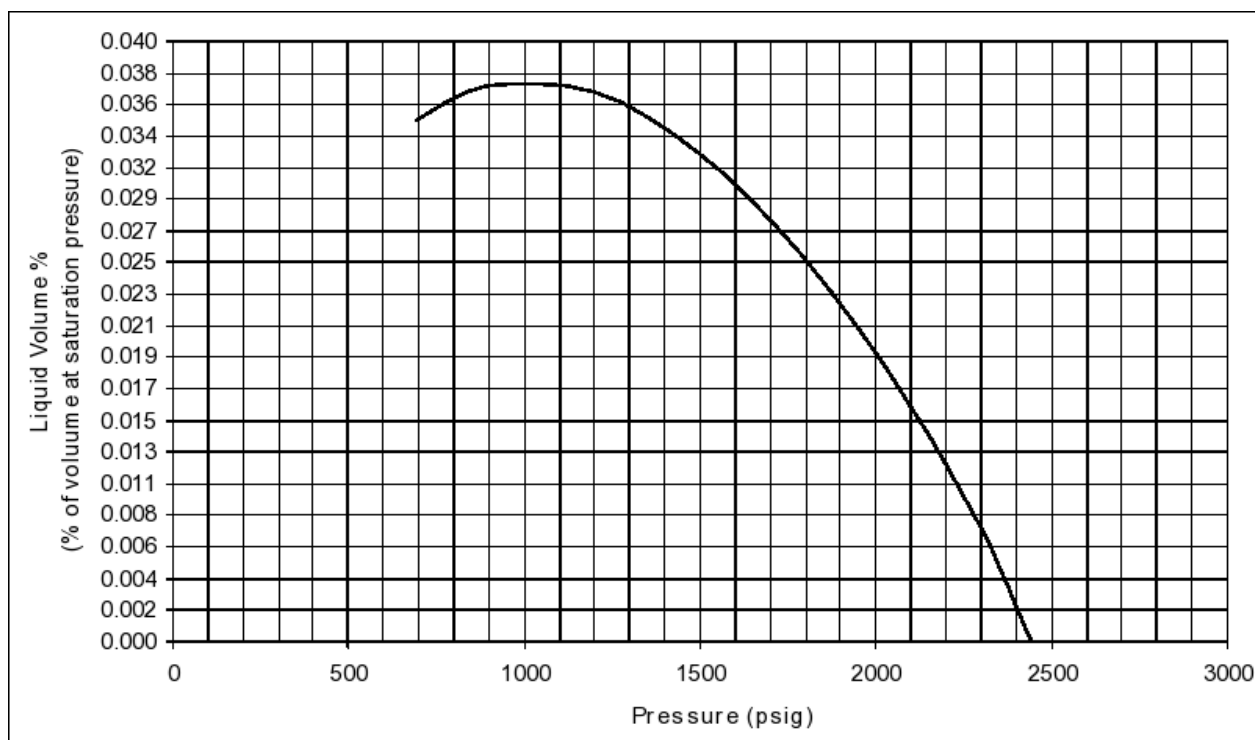
Notes: Evaluation Only. Created with Aspose.Cells for Python via .NET. Copyright 2003 - 2025 Aspose Pty Ltd.

- 1) * Measured reservoir fluid composition.
- 2) ** Recovered material balance composition
- 3) *** Insufficient liquid at 700psig to perform an equilibrium liquid composition. Hence the sample was flashed to atmospheric conditions and the residual liquid composition determined.

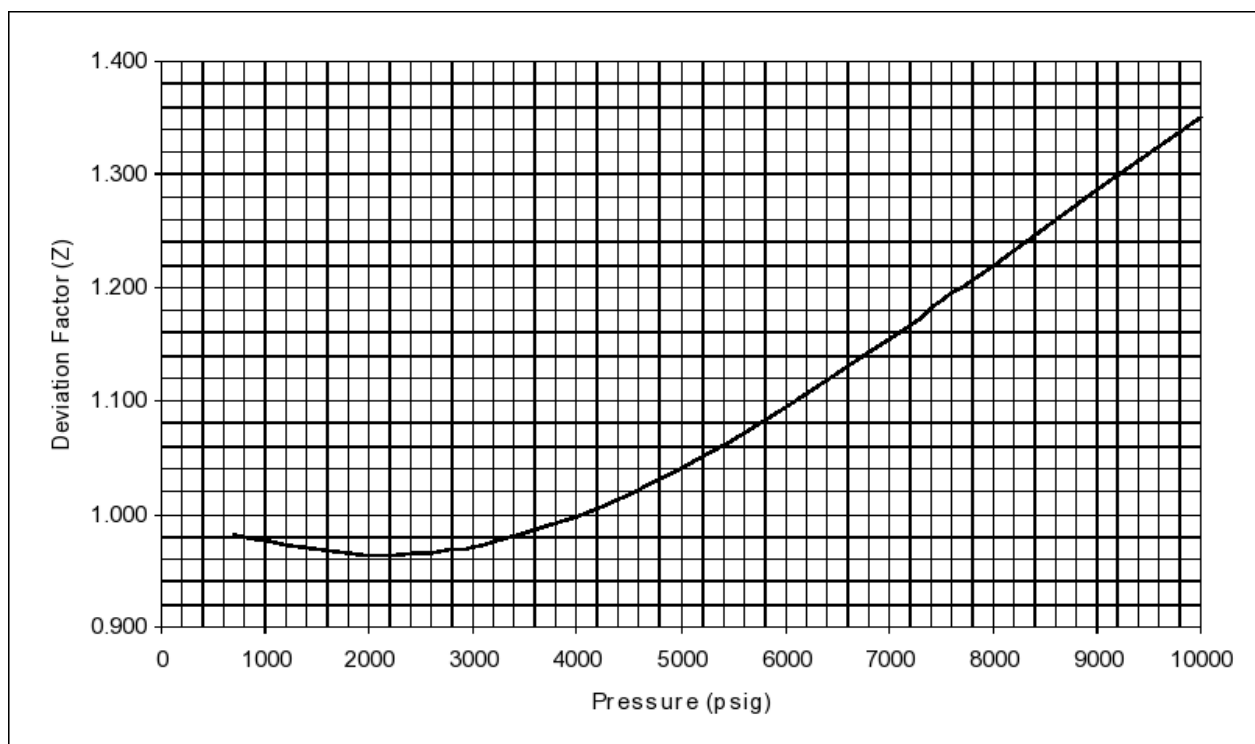
Conoco Phillips Australia
Pharos-1**AFL 2014028****Graphs of Constant Volume Depletion Data at 165.0°C**
Slickline (18969-QA); depth: 4986.0m MDBRT**Pressure v Cumulative Mole% Produced****Pressure v Calculated Gas Viscosity**

Graphs of CVD and CCE Data at 165.0°C
Slickline (18969-QA); depth: 4986.0m MDBRT

Pressure v Liquid Volume%



Pressure v Deviation Factor Z



Section L - Appendix

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Data Used in Gas Compositional Calculations

Component		Mole Weight (g mol-1)	Density g cm-3 at 60°F)	Component		Mole Weight (g mol-1)	Density (g cm-3 at 60°F)
Hydrogen	*	2.016	N/A	33DMC5	*	100.20	0.6954
Oxygen/(Argon)	**	31.999	1.1410	Cyclohexane	*	84.16	0.7827
Nitrogen (Corrected)	**	28.013	0.8086	2MC6/23DMC5	*	100.20	0.6917
Methane	**	16.043	0.2997	11DMCYC5/3MC6	*	99.20	0.7253
Carbon Dioxide	**	44.010	0.8172	t13DMCYC5	*	98.19	0.7528
Ethane	**	30.070	0.3558	c13DMCYC5/3EC5	*	99.20	0.7262
Hydrogen Sulphide	**	34.080	0.8006	t12DMCYC5	*	98.19	0.7554
Propane	**	44.097	0.5065	Heptanes (nC7)	*	100.20	0.6875
i-Butane	**	58.123	0.5623	22DMC6	*	114.23	0.6994
n-Butane	**	58.123	0.5834	MCYC6	*	98.19	0.7740
Neo-Pentane	*	72.15	0.5968	ECYC5	*	98.19	0.7704
i-Pentane	**	72.150	0.6238	223TMC5/24&25DMC6	*	114.23	0.7060
n-Pentane	**	72.150	0.6305	ctc124TMCYC5	*	112.21	0.7511
22DMC4	*	86.18	0.6529	ctc123TMCYC5	*	112.21	0.7574
23DMC4/CYC5	*	78.16	0.7129	Toluene	*	92.14	0.8734
2MC5	*	86.18	0.6572	Octanes (nC8)	*	114.23	0.7063
3MC5	*	86.18	0.6682	E-Benzene	*	106.17	0.8735
Hexanes (nC6)	*	86.18	0.6631	M/P-Xylene	*	106.17	0.8671
22DMC5	*	100.20	0.6814	O-Xylene	*	106.17	0.8840
M-C-Pentane	*	84.16	0.7533	Nonanes (nC9)	*	128.26	0.7212
24DMC5	*	100.20	0.6757	Decanes	***	134	0.778
223TMC4	*	100.20	0.6947	Undecanes	***	147	0.789
Benzene	*	78.11	0.8820	Dodecanes	***	161	0.800

Data Source Refs :

* ASTM Data Series Publication DS 4B (1991) - Physical Constants of Hydrocarbon and Non-Hydrocarbon Compounds.

** GPA Table of Physical Constants of Paraffin Hydrocarbons and Other Components of Natural Gas, GPA 2145-96.

*** Journal of Petroleum Technology, Nov 1978, Pages 1649-1655.
Predicting Phase Behaviour of Condensate/Crude Oil Systems Using Methane Interaction Coefficients
- D.L. Katz & A. Firoozabadi.

Note:

The gas mole % compositions were calculated from the measured weight % compositions using the most detailed analysis results, involving as many of the above components as were identified. The reported component mole % compositions were then sub-grouped into the generic carbon number components.

Conoco Phillips Australia
Pharos-1 MDT
AFL 2014028
Data Used in Liquid Compositional Calculations

Component		Mole Weight (g mol-1)	Density g cm-3 at 60°F)	Component		Mole Weight (g mol-1)	Density (g cm-3 at 60°F)
Hydrogen	*	2.016	N/A	Undecanes	***	147	0.789
Hyd. sulphide	**	34.080	0.8006	Dodecanes	***	161	0.800
Carbon Dioxide	**	44.010	0.8172	Tridecanes	***	175	0.811
Nitrogen	**	28.013	0.8086	Tetradecanes	***	190	0.822
Methane	**	16.043	0.2997	Pentadecanes	***	206	0.832
Ethane	**	30.070	0.3558	Hexadecanes	***	222	0.839
Propane	**	44.097	0.5065	Heptadecanes	***	237	0.847
i-Butane	**	58.123	0.5623	Octadecanes	***	251	0.852
n-Butane	**	58.123	0.5834	Nonadecanes	***	263	0.857
i-Pentane	**	72.150	0.6238	Eicosanes	***	275	0.862
n-Pentane	**	72.150	0.6305	Heneicosanes	***	291	0.867
Hexanes	**	86.177	0.6634	Docosanes	***	305	0.872
Me-cyclo-pentane	*	84.16	0.7533	Tricosanes	***	318	0.877
Benzene	*	78.11	0.8820	Tetracosanes	***	331	0.881
Cyclo-hexane	*	84.16	0.7827	Pentacosanes	***	345	0.885
Heptanes	**	100.204	0.6874	Hexacosanes	***	359	0.889
Me-cyclo-hexane	*	98.19	0.7740	Heptacosanes	***	374	0.893
Toluene	*	92.14	0.8734	Octacosanes	***	388	0.896
Octanes	**	114.231	0.7061	Nonacosanes	***	402	0.899
Ethyl-benzene	*	106.17	0.8735	triacontanes	***	416	0.902
Meta/Para-xylene	*	106.17	0.8671	Hentriacontanes	***	430	0.906
Ortho-xylene	*	106.17	0.8840	Dotriacontanes	***	444	0.909
Nonanes	**	128.258	0.7212	Tritriacontanes	***	458	0.912
1-2-4-T-M-benzene	*	120.19	0.8797	Tetratriacontanes	***	472	0.914
Decanes	**	142.285	0.7334	Pentatriacontanes	***	486	0.917

Data Source Refs :

* ASTM Data Series Publication DS 4B (1991) - Physical Constants of Hydrocarbon and Non-Hydrocarbon Compounds.

** GPA Table of Physical Constants of Paraffin Hydrocarbons and Other Components of Natural Gas GPA 2145-96.

*** Journal of Petroleum Technology, Nov 1978, Pages 1649-1655.
Predicting Phase Behaviour of Condensate/Crude Oil Systems Using Methane Interaction Coefficients
- D.L. Katz & A. Firoozabadi.

Note:
The residue mole weight and density values (eg heptanes plus, undecanes plus, eicosanes plus) are calculated so that the calculated average mole weights and densities correspond with the measured values. This can lead to anomalous residue mole weights and densities where the Katz and Firoozabadi values may not be suitable for the isomer groups detected.

Conoco Phillips Australia
Pharos-1 MDT

AFL 2014028

Report prepared by

Report approved by

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Murray Macleod
Laboratory Supervisor